Adaptive Design of Clustered Surveys and Experiments

Carlos Gonzalez-Perez*

Click here for the most recent version

October 24, 2026

Abstract

When designing clustered surveys or experiments on a budget, researchers face a trade-off between the number of clusters and the number of units per cluster. Unfortunately, the variance-minimizing allocation depends on the intra-cluster correlation (ICC), a property of the data the researcher has not yet collected. So, in practice, designers guess. I show that guessing is first-order expensive: a design built on a misspecified ICC forfeits precision of the same order as the target variance itself. Instead, I propose two algorithms—one two-staged, one fully sequential—that learn the ICC while the experiment runs. They attain negligible excess variance compared to a designer who knows the true ICC, improve an order of magnitude upon any misspecified static design, and are rate optimal. Moreover, adaptivity need not invalidate conventional inference. Under a symmetry assumption on the errors, mean estimators remain asymptotically normal conditional on the realized design, with no data discarded and no correction. The theory on inference invariance extends beyond the cluster setting and validates standard inference for popular sampling procedures such as Neyman allocations with variances computed from pilot data. Revisiting Fink et al. (2020b), I find that my methods would have lowered the variance by up to 41% at equal cost or reduced the cost by up to 42% at equal precision. I further demonstrate the feasibility of the algorithms by using them to field an online survey in real time. The results show that the ICC can be learned at only second-order precision cost and without compromising inference.

Keywords: Adaptive experimental design, optimal experimental design, clustered sampling, intra-cluster correlation, adaptive inference.

1 Introduction

When researchers design clustered surveys or clustered experiments, they face two costs: fixed costs of sampling additional clusters—travel, training additional staff, negotiating with local authorities—and variable costs of sampling additional units. Therefore, when operating on a budget, researchers trade-off *width*, the number of clusters, against *depth*, the number of units per cluster. For instance, Fink et al. (2020b), henceforth FJM, designed an experiment where they offered subsidized loans to farming households in Zambia to ease liquidity constraints with treatment at the village level. Their design featured 175 villages and 18 households on average per village. Visiting a new village requires renting

*University of Oxford, Department of Economics. Email: carlos.gonzalezperez@economics.ox.ac.uk. I would like to thank Maximilian Kasy, Davide Viviano, Günther Fink, Anders Kock, Francis DiTraglia and numerous seminar participants for feedback on this manuscript.

a vehicle for the day, while treating an additional household carries the monetary cost of providing one extra loan. Similar trade-offs exist between hospitals and patients per hospital, schools and students per school or provinces and villages per province. In these settings, the variance-minimizing allocation is pinned down by a single statistical parameter: the intra-cluster correlation (ICC), which measures how similar observations are within clusters relative to observations in different clusters. When units of the same cluster are highly correlated, there is little informational value in sampling many of them, hence the researcher is better-off by implementing a wide design with many clusters and few units per cluster.[1] The solution, which has been known at least since Cochran (1977), is simple and intuitive.

We cannot use it. The ICC is a property of the data the researcher has not yet collected, therefore unknown at the time of the design. So, in practice, designers guess: some apply rules of thumb, some import ICC estimates from prior studies, and some run non-optimally sized pilots.[2] The consequences of guessing are not innocuous. I show that any design built on a misspecified ICC incurs excess variance of the same order as the target variance itself. For example, the static design implemented by FJM might display up to 70% higher variance than the cost-optimal allocation.

The ICC is known to vary significantly across settings and outcomes, what makes it difficult to guess. For instance, Knox and Chondros (2004) report variations in the ICC of one order of magnitude across covariates within the same population.[3] Learning this parameter from pilot data is also non-trivial. The ICC is a function of the within- and between-cluster variances, where the latter can only be estimated by sampling many clusters—the expenditure the design is trying to minimize. The designer then faces a *circularity*: selecting the optimal design requires knowing the ICC, but learning the ICC requires sampling many clusters, which may or may not be design-optimal. Cluster-level costs are often sunk once incurred, so mistakes that take place during learning cannot be undone.

This manuscript makes a threefold contribution. Firstly, I establish formal optimality guarantees in the joint allocation problem for two algorithms that escape the circularity without knowing the ICC. The algorithms induce negligible excess variance compared to the optimal design, are rate optimal, and improve an order of magnitude upon any guessing-like policy (Theorems 5.4 and 5.5). For instance, in the context of the FJM application, my algorithms remove nearly all excess variance, reducing total variance by up to 41% compared to the authors' fixed design at no additional cost. Second, under a symmetry assumption on the errors, I show that inference can proceed as if the design had been fixed in advance, i.e. after conditioning on the realized design, estimators of population means remain asymptotically normal with the usual variance (Theorem 6.8). The result rests on a general principle: under symmetry, error signs and error magnitudes are independent. *Even* designs—coordinate-wise sign-invariant functionals—only depend on the magnitudes, hence signs remain unrestricted after conditioning on the design, and so a central limit theorem can run through them without compromising inference. This invariance principle extends beyond the cluster setting and includes many popular sampling methods such as Neyman allocations with pilot-estimated variances (Appendix D). This means that the reduced variance obtained by the adaptive algorithms can enter the usual formula for clustered standard errors without modification or data exclusion. Third, I

[1] In the extreme case where observations are perfectly correlated within clusters, the analyst extracts all of a cluster's information from a single unit, hence she would sample one unit from as many clusters as the budget allows.

[2] The design in Fink et al. (2020b) used an ICC-informed power calculation estimated through a small pilot, leading to a selection of 175 villages with up to 22 households each. Yet, they did not optimize the allocation of sampling effort across villages and households against the relative costs of those margins. Their register only documents: "to be powered to detect an increase in farm production value of 10%" (Fink et al., 2014).

[3] Knox and Chondros (2004) report variation of 0.055 to 0.451 on demographic covariates and from 0.005 to 0.059 on morbidity covariates within the same target population.

demonstrate the method end to end: through simulations, through a semi-synthetic reconstruction of FJM, and through an online survey which I field in real time using one of my algorithms.

This paper connects several literatures. Survey theory has long studied cost-optimal clustered designs through an algorithmic approach (Bethel, 1989; Barcaroli et al., 2022), but has rarely integrated ICC estimation within design optimization. Biostatistics, by contrast, has a long tradition on learning nuisance parameters entering the optimal design through internal pilots dating back to Stein (1945). Within the internal-pilot literature (Wittes and Brittain, 1990), the closest papers to mine are Lake et al. (2002) and van Schie and Moerbeek (2014), where the authors re-estimate the ICC within a cluster trial, but each adjusts a single margin—width in the former, depth in the latter—and neither embeds a cost structure nor derives an optimality theory. In econometrics, my paper is closest to the two-wave tradition of Hahn et al. (2011) and, in structure, to Viviano (2022), from which it differs in decision space—clustered sampling rather than network experiments—and in requiring no sample splitting for inference. The inference results extend an invariance tradition in sample-size re-estimation, where, for asymptotically even policies, I can weaken normality to error symmetry.

I start this manuscript by characterizing the optimal design under known ICC as a function of budget using a minimal model with nested errors, independent clusters, linear costs and a hard budget, as in Cochran (1977). This allocation yields an oracle variance which decreases linearly with budget. I then propose two algorithms that implement the optimal design in the limit with no prior knowledge of the ICC. Both share the same architecture: run a carefully-sized pilot, learn the ICC, and allocate the remaining budget through plug-in rules. The first of the two, Explore-then-Commit Cluster Selection (`EtC-CS`), implements a simple two-stage design with a minimal pilot; the second, Adaptive Cluster Selection (`ACS`), samples clusters sequentially, stops under a data-dependent rule, and commits to a final design. The expected excess variance of the algorithms is negligible compared to the oracle variance, and a full order of the budget smaller than the excess variance of any static design built on a misspecified ICC, including the maximin optimal design of van Breukelen and Candel (2015). I further show that the adaptive algorithms are rate unimprovable by deriving matching lower bounds on the problem. Although both algorithms display bounds of the same rate, `ACS` outperforms `EtC-CS` in simulations at the expense of greater adaptivity requirements.

It is well-known that data-dependent designs can compromise conditional inference. Most results in this literature tackle this issue by excluding pilot data (Viviano, 2022), re-weighting the terminal estimator (Yin and Shen, 2005), imposing strong normality assumptions (Rinott, 1978), conditioning on designs only with degenerate limiting probabilities (Hahn et al., 2011), assessing distortions through simulation (Wittes et al., 1999), or some combination of the above. My algorithms also select the design adaptively, therefore they could be subject to the same limitations. They are not. Under a symmetry assumption on the errors, inference on sample averages is invariant to adaptive data collection whenever the design depends only on (asymptotically) even functions of the data, such as sample variances. Asymptotic normality is then preserved with no correction and without discarding pilot observations. Unlike Hahn et al. (2011), my theory allows for conditioning on design events with non-degenerate limiting probability, which is what makes conditional and unconditional inference meaningfully distinct. While the driving heuristic of the theory leverages exact evenness, feasible designs are never exactly even as the designer can only observe the squared residuals, not the squared errors. Still, I show that the same conclusions extend to the feasible class of asymptotically even policies by coupling them with unfeasible even surrogates at the relevant rate and imposing an

anti-concentration requirement on the conditioning events. Simulations verify the results and delineate their limits: once symmetry or (asymptotic) evenness fails, conditional inference can be distorted.

The results are also borne out in simulations and in a reconstruction of FJM. FJM approached the design with more care than is typical: they ran a small pilot survey, estimated the ICC from the sample, and implemented a design "sufficiently powered to detect an increase in farm production of 10%". Yet neither the pilot nor the main experiment explicitly optimized the width-depth allocation against its cost structure. Far from exceptional, this is the norm.[4] As a consequence, under a plausible grid of fixed-to-variable cost ratios, FJM could have achieved substantially lower variance for the same budget, or spent a substantially smaller budget at fixed precision. In particular, I show that both algorithms attain nearly zero excess variance, what reduces the variance of their estimator by up to 41% and the standard errors of their main specification by up to 19% at equal costs. Conversely, my algorithms could have achieved the same precision as FJM with 42% lower budget. The application also illustrates how to test for error symmetry—the load-bearing assumption entering the inference theory. Finally, I demonstrate that the methods are implementable by using `ACS` to field an adaptive online survey in real time. The survey measures people's ability to distinguish AI-generated from human-written text across languages. Each language forms a cluster, as outcomes are likely correlated within languages, and carries a genuine fixed cost, as materials must be produced and checked by paid native speakers. The variable cost is given by the participation fee paid to each survey respondent.[5]

2 Literature Review

Clustered Designs in Survey Theory. Designing optimal clustered surveys—also known as multi-stage designs—under a budget constraint is a classical problem in survey theory, dating back to Tschuprow (1923), Hansen and Hurwitz (1942) and Kish (1965).[6] In its simplest form, the researcher first partitions the population into non-overlapping clusters, then samples K clusters at random, and finally samples N units at random within each selected cluster. The objective of this literature is the one I adopt here: to minimize the variance of a population estimator through the allocation of sampling effort across clusters and units. My adaptive designs are benchmarked against the cost-optimal oracle allocation that knows the ICC as derived by Cochran (1977).[7] Since Bethel's algorithm (Bethel, 1989), this literature has extended Cochran's logic to multi-domain, multi-variable and multi-objective settings, with an emphasis on algorithmic performance (Cicchitelli et al., 1992; Campbell et al., 2004; Ballin and Barcaroli, 2013; Barcaroli et al., 2022). Despite the central role of the ICC in these algorithms, the literature has almost never attempted to estimate it as part of the optimization problem. For instance, Barcaroli et al. (2022) assume that it can be recovered from auxiliary data, Tam (2005) advocates for recruiting as many clusters as possible, and Sedgwick (2015) recommends fixing the number of clusters in advance. Neither prescription is optimal in general.

Adaptive Designs and ICC Re-estimation in Biostatistics. The idea of using an initial sample to learn a nuisance parameter that governs the design has a long history in statistics and biostatistics. It begins with Stein (1945), who proposes a two-stage rule for Gaussian data with unknown variance and preserves the t distribution of the second-stage statistic by choosing the total sample size as a

[4] If anything, by pre-estimating the ICC via a pilot, FJM display additional care relative to the large body of designs built on guesstimates borrowed from the literature or previous studies.

[5] All proofs in this paper, including a standalone section on the general inference theory, are collected in the Appendix.

[6] See Biemer and Lyberg (2003) for a modern textbook treatment.

[7] Raudenbush (1997) and Moerbeek et al. (2000) obtain analogous oracle allocations for cluster designs.

function of the first-stage variance estimate. Rinott (1978) and Dudewicz and Dalal (1975) extend this logic to best-group identification, and Yin and Shen (2005) embed ICC re-estimation in a fully sequential GEE trial. More recently, this idea was revived by the *internal pilot* literature. Wittes and Brittain (1990) and Wittes et al. (1999) argue that nuisance parameters can be estimated mid-trial and the design revised accordingly, with the final analysis conducted on all data as if the design had been fixed ex ante. Much of this literature assesses performance and inference through simulation under Gaussianity, reporting type I error inflation that varies with the timing and size of the pilot, but without formal guarantees. Lake et al. (2002) and van Schie and Moerbeek (2014) are the closest antecedents of my paper as they specialize the internal pilots thesis to ICC estimation in the context of cluster-randomized medical trials. Two differences stand out: they optimize a single margin—width in Lake et al. (2002), depth in van Schie and Moerbeek (2014)—whereas I optimize both jointly. Also, they do not impose a cost structure and do not derive an optimality theory. A parallel literature acknowledges ICC uncertainty but holds the design fixed, either by placing Bayesian priors on the ICC (Spiegelhalter, 2001; Turner et al., 2005) or by adopting maximin designs (van Breukelen and Candel, 2015, 2018, 2021). These approaches optimize against ICC uncertainty within a static design class, but do not benchmark the resulting loss against the known-ICC oracle. I establish worst-case excess variance bounds of $O(1/B^2)$ that cannot be improved in rate, and I show that my algorithms asymptotically dominate every misspecified static policy over the ICC range—including the maximin design of van Breukelen and Candel (2015), whose worst-case excess variance is $\Theta(1/B)$ (Lemma 5.2). To my knowledge, this is the first paper to provide formal optimality guarantees for an adaptive design in the joint clusters-versus-units allocation problem.

The Econometrics of Experimental Design. My work relates thematically to the econometric literature on optimal but non-adaptive design through stratification and covariate adjustment (Cytrynbaum, 2021; Bai, 2022; Cytrynbaum, 2024). That literature shares my variance-minimization objective, and occasionally my attention to sampling costs, but neither address the cluster-versus-unit margin nor the learning problem. Closer in structure is the two-wave tradition initiated by Hahn et al. (2011), in which a pilot estimates the parameters that enter the optimal design: treatment probabilities in Hahn et al. (2011), and optimal stratification in Chambaz et al. (2015) and Tabord-Meehan (2023). Nearest to my setting is Viviano (2022), who designs two-wave experiments under network interference by minimizing estimator variance using pilot-estimated variance components, and derives regret guarantees, asymptotic normality, and a characterization of the regret-minimizing pilot size. I share the pilot-then-optimize architecture and the regret criterion, but my decision space is the clustered sampling design and my inference results require no data exclusion: Viviano (2022) must remove both the pilot units and their network neighbors from the main experiment to keep the design unconfounded, whereas I use the full sample. My work also connects to adaptive experimentation and policy learning in economics (Kasy and Sautmann, 2021; Adusumilli, 2026; Cesa-Bianchi et al., 2025; Li et al., 2026).

Neyman Regret in Sequential Experiments. My performance criterion—expected excess variance relative to the variance-minimizing oracle design—is the clustered-sampling analogue of *Neyman regret* introduced by Dai et al. (2023) for adaptive treatment allocation. Working in a design-based framework, they attain $\tilde{O}(\sqrt{T})$ Neyman regret through clipped online gradient descent and show that explore-then-commit designs can suffer linear Neyman regret when early units are unrepresentative.[8] Subsequent work attains logarithmic rates in stochastic environments (Neopane et al., 2024), and does

[8] Their linear-regret example is specific to the design-based environment they study. I work in the stochastic *iid* setting, in which explore-then-commit designs such as `EtC-CS` do not incur linear regret.

so even under the stronger notion of Neyman regret in Neopane et al. (2025). Both rates are consistent with my guarantees.

Inference with Adaptive Designs. Adaptive data collection frequently invalidates conventional inference. In particular, adaptivity can bias sample means (Nie et al., 2018) and destroy asymptotic normality (Hadad et al., 2021).[9] The remedies fall into two traditions: an *adjustment* tradition, which reweights the observations entering the estimator to restore martingale central limit behavior (Hadad et al., 2021; Zhang et al., 2021; Shen and Fisher, 1999; Yin and Shen, 2005); and an *invariance* tradition, which shows that, under normality assumptions, designs depending on the sample variance are independent of the sample mean (Stein, 1945; Rinott, 1978; Dudewicz and Dalal, 1975). More generally, designs driven by policies depending on the vector of residuals are independent of the sample mean under Gaussianity. The same invariance is used implicitly in the internal pilot and blinded sample size re-estimation literatures (Gould and Shih, 1992; Kieser and Friede, 2003). Proposition 6.6 and Theorem 6.8 in this paper belong to the latter. By restricting the policy class from policies depending on the residuals to policies depending (asymptotically) on the squared errors, I can substantially weaken the normality assumption to error symmetry. The resulting class contains my algorithms, but also many popular sampling procedures like Neyman allocations computed from pilot variances, and two-stage propensity scores as in Hahn et al. (2011).[10] In addition, the contrast with the class of conditioning events in Hahn et al. (2011) is instructive. While their validity rests on the induced design converging to a deterministic limit, my theory allows for conditioning on events with non-degenerate probability, like width quantiles. The symmetry assumption plays a role reminiscent of group invariance in randomization inference (Canay et al., 2017). The boundary of these results, however, remains equally sharp: adaptation through *odd* functionals of the data, like estimated effects or futility rules, or error asymmetry reintroduces the pathologies that motivate the adjustment tradition.

Other Connections. My work is more distantly related to the treatment of nested errors in panel data (Wooldridge, 2003), to standard error adjustment in clustered experiments (Abadie et al., 2023), and to the optimal-design tradition of Kiefer and Wolfowitz (1960), which underpins modern algorithms for linear bandits.[11] These connections are conceptual rather than methodological.

3 Cluster Design Problem

3.1 Model

Consider a population partitioned into equally sized clusters. The designer samples K clusters, indexed by $k = 1, \dots, K$, and N^k units from cluster $k \leq K$, indexed by $i = 1, \dots, N^k$.[12]

Assumption 3.1. *Linear Additive Model and Error Distributions. Let the outcome of interest y_{ki} follow an additive model with nested errors,*

$$y_{ki} = \mu + u_k + \varepsilon_{ki} \tag{1}$$

[9] See Wager (2024) for a textbook treatment.

[10] Propensity scores are ratios of pilot-estimated conditional standard deviations, therefore even. Conditional-mean heterogeneity enters the semiparametric efficiency bound additively and does not depend on the assignment probability, so it drops out of the optimization, and the score depends on the pilot only through second moments.

[11] See Lattimore and Szepesvári (2020) for a textbook treatment.

[12] The model implicitly assumes the existence of a super-population with an infinite amount of clusters and units.

where μ is the population mean and estimand, u_k is the cluster-level error and ε_{ki} is the unit-level error. $(u_k)_k$ is an iid sequence with $\mathbb{E}[u_k] = 0$, $Var\,(u_k) = \tau^2 > 0$; $(\varepsilon_{ki})_{k,i}$ is an iid sequence with $\mathbb{E}[\varepsilon_{ki}] = 0$, $Var\,(\varepsilon_{ki}) = \sigma^2 > 0$; and $(u_k)_k \perp\!\!\!\perp (\varepsilon_{ki})_{k,i}$.[13]

Define the variance ratio $\lambda_0 = \sigma^2/\tau^2$. λ_0 maps one-to-one to the intra-cluster correlation, ICC $= \tau^2/(\tau^2 + \sigma^2) = 1/(1 + \lambda_0)$, and is unknown to the designer.

Assumption 3.2. *Known Compact Parameter Space. Let $\omega = (\tau^2, \lambda_0) \in \Omega$, where $\Omega = [\tau^2_m, \tau^2_M] \times [\lambda_m, \lambda_M]$ is known to the designer, and $\tau^2_m > 0, \lambda_m > 0$.*

3.2 Variance Minimization in Survey Design

In the homogeneous-variance setting given by Assumption 3.1, the natural estimator of μ is given by the average of averages $\hat{\mu}$, i.e.

$$\hat{\mu} = \frac{1}{K} \sum_{k:N^k>0} \hat{\mu}^k_{N^k}, \qquad \hat{\mu}^k_{N^k} = \frac{1}{N^k} \sum_{i=1}^{N^k} y_{ki}, \quad k : N^k > 0$$

where $K = \sum \mathbb{1}(N^k > 0)$ is the number of sampled clusters and $\hat{\mu}^k_{N^k}$ is the sample average of cluster k. The designer selects the vector $(N^k)_k$ to minimize the variance of the mean estimator $\hat{\mu}$. Appendix A shows that, for any fixed number of sampled clusters, the variance-minimizing allocation is balanced, i.e. $N^k \in \{0, N\}$ for all k and so $\hat{\mu} = \hat{\mu}_{K,N}$.[14] Under unit balance, the variance of the estimator is given by $\mathrm{Var}(\hat{\mu}_{K,N}) = \frac{\tau^2}{K}(1 + \lambda_0/N) = W(K, N)$.

Assumption 3.3. *Linear Costs. A balanced design (K, N) incurs costs $K(F + VN)$ where F is the fixed cost of sampling an additional cluster and V is the variable cost of sampling an additional unit.*

The goal of the designer is to select (K, N) to minimize the estimator's variance subject to a hard budget constraint B that must hold almost surely,[15]

$$\min_{K\geq 2, N\geq 2} W(K, N) \qquad \text{s.t.} \quad K(F + VN) \leq B \ \text{ a.s.} \tag{2}$$

For the performance analysis, I work with the continuous relaxation of (2), treating (K, N) as real-valued.[16] I further rule out corner solutions through an additional assumption.

Assumption 3.4. *Interior Solution. $\lambda_m F/V \geq 4$ and $B \geq 2F + 2\max\left\{V, \sqrt{\lambda_M FV}\right\}$.*[17]

Under Assumptions [3.1– 3.4], the first-order conditions delivers the oracle allocation in Cochran (1977),

$$N^*(\lambda_0) = \sqrt{\lambda_0\ F/V}, \qquad K^*(\lambda_0) = \frac{B}{F + VN^*(\lambda_0)}, \qquad W_0 = \frac{\tau^2\left(\sqrt{V\lambda_0} + \sqrt{F}\right)^2}{B} = \Theta(1/B) \tag{3}$$

[13] Section 9 introduces a more general model with heterogeneous cluster variances τ^2_k.

[14] Unit balance is specific to the unweighted average of averages under homogeneous variances. It fails for the GLS-weighted average with heterogeneous cluster variances studied in Section 9.

[15] The constraint $K \geq 2$, $N \geq 2$ is imposed such that the estimators in Section 4 are well-defined. This restriction is with little loss, as nearly all clustered surveys and experiments consider larger samples anyway.

[16] In simulations, all algorithms are mapped to integer allocations through $(\lfloor K^\pi \rfloor, \lfloor N^\pi \rfloor)$. This rounding criterion guarantees implementability, eases cross-algorithm comparison, and removes differences stemming from integer approximation. In practice, alternative terminal budget allocations may achieve superior performance.

[17] The assumption is stated in primitives and it is equivalent to requiring $N^*(\lambda) \geq 2$ and $K^*(\lambda) \geq 2$ for every $\lambda \in [\lambda_m, \lambda_M]$, with $K^*(\lambda), N^*(\lambda)$ defined in (3). The assumption further guarantees that the minimal design $(K = 2, N = 2)$ is implementable. The second part of the assumption is with little loss as the manuscript focuses on the $B \to \infty$ regime.

where $W_0 = W(K^*(\lambda_0), N^*(\lambda_0))$ is the oracle variance. The term *oracle* is used because it depends on the unknown quantity λ_0. Finally, for $\lambda \in [\lambda_m, \lambda_M]$ define the optimal map as $\lambda \mapsto (K^*(\lambda), N^*(\lambda))$.

Clustered Experimental Design. The survey design problem in (2) extends mechanically to two prevalent clustered experimental designs: treatment assigned at the cluster level, where half the clusters are assigned to treatment and half to control; and clusters as blocks, where within each cluster half the units are assigned to treatment and half to control, usually assuming no interference. To do so, the model in (1) is extended to the potential outcomes framework via $y_{ki}(d_{ki}, \boldsymbol{d}_{k,-i})$, where $d_{ki} \in \{0, 1\}$ is unit i's own treatment and $\boldsymbol{d}_{k,-i}$ collects the assignments of the remaining units in cluster k. In both cases, the design variance takes the form $W(K, N) = \frac{c}{K}\left(\tau^2 + \sigma^2/N\right)$ for some constant c, under the substitutions in Table 1. Appendix A introduces both settings in greater detail.

Problem	**Estimand**	**Estimator**	c	τ^2	σ^2
Survey	$\mathbb{E}[y_{ki}]$	$\frac{1}{K}\sum_k^K \hat{\mu}_N^k$	1	τ^2	σ^2
Cluster-Level Treatment	$\mathbb{E}[y_{ki}(1, \mathbf{1}) - y_{ki}(0, \mathbf{0})]$	$\frac{2}{K}\left(\sum_{k:d_k=1} \hat{\mu}_N^k - \sum_{k:d_k=0} \hat{\mu}_N^k\right)$	2	$\tau_1^2 + \tau_0^2$	$\sigma_1^2 + \sigma_0^2$
Clusters as Blocks	$\mathbb{E}[y_{ki}(1, \cdot) - y_{ki}(0, \cdot)]$	$\frac{2}{KN}\sum_k^K \left(\sum_{i:d_{ki}=1} y_{ki} - \sum_{i:d_{ki}=0} y_{ki}\right)$	1	$\tau_1^2 + \tau_0^2 - 2\tau_{01}$	$2(\sigma_1^2 + \sigma_0^2)$

Table 1: Canonical clustered design problems

Notes: (τ_d^2, σ_d^2) are the cluster- and unit-level variances of potential outcome d, and $\tau_{01} = \mathrm{Cov}(u_k^1, u_k^0)$. Under cluster-level assignment $d_{ki} = d_k$ and the estimand is the TATE; under clusters as blocks, assuming no interference, the estimand is the ATE. The blocks design admits a degenerate corner: when $u_k^1 = u_k^0$, within-cluster differencing removes the cluster-level noise entirely ($\tau^2 = 0$), and the variance is minimized by sampling a single cluster. Assumption 3.2 excludes this case by requiring $\tau_m^2 > 0$. Finally, unlike the survey design problem, the balanced split across treatment and control may not be optimal. In general, the optimal allocation features an equal number of treated and control clusters and units iff relative cost to variance ratios are homogeneous across potential outcomes (see Remark A.4).

4 Adaptive Sampling under Unknown ICC

4.1 Estimating the Variance Ratio

When the variance ratio is unknown, the designer can still learn the key parameters through standard estimation. For any fixed $K \geq 2, N \geq 2$, define

$$S_{k,N}^2 = \frac{1}{N-1}\sum_{i=1}^N (y_{ki} - \hat{\mu}_N^k)^2, \qquad \hat{w}_{K,N}^2 = \frac{1}{K}\sum_{k=1}^K S_{k,N}^2, \qquad \hat{b}_{K,N}^2 = \frac{1}{K-1}\sum_{k=1}^K (\hat{\mu}_N^k - \hat{\mu})^2,$$
$$\hat{\tau}_{K,N}^2 = \Pi_{[\tau_m^2, \tau_M^2]}\left(\hat{b}_{K,N}^2 - \hat{w}_{K,N}^2/N\right), \qquad \hat{\lambda}_{K,N} = \Pi_{[\lambda_m, \lambda_M]}\left(\hat{w}_{K,N}^2/\hat{\tau}_{K,N}^2\right)$$

where $\hat{w}^2$ is the within-cluster sample variance, $\hat{b}^2$ is the between-cluster sample variance, $\Pi_{[a,b]}(x) = \min\{b, \max\{a, x\}\}$ is the clip operator and $\hat{\lambda}$ is the (clipped) sample estimator of λ. Lemma 4.1 shows consistency of the estimators under K and N asymptotics.

Lemma 4.1. *Consistency and Asymptotic Regimes. Let $\theta_N^2 := \tau^2 + \frac{\sigma^2}{N}$ be the variance of a cluster sample mean, and let $\xrightarrow[T]{x}$ denote convergence in x under $T \to \infty$ asymptotics. Under Assumptions 3.1 and 3.2: (i) $\hat{w}_{K,N}^2 \xrightarrow[N]{a.s.} \sigma^2$ (with $K \geq 1$ fixed) and $\hat{w}_{K,N}^2 \xrightarrow[K]{a.s.} \sigma^2$ (with $N \geq 2$ fixed); (ii) $\hat{b}_{K,N}^2 \xrightarrow[K]{a.s.} \theta_N^2$ (with $N \geq 2$ fixed); and (iii) $\hat{\lambda}_{K,N} \xrightarrow[K]{a.s.} \lambda_0$ (with $N \geq 2$ fixed).*

Width is the binding margin for design and for learning alike. Both optimal design and learning require asymptotic sampling of clusters, an observation that shapes the adaptive algorithms

and explains their success. *Design.* For any $\tau^2 > 0$, $\hat{\mu}_{K,N}$ is consistent for μ if and only if $K \to \infty$. Intuitively, increasing N averages away the individual-level noise ε_{ki}, but only additional clusters average away the cluster-level noise u_k. The optimal map encodes this asymmetry, while $N^*(\lambda)$ does not depend on B, $\lim_B K^*(\lambda) = \infty$ for any fixed λ. *Learning* features a similar asymmetry. According to Lemma 4.1, the within-cluster variance can be estimated consistently under both N- or K-asymptotics, however, the between-cluster variance—and therefore λ_0—can only be estimated consistently under K-asymptotics. Three consequences follow. First, in the limit, any candidate to optimal sampling policy must set K to infinity and keep N bounded; second, the asymptotic alignment between design and learning makes the $O(1/B^2)$ excess-variance rate attainable in a way that I make explicit in Section 5; and third, despite the asymptotic alignment, learning and design compete at finite budgets. Competition is explained by the fact that sampling more units or clusters will always improve the estimation of λ_0 but risks committing resources beyond what the oracle allocation would prescribe. I expand on this issue below.

4.2 Pilot-Based Policies and Implementability

For the remaining of this paper, I focus on a policy class called pilot-based policies. A *pilot* is a dataset with size (K_{pB}, N_p) collected during an initial stage. Only the width of the pilot is allowed to vary with B. Additionally, a *terminal design* (K_B, N_B) is the total sample collected by the designer, including the initial pilot data.

Definition 4.2. *Pilot-Based Policies. A policy π is pilot-based if it collects a pilot of (possibly random) size (K^π_{pB}, N^π_p), estimates $\hat{\lambda}_{pB} = \hat{\lambda}_{K^\pi_{pB}, N^\pi_p}$, and implements a terminal design (K^π_B, N^π_B) such that*

$$K^\pi_B = K^*(\hat{\lambda}_{pB}), \qquad N^\pi_B = N^*(\hat{\lambda}_{pB})$$

In essence, pilot-based policies collect an initial set of observations that are used for estimating λ_0, and then, they select the terminal design by applying the optimal map to the pilot estimate. Crucially, not all pilot-based policies can be run. Clusters and units cannot be unsampled, so pilot observations must be retained in the terminal sample. Definition 4.3 captures this idea through the notion of *implementability* and Lemma 4.4 argues that implementable pilot-based policies hold nice properties. Yet, they feature a clear tension between learning and optimal design implementation.

Definition 4.3. *Implementability. A pilot-based policy π is implementable if its pilot is almost surely nested within its terminal design,*

$$K_{pB} \leq K^\pi_B = K^*(\hat{\lambda}_{pB}), \qquad N_p \leq N^\pi_B = N^*(\hat{\lambda}_{pB}) \qquad a.s.$$

Lemma 4.4. *Implementable Pilot-Based Policies with Diverging K Converge to the Oracle. Define the scaled width $\nu^\pi_B = K^\pi_B/B$, and the oracle constants $\nu_0 = K^*(\lambda_0)/B = (F + \sqrt{FV\lambda_0})^{-1}$ and $N_0 = N^*(\lambda_0)$. Let π be an implementable pilot-based policy with pilot (K^π_{pB}, N^π_p), and suppose $K^\pi_{pB} \xrightarrow[B]{p} \infty$. Then, under Assumptions 3.1, 3.2, 3.3 and 3.4,*[18]

$$\hat{\lambda}_{pB} \xrightarrow[B]{p} \lambda_0, \qquad |\nu^\pi_B - \nu_0| \xrightarrow[B]{p} 0, \qquad |N^\pi_B - N_0| \xrightarrow[B]{p} 0$$

[18] Write $N_0 = \lfloor N_0 \rfloor$. If the terminal depth is rounded to an integer and $N_0 \notin \mathbb{N}$, the second statement is equivalent to $\mathbb{P}(\lfloor N^\pi_B \rfloor = N_0) \to 1$.

Two remarks. First, the takeaway of Lemma 4.4 is that implementable pilot-based policies which sample a diverging number of clusters within the pilot induce the optimal design in the limit. Convergence to the oracle design, however, takes place at two different scales. For the oracle depth, convergence can be established at the unit scale, as it does not depend on the budget. By contrast, the oracle width $K^*(\lambda_0)$ diverges linearly with B, therefore convergence is only possible at the budget-normalized scale. Second, Lemma 4.4 concretizes the anticipated tension between learning and optimal design implementation for pilot-based policies. The implementation of the optimal design takes place through convergence of the estimator $\hat{\lambda}_{pB} \underset{K_{pB}}{\to} \lambda_0$, which benefits from a large number of clusters entering the pilot. However, the implementability requirement entering the lemma caps the pilot size by the very design the pilot produces, i.e. $K_{pB} \leq K^*(\hat{\lambda}_{pB})$. $\hat{\lambda}_{pB}$ is unavailable until the pilot has been finalized, so a designer who explores many clusters improves the precision of $\hat{\lambda}_{pB}$ but risks committing resources beyond the prescribed terminal design. The two algorithms below resolve this circularity in different ways—the first by selecting a small enough pilot that implementability can never bind, the second by monitoring the online estimator of the variance ratio and stopping just before oversampling.

4.3 Explore-then-Commit Cluster Selection

For convenience define the extreme widths and depths,

$$K_M = \lfloor K^*(\lambda_M) \rfloor, \qquad K_m = \lfloor K^*(\lambda_m) \rfloor, \qquad N_m = \lfloor N^*(\lambda_m) \rfloor, \qquad N_M = \lfloor N^*(\lambda_M) \rfloor \tag{4}$$

Since $K^*(\cdot)$ is strictly decreasing and $N^*(\cdot)$ strictly increasing in λ, K_M is the narrowest and N_m the shallowest design the optimal map can possibly prescribe on $[\lambda_m, \lambda_M]$, while K_m is the widest and N_M the deepest. The simplest resolution to the circularity above is to choose a pilot small enough in both margins that implementability holds regardless of what the pilot finds. Since the clipped estimator satisfies $\hat{\lambda}_{pB} \in [\lambda_m, \lambda_M]$, the narrow-and-shallow pilot $(K_{pB}, N_p) = (K_M, N_m)$ is a.s. implementable,

$$K_{pB} = K_M \leq K^*(\lambda_M) \leq K^*(\hat{\lambda}_{pB}) = K_B^{\pi}, \qquad N_p = N_m \leq N^*(\lambda_m) \leq N^*(\hat{\lambda}_{pB}) = N_B^{\pi}$$

I refer to this two-stage policy as Explore-then-Commit Cluster Selection (`EtC-CS`).

Algorithm 1: `EtC-CS`

Input B, F, V, Ω

Sample N_m units from each of K_M clusters

  observe $(y_{ki})_{i \leq N_m,\, k \leq K_M}$ and **compute** $\hat{\lambda}_p = \hat{\lambda}_{K_M, N_m}$

Set $\left(K_B^{\texttt{EtC-CS}}, N_B^{\texttt{EtC-CS}}\right) = \left(K^*(\hat{\lambda}_p),\, N^*(\hat{\lambda}_p)\right)$

While theorists emphasize the superior performance of fully adaptive designs, practitioners often face logistical obstacles that prevent their deployment in the field (Chow and Corey, 2011). In that sense, the simplicity of `EtC-CS` is an asset: it attains rate-optimal performance guarantees with a single design update (Theorem 5.4). Nevertheless, the reader may wonder what could be achieved under further adaptivity.

4.4 Adaptive Cluster Selection

Example 4.5. *Imagine a researcher choosing a sampling width in* $[20, 60]$*, i.e. for the given budget,* $K_M = 20$ *and* $K_m = 60$*.* ***EtC-CS*** *would then collect a pilot of* 20 *clusters, compute* $\hat{\lambda}_p = \hat{\lambda}_{20, N_m}$*, and*

implement $(K^*(\hat{\lambda}_p), N^*(\hat{\lambda}_p))$*. Suppose the design prescribed by the pilot is* $K^*(\hat{\lambda}_p) = 50$*. Committing immediately would be wasteful. The researcher knows she will recruit thirty more clusters in any case, so she could use at least some of them to improve her estimate of* λ_0 *before committing to a terminal design, at negligible risk of oversampling.*

Building on this heuristic, I propose an algorithm called Adaptive Cluster Selection (`ACS`). The algorithm samples clusters sequentially until it commits to a terminal design according to a stopping rule informed by the online estimator of the variance ratio. Mathematically, the stopping time s of `ACS` is given by

$$s = \inf\left\{k \in \mathbb{N} : k \geq K_M \quad \text{and} \quad k+1 > K^*(\hat{\lambda}_{k,N_m})\right\} \tag{5}$$

with $s \in [K_M, K_m]$ a.s..[19] In words, in every period the algorithm samples $N_p = N_m$ units from one new cluster, until recruiting an additional cluster would constitute oversampling according to the estimate $\hat{\lambda}_k$, computed from the previous $N_p \times k$ observations. The reader may then wonder if instead of fixing depth and exploring only along the width margin, the designer could have considered richer exploration paths. In particular, at each step a designer could sample an additional unit from a previously visited cluster or from a fresh cluster. Nonetheless, such a path is not only computationally intractable but also impractical to administer in the field. Furthermore, concentrating exploration along the width margin is the right simplification. Depth and width are not symmetric inputs to learning: increasing N_p lowers the asymptotic variance of $\hat{\lambda}$ only through a constant, whereas increasing K_p improves the rate at which $\hat{\lambda}$ concentrates.[20] Therefore, holding N_p fixed and exploring along the width margin keeps the rule simple while preserving the convergence guarantees of Lemma 4.4.

Algorithm 2: `ACS`

Input B, F, V, Ω
Sample N_m units from each of K_M clusters
  observe $(y_{ki})_{i \leq N_m,\, k \leq K_M}$
  set $k = K_M$ and **compute** $\hat{\lambda}_k = \hat{\lambda}_{k,N_m}$
while $k+1 \leq K^*(\hat{\lambda}_k)$
  sample N_m units from cluster $k+1$ and **observe** $(y_{k+1,i})_{i \leq N_m}$
  update $k = k+1$ and **recompute** $\hat{\lambda}_k = \hat{\lambda}_{k,N_m}$
Set $s = k$ and **recover** $\hat{\lambda}_{\texttt{ACS}} = \hat{\lambda}_{\max\{s-1,\, K_M\},\, N_m}$
Set $\left(K_B^{\texttt{ACS}}, N_B^{\texttt{ACS}}\right) = \left(K^*(\hat{\lambda}_{\texttt{ACS}}),\, N^*(\hat{\lambda}_{\texttt{ACS}})\right)$

`ACS` begins with the same worst-case pilot as `EtC-CS`, then recruits clusters one at a time and stops just before oversampling according to the online estimate $\hat{\lambda}_k$. It therefore uses weakly more information than `EtC-CS` without compromising implementability. In fact, while `EtC-CS` may recruit a large number of clusters after committing, `ACS` concentrates nearly all cluster exploration within the pilot. As mentioned repeatedly, the estimation of λ_0 depends critically on the width margin, hence moving clusters into the pilot improves estimation of the optimal design at no additional sampling cost. The increased information set entering `ACS` manifests through lower excess variance at small budgets, yet the rate of the excess variance of both algorithms remains of the same order. In fact,

[19] The lower bound is immediate from the index set in (5). For the upper bound, suppose $s > K_M$, so that cluster s was actually explored; but then, the stopping condition must have failed at $k = s-1$, giving $s \leq K^*(\hat{\lambda}_{s-1,N_m}) \leq K^*(\lambda_m)$. Since s is an integer, $s \leq K_m$. When $s = K_M$ the upper bound holds trivially under Assumption 3.4.

[20] For any fixed N_p, the limit law of $\sqrt{K}(\hat{\lambda}_{K,N_p} - \lambda_0)$ has finite variance. See Lemma E.15 for a complete treatment.

this improvement in performance at finite budgets does not come for free: ACS requires the designer to revise the sample size after every cluster, so the designer must weigh the logistic inconvenience against the expected gain when choosing between the two.

Remark 4.6. *Committing One Cluster Early.* ***ACS*** *estimates* λ_0 *from* $\max\{s-1, K_M\}$ *clusters rather than all* s*. By definition of the stopping time,* $s+1 > K^*(\hat{\lambda}_{s,N_p})$*; moreover, because cluster* s *is always explored,* $s \le K^*(\hat{\lambda}_{s-1,N_p})$*. Together these give* $K^*(\hat{\lambda}_{s,N_p}) - 1 < s \le K^*(\hat{\lambda}_{s-1,N_p})$*, from which the implementability requirement* $s \le K^*(\hat{\lambda}_{s,N_p})$ *cannot be inferred—except when* $s = K_M$*, where it holds under Assumption 3.4. Committing at* $\max\{s-1, K_M\}$ *instead delivers an almost surely implementable design at the cost of one cluster's worth of information. For large pilots* $\hat{\lambda}_{s,N_p} \approx \hat{\lambda}_{s-1,N_p}$*, so the implications for the terminal design are immaterial by continuity of the optimal map, and at most one cluster is explored outside the pilot.*

4.5 Convergence to the Oracle Design

Both algorithms satisfy the premises of Lemma 4.4, hence, inducing the optimal design in the limit.

Corollary 4.7. ***EtC-CS*** *and* ***ACS*** *Converge to the Oracle Design. Both algorithms are pilot-based with fixed pilot depth* $N_p = N_m \ge 2$ *and pilot widths* $K_{pB} = K_M$ *for* ***EtC-CS*** *and* $K_{pB} = \max\{s-1, K_M\}$ *for* ***ACS****. Both are implementable—by construction for* ***EtC-CS****, and by committing one cluster early for* ***ACS*** *(Remark 4.6)—and both satisfy* $K_{pB} \ge K_M \to \infty$ *almost surely, since* $K^*(\lambda)$ *is monotonic in* B *for every fixed* λ*. The conditions of Lemma 4.4 therefore hold, and*

$$\left(\nu_B^\pi, N_B^\pi\right) \xrightarrow[B]{p} (\nu_0, N_0), \qquad \pi \in \{\textbf{\textit{EtC-CS}}, \textbf{\textit{ACS}}\}$$

5 Excess Variance and Optimality

This section evaluates the performance of EtC-CS and ACS against the oracle and other static policies. The performance criterion is the excess variance of the policy, sometimes referred to in the literature as *Neyman regret* (Dai et al., 2023): the difference between the variance induced by the terminal design (K_B^π, N_B^π) selected by some policy π and the oracle variance W_0,

$$\Delta_\pi = W\left(K_B^\pi, N_B^\pi\right) - W_0 \ge 0 \tag{6}$$

This criterion isolates the efficiency loss coming from terminal-design selection, while holding fixed the sampling variance common to any given design. Since an adaptive policy selects (K_B^π, N_B^π) using stochastic observations, Δ_π is itself random. I therefore evaluate policies by their worst-case expected regret

$$\min_\pi \max_{\omega \in \Omega} \mathbb{E}_\omega[\Delta_\pi]$$

where the expectation is taken over the data-generating process ω, and the cost structure is kept fixed. For EtC-CS and ACS, the randomness is induced by the realized sequence of cluster errors $(u_k)_k$ and unit errors $(\varepsilon_{ki})_{k,i}$ encountered along the sampling path.[21] For static policies the design is non-random and the expectation is simply redundant. Every policy considered in this paper, adaptive or not, selects

[21] For a randomized policy, the expectation would additionally integrate over the policy's internal randomization. More generally, expectations are always taken over the filtration generated by the interaction between the policy and the environment. See Lattimore and Szepesvári (2020) for a textbook treatment.

its terminal design by feeding a value of λ into the optimal map (3). In these cases, the excess variance can be written in closed form as

Lemma 5.1. *Excess Variance for Optimal Maps. Under Assumptions 3.1, 3.2, 3.3, 3.4, any policy π with terminal design $(K_B^\pi, N_B^\pi) = (K^*(\lambda), N^*(\lambda))$ for some input λ has an excess variance of the form*

$$\Delta_\pi = W\big(K^*(\lambda), N^*(\lambda)\big) - W_0 \;=\; \frac{\tau^2\sqrt{FV}}{B} \cdot \frac{\big(\sqrt{\lambda} - \sqrt{\lambda_0}\big)^2}{\sqrt{\lambda}} \qquad \forall \lambda \in [\lambda_m, \lambda_M] \tag{7}$$

Two features of (7) are worth discussing. First, misspecification errors affect variance only to second order on λ because the oracle satisfies a first-order condition at λ_0. Second, the entire expression carries a factor $1/B$, so any policy whose input is a fixed distance from λ_0 will inherit the order of the oracle variance itself. It immediately follows that any policy which plugs-in a misspecified λ_0 guess, including the minimax optimal, will incur excess variance of order $\Theta(1/B)$. By contrast, pilot-based algorithms like `EtC-CS` and `ACS` input a converging estimator $\hat{\lambda}_p$ into the optimal map, what allows them to significantly improve upon static policies. In particular, I derive upper bounds on their expected excess variance of order $1/B^2$—a magnitude of B below the oracle variance itself, hence asymptotically negligible. Finally, I show that the algorithms are rate optimal by deriving matching lower bounds of order $\Omega(1/B^2)$ on a generalized problem. Table 2 summarizes these rates.

Policy	**Metric**	**Rate**	**Result**
Oracle	Variance	$\Theta(1/B)$	Equation (3)
Misspecified static policies including minimax	Excess variance	$\Theta(1/B)$	Lemma 5.2
`EtC-CS` and `ACS`	Excess variance	$O(1/B^2)$	Theorem 5.4
Minimax lower bound	Excess variance	$\Omega(1/B^2)$	Theorem 5.5

Table 2: Performance rates

Notes: Results hold under the usual Assumptions 3.1, 3.3, 3.2 and 3.4, and the additional Assumption 5.3.

5.1 Static Policies and the Cost of Guessing

It is common practice to insert an ICC guess from the literature or a previous survey into the optimal map. Figure 1 illustrates the risks of blind imputation by plotting the excess variance as a function of λ for different F/V ratios. The excess variance presents a characteristic U-shape with a minimum at $\lambda = \lambda_0$. Lemma 5.2 quantifies this risk: any design built on a misspecified ICC incurs worst-case excess variance of order $\Theta(1/B)$. In other words, ICC misspecification is as costly as the sampling noise the design sets out to minimize, what makes it a first-order concern in clustered design. The same result holds true even when selecting the minimax λ input over the parameter space as in van Breukelen and Candel (2015, 2018, 2021).

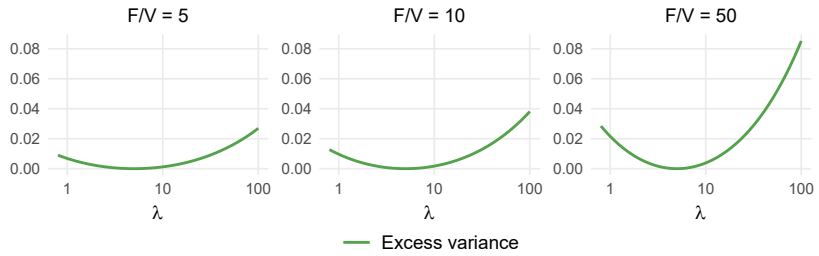


Figure 1: Design and excess variance under λ misspecification

Notes: Log-scale for λ. $\mu = 0$, $\omega = (1, 5)$, $B = 1000, V = 2$.

Lemma 5.2. *Misspecified and Minimax-Optimal Static Policies. Let $\lambda_0, \lambda_b \in [\lambda_m, \lambda_M]$ with $\lambda_b = \lambda_0 + b$ for some sufficiently small $b \neq 0$. Under Assumptions 3.1, 3.2, 3.3 and 3.4,*

(i) Any misspecified static input. *Define the static policy $\pi_b : (K_B^{\pi_b}, N_B^{\pi_b}) = (K^*(\lambda_b), N^*(\lambda_b))$. Then, the excess variance of π_b satisfies*

$$\frac{\tau^2\sqrt{FV}}{4\lambda_M^{3/2}} \cdot \frac{b^2}{B} \leq \Delta_{\pi_b} \leq \frac{\tau^2\sqrt{FV}}{4\lambda_m^{3/2}} \cdot \frac{b^2}{B} \qquad \Delta_{\pi_b} = \Theta(1/B) \;\; \forall b \neq 0$$

(ii) The minimax-optimal input. *Let*

$$\lambda_{mM} = \underset{\lambda \in [\lambda_m, \lambda_M]}{\arg\min} \max_{\omega = (\tau^2, \lambda_0) \in \Omega} \Big[W\big(K^*(\lambda), N^*(\lambda); \omega\big) - W\big(K^*(\lambda_0), N^*(\lambda_0); \omega\big) \Big]$$

Define the minimax-optimal policy $\pi_{mM} : (K^*(\lambda_{mM}), N^*(\lambda_{mM}))$. Then, the minimax input and its worst-case excess variance are given by

$$\lambda_{mM} = \left(\frac{\sqrt{\lambda_m} + \sqrt{\lambda_M}}{2} \right)^2, \qquad \max_{\omega \in \Omega} \Delta_{\pi_{mM}} = \frac{\tau_M^2 \sqrt{FV}}{2B} \cdot \frac{\big(\sqrt{\lambda_M} - \sqrt{\lambda_m}\big)^2}{\sqrt{\lambda_m} + \sqrt{\lambda_M}} \;=\; \Theta(1/B)$$

Part (ii) is worth emphasizing. Optimizing over the static class does not change the order of the penalty, it only pins down the constant. This constant is governed by the width of the designer's prior uncertainty $(\sqrt{\lambda_M} - \sqrt{\lambda_m})$, such that, more uncertain designers pay a higher price for their lack of adaptivity. $\Theta(1/B)$ is therefore the benchmark rate to improve upon.

5.2 Upper Bounds for the Learning Algorithms

I now show that learning improves on the static benchmark by a full order of B. Evaluating (7) at the pilot estimate and using some basic algebra,

$$\Delta_\pi \;=\; \frac{\tau^2\sqrt{FV}}{B} \cdot \frac{\big(\sqrt{\hat{\lambda}_{pB}} - \sqrt{\lambda_0}\big)^2}{\sqrt{\hat{\lambda}_{pB}}} \implies \max_{\omega \in \Omega} \mathbb{E}[\Delta_\pi] \leq \frac{\tau_M^2 \sqrt{FV}}{4B\lambda_m^{3/2}} \cdot \mathbb{E}\Big[(\hat{\lambda}_{pB} - \lambda_0)^2\Big] \tag{8}$$

so the problem immediately reduces to bounding $\mathbb{E}\big[(\hat{\lambda}_{pB} - \lambda_0)^2\big]$. To that end, I impose a standard subgaussianity condition, which delivers sufficient concentration of the input estimators.[22]

Assumption 5.3. *Errors Subgaussianity. u_k is κ_u-subgaussian and ε_{ki} is κ_ε-subgaussian.*

Theorem 5.4. *Expected Excess Variance of `EtC-CS` and `ACS`. Under Assumptions 3.1, 3.2, 3.3, 3.4 and 5.3, there exist finite constants $C_{\texttt{EtC-CS}}$ and $C_{\texttt{ACS}}$ not depending on B, such that*

$$\max_{\omega \in \Omega} \mathbb{E}\big[\Delta_{\texttt{EtC-CS}}\big] \leq \frac{C_{\texttt{EtC-CS}}}{B^2}, \qquad \max_{\omega \in \Omega} \mathbb{E}\big[\Delta_{\texttt{ACS}}\big] \leq \frac{C_{\texttt{ACS}}}{B^2}$$

The theorem states that both `EtC-CS` and `ACS` incur a variance penalty that vanishes relative to the oracle variance itself, what in turn implies that the ICC can be learned only at second-order

[22] Analogous results hold under weaker conditions such as subexponential tails of the errors.

cost. As anticipated in Section 4, ACS delivers superior performance in simulations, since it uses a weakly larger set of clusters for estimating λ_0. However, the rate of both algorithms remains the same because their pilot widths are also of the same order, i.e. $K_{pB}^{\texttt{ACS}}/K_{pB}^{\texttt{EtC-CS}} = O(1)$ almost surely.[23]

Why $1/B^2$? The result rests on a $1/B \times 1/B$ mechanism. The first term follows directly from the order of the oracle variance itself, as discussed in (7). The second term is nonetheless fortuitous and can be explained by the asymptotic alignment between learning and optimal design implementation presented earlier. Mathematically, Lemma C.5 shows that both algorithms satisfy $\max_{\omega\in\Omega} \mathbb{E}\big[(\hat{\lambda}_{pB} - \lambda_0)^2\big] \leq C/K_M$ for some constant C. Since $K_M = \Theta(B)$, the result follows, but this observation requires further attention. While setting $K_{pB} = \Theta(B)$ is unequivocally good for estimation, the compatibility between estimation and optimal design remains surprising. Were the optimal design to require only $\Theta(1)$ clusters, a pilot of width $\Theta(B)$ would be infeasible, consistent learning of λ_0 could fail, and the B^{-2} rate would no longer follow.

5.3 Lower Bound

Finally, I show that the two algorithms are essentially unimprovable by deriving a matching lower bound of order $\Omega(1/B^2)$. In particular, for every sampling policy π there exists an environment $\omega \in \Omega$ in which it incurs expected excess variance of that order. EtC-CS and ACS are therefore minimax rate optimal in the joint (K, N) design problem, and can be improved by no more than a constant factor. The bound is derived for the generalized problem of Appendix A, in which the learner may sample a different number of units N^k from each cluster and revisit clusters at will.

Theorem 5.5. *Lower Bound on Expected Excess Variance. There exist constants $C > 0$ and $B_0 < \infty$, not depending on B or the policy, such that for every $\pi \in \Pi$ and every $B \geq B_0$,*[24]

$$\max_{\omega\in\Omega} \mathbb{E}_\pi[\Delta_\pi] \;\geq\; C/B^2$$

The proof proceeds in three steps. *First*, I work with an easier version of the problem in which the cluster error u_k is revealed to the learner the first time she visits cluster k. In the cluster setting, observations within a cluster are correlated, so once a cluster has been visited its subsequent observations are not independent conditional on the history. Revealing u_k on the first visit avoids this nuisance and restores the conditional independence on which standard divergence-decomposition arguments rely.[25] This "augmented experiment" gives the learner weakly more information than the benchmark experiment, so any lower bound established there carries over to the original problem. *Second*, within the augmented environment I consider two data generating processes y_- and y_+ of the form (1), indexed by

$$\omega_- = (\tau_-^2, \lambda_-), \qquad \lambda_- = \sigma^2/\tau_-^2, \qquad \omega_+ = (\tau_+^2, \lambda_+), \qquad \lambda_+ = \sigma^2/\tau_+^2$$

The two environments share the same mean μ and the same unit-level variance σ^2, and so they only

[23] Recovering the rate is technically more demanding under sequential sampling, since the deviations must be bounded for a pilot of random width, rather than deterministic. Appendix C circumvents this technicality by noting that $\hat{\lambda}_k - \lambda_0$ can be decomposed asymptotically as a function of martingales—which are amenable to a maximal version of Bernstein's inequality—and that K_m/K_M is bounded uniformly in B.

[24] The policy class Π is defined in Appendix C. The restriction excludes only policies that fail to terminate in finite time or violate the budget constraint with positive probability. Policies in Π may sample a different number of units from different clusters, revisit clusters freely, and sample a different number of units on each visit.

[25] See Chapter 15 in Lattimore and Szepesvári (2020).

differ in the cluster-level variance τ^2. For a small $\delta_B > 0$, the cluster variances are chosen so that $\lambda_- = \lambda - \delta_B$ and $\lambda_+ = \lambda + \delta_B$. Recall from Lemma 5.2 that optimizing against the wrong instance is costly; for instance, if implementing $(K^*(\lambda_+), N^*(\lambda_+))$ when the truth is ω_-, the designer would have implemented a design too narrow and too deep compared to the oracle. See Figure 2.

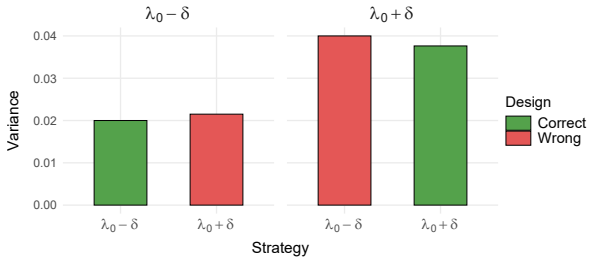


Figure 2: Optimal policies across environments (y_+, y_-).

Notes: $W(K^*(\cdot), N^*(\cdot))$ under ω_- (columns 1, 2) and ω_+ (columns 3, 4) for policy $(K^*(\lambda_-), N^*(\lambda_-))$ (columns 1, 3) and policy $(K^*(\lambda_+), N^*(\lambda_+))$ (columns 2, 4). $\mu = 0$, $\omega = (1 \pm \delta, 5/(1 \pm \delta))$, $\delta = 0.5$, $B = 2000$, $F = 20$, $V = 2$.

Third, δ_B is chosen to balance the trade-off between identification and punishment. A large δ_B makes the two environments easy to tell apart, so the learner optimizes correctly most of the times, but is penalized heavily when she does not. Conversely, a small δ_B makes them hard to distinguish, but shrinks the cost of choosing the wrong design. The two forces are balanced at $\delta_B \asymp B^{-1/2}$, as is customary in learning problems, at which point the resulting bound is maximized at order $1/B^2$.

6 Inference

Adaptive data collection generally invalidates conventional inference. Hence, if the reduction in variance cannot be paired with suitable inference, the result could end up being of limited use. This section shows that, under a symmetry restriction on the errors, there is no trade-off between adaptivity—thus, variance reduction—and inference. Inference on population means remains invariant after conditioning on the realized design induced by an *even* policy, and remains asymptotically invariant after conditioning on the realized design induced by an *asymptotically even* policy. Informally, a policy is even if its terminal design depends on the data only through even functionals of the errors, such as squared errors; and asymptotically even if its terminal design is close to the terminal design that would have been generated by a *twin* even policy. For instance, this will be the case when a policy depending on the sample variance is replaced by the exact same policy but depending on the population variance. Under (asymptotic) evenness, average estimators remain asymptotically normal under symmetry, with no correction and no exclusion of pilot data. This principle covers both `EtC-CS` and `ACS` as they are functions of sequences of (sample) variance ratios.

Why design-conditional inference? I study inference conditional on the realized design, rather than averaged over counterfactual designs for four reasons. First, it is, implicitly, the prevalent notion of validity in fixed clustered designs. Fixed designs are usually presented as non-random, however, they are typically the outcome of a deliberation over alternatives. So, when *fixing the design*, the researcher is implicitly conditioning on the design and assuming that the selection process was independent of the realized data. In the case of adaptive designs, independence need not to hold, yet, by keeping the conditioning step, adaptive and static designs are being compared on the same inferential footing. This is the selective-inference stance in Fithian et al. (2014) which suggests that

designers must condition on what they learned and acted on. Second, design-conditional inference is the honest guarantee. A reader who sees an interval reported alongside the realized design cares whether that interval covers the target for the experiment actually run. An interval with correct unconditional coverage can systematically under-cover some realized designs and over-cover others. Third, the designs induced by `EtC-CS` and `ACS` remain genuinely random at the relevant scale, so the conditioning events are non-degenerate and the conditional statement has content (Section 6.3). Finally, design-conditional validity is the stronger result: averaging over designs recovers unconditional validity, and so my results are robust to the inference convention adopted by the reader. Conditional inference usually comes at the expense of Gaussianity assumptions, pilot-data exclusion or *data carving* (Fithian et al., 2014). In the cluster setting, I show that design-conditional inference only costs a mild symmetry assumption on the errors.

The section proceeds in four steps. Section 6.1 first discusses the Gaussian benchmark, where the result is exact and finite-sample. Section 6.2 weakens Gaussianity into symmetry and establishes an asymptotic result for exactly even designs at every design scale. Section 6.3 extends the results to the feasible class, and identifies the scale at which conditioning events are both well defined and non-degenerate. Finally, Section 6.4 generalizes the symmetry principle beyond cluster sampling. A general theory of inference invariance can be found in Appendix D.

6.1 Exact Design Independence under Gaussian Errors

It is well known that, under Gaussianity, sample averages are independent of the vector of residuals. This result has been exploited repeatedly in the sample size re-estimation and the internal pilot literatures.[26] It follows that any design which depends on the data only through the residuals is independent of the sample average, and so design-conditional inference reduces to unconditional inference at every finite sample size.[27] Policies depending on sample variances would just be a subset of this policy class, thus, this result covers both `EtC-CS` and `ACS`, which are functions of sequences of variance ratios. For some policy π, define a (terminal) design event $A_B := \{(K_B^\pi, N_B^\pi) \in C\}$ for some Borel set C.

Proposition 6.1. *Exact Conditional Normality under Gaussian Errors. Under Assumptions 3.1, 3.2 and 3.4, let $u_k \sim N(0, \tau^2)$ and $\varepsilon_{ki} \sim N(0, \sigma^2)$. Then, for $\pi \in \{$`EtC-CS`, `ACS`$\}$, every budget B, and every design event $A_B \in \mathscr{P}^\pi$ with $\mathbb{P}(A_B) > 0$,*[28]

$$\mathcal{L}\left(\frac{\sqrt{K_B^\pi}\big(\hat{\mu}_{K_B^\pi, N_B^\pi} - \mu\big)}{\theta_{N_B^\pi}} \;\middle|\; A_B \right) = N(0,1) \quad \textit{exactly}$$

Three notes on Proposition 6.1. First, exactness holds only for the infeasible standardization $\theta_{N_B^\pi}$. When replaced by the sample standard deviation $\hat{b}_{K_B^\pi, N_B^\pi}$, the theory requires an asymptotic argument because the numerator and denominator depend on the same cluster means. Second, the argument relies on the pilot being nested within the terminal design—$K_{pB} \leq K_B^\pi$ and $N_p \leq N_B^\pi$—, guaranteed under implementability. Nesting makes the terminal average orthogonal to every residual

[26] See Stein (1945); Dudewicz and Dalal (1975); Rinott (1978) for the former and Wittes and Brittain (1990); Wittes et al. (1999); Lake et al. (2002); van Schie and Moerbeek (2014) for the latter.

[27] Equivalently, the sample average is sufficient for μ while the residual vector is ancillary, so independence also follows from Basu's theorem.

[28] Mathematically, the relevant policy class is the set of policies which are measurable wrt the (stopped) σ-algebra generated by the residuals. Appendices D.2 and E.1 collect the formal definitions of the residual σ-algebra R_j and the stopped σ-algebras $\mathscr{P}^{\texttt{EtC-CS}}$ and $\mathscr{P}^{\texttt{ACS}}$ entering Proposition 6.1.

generating the (stopped) residual σ-algebra, so that pilot observations need not be discarded. Third, the proposition delineates what Gaussianity can and cannot buy. Under Gaussianity the invariant class is large. The design may depend arbitrarily on the residuals—not just through the sample variances but also through any *odd* functional of the residuals—and still deliver exact conditional inference. When Gaussianity is weakened to symmetry, the invariance policy class also shrinks: from policies which read the residuals, to policies which only read their magnitudes (even policies). What no assumption can rescue are policies that condition on the *level* of the data rather than on demeaned outcomes. A rule that reads the pilot mean—such as futility rules or most bandit algorithms—conditions on the direction removed through demeaning, its designs lie outside the invariance class, and conditional normality fails even under Gaussianity (Figure 8).

6.2 Inference for Even Designs under Symmetry

Exact Gaussianity is a strong restriction. By reducing the invariance policy class from functionals of the residuals to (asymptotically) even functionals of the data, I can significantly weaken the assumption on the errors from Gaussianity to symmetry. Moreover, my results are robust to studentization.

Assumption 6.2. *Error Symmetry. Let* $C_k = (u_k, \varepsilon_{k1}, \varepsilon_{k2}, \dots)$. *Then* $C_k \overset{d}{=} -C_k$.

Because of independence of u_k and ε_{ki}, a simple sufficient alternative for Assumption 6.2 is that u_k and ε_{ki} are each symmetric about zero. The symmetry assumption is familiar and comparatively benign, as it underlies most of sign and permutation tests, as well as symmetry-based randomization inference (Canay et al., 2017). Additionally, it is far weaker than the parametric assumptions practitioners routinely impose in power calculations, and it is partially testable.[29] The restriction however excludes important error distribution families such as binary and count outcomes, as well as censored and one-sided distributions.

As hinted in the introduction, Assumption 6.2 permits a decomposition of the errors into independent signs and magnitudes. Conditioning on any event that depends on the magnitudes alone—such as events generated by even functionals of the data—leaves the signs unrestricted, so that a central limit theorem can run through them independently. Formally,

Lemma 6.3. *Cluster Sign Factorization. Fix* N *and let* $(\zeta_k)_k$ *be an iid Rademacher sequence independent of* $\big((C_k)_k, (U_k)_k\big)$. *Define*

$$M_{k,N} = u_k + \frac{1}{N}\sum_{i=1}^{N} \varepsilon_{k,i}, \qquad \chi_{k,N} = \mathbb{1}(M_{k,N} \neq 0)\,\mathrm{sign}(M_{k,N}) + \mathbb{1}(M_{k,N} = 0)\,\zeta_k, \qquad R_{k,N} = |M_{k,N}|$$

so that $M_{k,N} = \chi_{k,N} R_{k,N}$, *where* $M_{k,N}$ *is the average error of cluster* k. *Under Assumptions 3.1 and 6.2,* $(\chi_{k,N})_k$ *is iid Rademacher and* $(\chi_{k,N})_k \perp\!\!\!\perp (R_{k,N})_k$.

Corollary 6.4. *Signs are Independent of the Even* σ*-algebra. For a fixed* N_p *and an iid sequence* $(U_k)_k \sim \mathrm{Unif}[0,1]$ *independent of* $\big((C_k)_k, (\zeta_k)_k\big)$, *define the even* σ*-algebra*

$$\mathcal{G}_\infty := \sigma\Big(\big(S^2_{k,N_p}\big)_{k\geq 1},\ \big(M^2_{k,N}\big)_{k\geq 1,\, N\geq 1},\ (U_k)_{k\geq 1}\Big)$$

Then, under Assumptions 3.1 and 6.2, for every fixed $N \geq 1$ *the sign sequence* $(\chi_{k,N})_{k\geq 1}$ *is iid*

[29] See Section 8 for an extended discussion and test implementation with real data.

Rademacher and independent of $\mathcal{G}_\infty$, hence independent of every event $A \in \mathcal{G}_\infty$.

Next, I introduce the class of policies which are measurable with respect to the even σ-algebra.

Definition 6.5. *A policy π is*

(i) ***Even*** *if the terminal design (K_B^π, N_B^π) is $\mathcal{G}_\infty$-measurable;*

(ii) ***Non-starving*** *if $\mathbb{P}(\nu_B^\pi \geq c) \to 1$ for some constant $c > 0$;*

(iii) ***N-consistent*** *if $N_B^\pi \xrightarrow[B]{p} N_{\underline{0}} = \lfloor N^*(\lambda_0) \rfloor \in \mathbb{N}$.*

Evenness is the substantive requirement: an even policy induces design events in $\mathcal{G}_\infty$, so conditioning on the even σ-algebra freezes the design and the magnitudes but leaves the signs unrestricted. The other two conditions are technical but do real work. Any asymptotic argument requires a divergent sequence, which is guaranteed by non-starvation. As such, it is also present in the general theory of Appendix D. The linear form of the divergence is immaterial for exactly even policies, but becomes necessary when transferring results to the feasible class. N-consistency is specific to the cluster problem: it stabilizes the depth margin so that the analysis can proceed along K-asymptotics alone. By Corollary 4.7, `EtC-CS` and `ACS` are both non-starving and N-consistent—but neither is exactly even.

Proposition 6.6. *Uniform Mixing for Even Designs. Let $\big((K_B^\pi, N_B^\pi), (A_B)_B\big)$ be a design and a sequence of design events induced by an even, non-starving, N-consistent policy π. Define*

$$Z_B^\pi = \sqrt{K_B^\pi}\,(\hat{\mu}_{K_B^\pi, N_B^\pi} - \mu) = \frac{1}{\sqrt{K_B^\pi}} \sum_{k=1}^{K_B^\pi} M_{k, N_B^\pi}$$

and the associated conditional c.f. $\phi_B(t) = \mathbb{E}\big[\exp(itZ_B^\pi) \mid \mathcal{G}_\infty\big]$. Let $\phi(t) = \exp(-\theta_{N_{\underline{0}}}^2 t^2/2)$ be the Gaussian c.f. evaluated at the oracle depth $N_{\underline{0}}$. Then, under Assumptions 3.1, 3.2, 3.4 and 6.2,

$$\Big|\mathbb{E}\big[\phi_B(t)\,\mathbb{1}_{A_B}\big] - \phi(t)\,\mathbb{P}(A_B)\Big| \to 0 \qquad \forall t \in \mathbb{R},$$

uniformly over $A_B \in \mathcal{G}_\infty$. Moreover, for any such sequence with $\liminf_B \mathbb{P}(A_B) > 0$,[30]

$$\mathcal{L}\big(Z_B^\pi \mid A_B\big) \implies N\big(0, \theta_{N_{\underline{0}}}^2\big), \qquad \mathcal{L}\left(Z_B^\pi \,/\, \hat{b}_{K_B^\pi, N_B^\pi} \,\middle|\, A_B\right) \implies N(0,1)$$

Proposition 6.6 displays a very strong mode of convergence: the scaled sums converge $\mathcal{G}_\infty$-uniformly mixing, thus asymptotic normality is preserved after conditioning on any $\mathcal{G}_\infty$-measurable event, at any scale. Nonetheless, this result is of limited use on its own as policies cannot observe the true squared errors. At best, policies may sample according to quasi-even statistics such as sample variances. Recall that sample variances demean the observations by the sample average, so a policy depending on a sequence of sample variances could induce designs depending on the relative signs, hence not in $\mathcal{G}_\infty$.[31] The next section shows that the results extend to the feasible class, once the set of conditioning events is restricted to the appropriate scale.

[30] For exactly even designs, symmetry buys more than asymptotic normality. Conditional on $\mathcal{G}_\infty$ the signs $(\chi_{k,N})_k$ are *iid* Rademacher (Corollary 6.4), so sign-flip randomization yields an exact finite-sample test of $H_0 : \mu = \mu_0$. This finite-sample property however does not extend to feasible policies. `EtC-CS` and `ACS` are only asymptotically even, so for them the sign-flip test is only asymptotically valid, just like the normal approximation pursued here.

[31] Concretely, for a simple model $X_s = \mu + \varepsilon_s$, the sample variance can be written as $\hat{\sigma}_m^2 = \frac{m-1}{m^2}\sum_{s \leq m} \varepsilon_s^2 - \frac{2}{m^2} U_m$, where $U_m = \sum_{1 \leq i < j \leq m} \varepsilon_i \varepsilon_j$. For $m = 2$, $U_2 = |\varepsilon_1 \varepsilon_2| \text{sign}(\varepsilon_1)\text{sign}(\varepsilon_2)$ which is clearly not even. Therefore, a policy depending on $\hat{\sigma}_2^2$ may induce designs outside $\mathcal{G}_\infty$. The leakage is nonetheless asymptotically negligible relative to the even component, which is what licenses the notion of asymptotic evenness. See Appendix D for the general statement.

6.3 Inference for Asymptotically Even Designs under Symmetry

Definition 6.7. *Asymptotically Even Policies. Let* $K_{0B} = K^*(\lambda_0)$ *and define* $\Xi_B^\pi = (K_B^\pi - K_{0B})/\sqrt{B}$. *A policy* π *is asymptotically even if there exists a* $\mathcal{G}_\infty$*-measurable pair* $(\tilde{K}_B, \tilde{N}_B)$*, with* $\tilde{\Xi}_B = (\tilde{K}_B - K_{0B})/\sqrt{B}$*, such that*

$$\left|\Xi_B^\pi - \tilde{\Xi}_B\right| \xrightarrow[B]{p} 0, \qquad \left|N_B^\pi - \tilde{N}_B\right| \xrightarrow[B]{p} 0$$

In essence, a policy is asymptotically even if its design is close in probability to one induced by an even policy, with closeness measured on the $\sqrt{B}$ scale for width and on the unit scale for depth. Observe that the definition places no restriction on the surrogate. The even design $(\tilde{K}_B, \tilde{N}_B)$ need be neither feasible nor related to π in any substantive way. However, in practice, the analyst will construct the even surrogate by replacing the squared residuals entering π with the true squared errors, and then prove convergence at the relevant scale. The transition from exact to asymptotic evenness further restricts the class of admissible conditioning events to the set of events that agree asymptotically under the two policies—in the sense that $\mathbb{P}(A_B \triangle \tilde{A}_B) \to 0$. When events are expressed on the same scale as the required convergence of the designs, i.e.

$$(A_B)_B : A_B = \left\{(\Xi_B^\pi, N_B^\pi) \in C\right\}, \qquad C \subseteq \mathbb{R} \times \mathbb{N} \text{ Borel} \tag{9}$$

the transition is warranted under a mild margin condition.

Which scale should the analyst condition on? Three scales are *a priori* available in the width dimension: unit, $1/\sqrt{B}$ and $1/B$. For exactly even policies the choice is irrelevant because the limiting law of Z_B^π is unchanged after conditioning on any event of any scale in $\mathcal{G}_\infty$ (Proposition 6.6). For asymptotically even policies the choice is decisive. K_B^π is proportional to B and fluctuates over a window of width $O_p(\sqrt{B})$ around the oracle (Lemma E.11), therefore the feasible and surrogate designs cannot couple at the unit scale, $\mathbb{P}(A_B \triangle \tilde{A}_B) \not\to 0$, and inference results do not transfer. At the $1/B$ scale, conditioning is possible but asymptotically vacuous. By Lemma 4.4, $\nu_B^\pi \xrightarrow[B]{p} \nu_0$, so $\mathbb{P}(\nu_B^\pi \in C) \to \mathbb{1}\{\nu_0 \in C\} \in \{0, 1\}$ for every C whose boundary ν_0 avoids. At that scale, the conditional law degenerates to the unconditional one, and a random-index CLT in the spirit of Anscombe (1952) and Chow and Robbins (1965) will suffice without ever invoking symmetry or policy evenness. At the intermediate $1/\sqrt{B}$ scale, the designer can obtain both: the coupling survives and the conditioning events remain non-degenerate. $1/\sqrt{B}$-scale events are particularly relevant as they include events involving quantiles of the realized width. These events lie outside the reach of Anscombe (1952), and so require the special treatment discussed in this manuscript. Table 3 summarizes the scale discussion.

Scale	Does Surrogate Coupling Survive?	Non-degenerate Events?	Tools
Unit	No	–	–
$1/B$	Yes	No	Anscombe
$1/\sqrt{B}$	Yes	Yes	Symmetry and Evenness

Table 3: Scale of conditioning events

Theorem 6.8. *Inference for Asymptotically Even Designs. Let* π *be an asymptotically even, non-starving,* N*-consistent policy, and let* $(A_B)_B$ *be a sequence of events as in* (9) *satisfying the margin condition*

$$\lim_{\eta \downarrow 0} \limsup_B \mathbb{P}\Big(\operatorname{dist}\big((\Xi_B^\pi, N_B^\pi), \partial C\big) \le \eta\Big) = 0 \tag{10}$$

Then, under Assumptions 3.1, 3.2, 3.4 and 6.2,[32]

$$\left|\mathbb{E}[\exp(itZ_B^\pi)\mathbb{1}_{A_B}] - \phi(t)\mathbb{P}(A_B)\right| \to 0 \quad \forall t \in \mathbb{R}$$

If in addition $\liminf_B \mathbb{P}(A_B) > 0$*, then*

$$\mathcal{L}\left(Z_B^\pi \,|\, A_B\right) \implies N(0, \theta^2_{N_0}), \qquad \mathcal{L}\left(Z_B^\pi \,/\, \hat{b}_{K_B^\pi, N_B^\pi} \,\middle|\, A_B\right) \implies N(0,1)$$

Condition (10) is an anti-concentration requirement on the set C. The scaled design must not pile up on the boundary of the conditioning set, since this is where the feasible and surrogate designs can disagree. In practice this condition is mild, and can be established through a central limit theorem over the statistic governing the design—$\hat{\lambda}$ in cluster context—under finite fourth moments. The only exception arises when λ_0 lies on the boundary $\{\lambda_m, \lambda_M\}$. The clip would then bind with positive probability and the scaled width will gain at atom at zero. Only in that context, (10) excludes sets C whose boundary contains zero, see Lemma E.15.

It remains to verify that the two algorithms are asymptotically even. Propositions E.12 and E.13 do so constructively, building surrogates `EtC̃-CS` and `AC̃S` that replace the nearly even statistic $\hat{b}^2_{K,N}$ with its even counterpart $K^{-1}\sum_{k\le K} M^2_{k,N}$.[33] For `EtC-CS` the coupling is immediate: the design depends on the single statistic $\hat{\lambda}_p$, whose versions differ by $O_p(1/B)$. The induced widths then differ by $O_p(1)$, an order below the required $o_p(\sqrt{B})$. `ACS` is more delicate. Replacing residuals by errors would still leave the analyst with the difficult task of coupling the two estimators at two different random stopping times s and $\tilde{s}$. I avoid the difficulty by giving the surrogate a deterministic stopping rule $\tilde{s} = K_{0B}$. The proof is completed by localizing the stopping time of `ACS` in an $O_p(\sqrt{B})$ window around K_{0B}, and applying a maximal inequality.

Theorem 6.9. *Design-Robust Inference for `EtC-CS` and `ACS`. Let* (K_B^π, N_B^π) *be the design induced by* $\pi \in \{$`EtC-CS`, `ACS`$\}$*, and let* $A_B = \{(\Xi_B^\pi, N_B^\pi) \in C\}$ *for any* C *satisfying* (10) *and* $\liminf_B \mathbb{P}(A_B) > 0$*. Then, under Assumptions 3.1, 3.2, 3.4 and 6.2,*

$$\mathcal{L}\left(Z_B^\pi \,/\, \hat{b}_{K_B^\pi, N_B^\pi} \;\middle|\; A_B\right) \implies N(0,1)$$

The takeaway from Theorem 6.9 is that nothing needs to change. The estimator is the ordinary average of cluster averages computed on the full sample—pilot observations included—, the standard errors are the usual clustered ones, and the normal approximation is valid. No reweighting, no sample splitting, and no discarding of pilot data is required. The results are confirmed in simulations.

The boundary of the result. Policy evenness and error symmetry are not merely sufficient. The simulation evidence shows that relaxing either can destroy the invariance. A policy that adapts through odd functionals of the data leaks information about the signs and breaks the sign–magnitude factorization of Lemma 6.3. For such policies the design-conditional mean of Z_B^π need not vanish, and the pathologies documented by Nie et al. (2018) and Hadad et al. (2021) reappear, along with the necessity to introduce the reweighting corrections developed in those papers. Similarly, error asymmetry

[32] Clearly, the uniform mixing statement cannot be established for Z_B^π, as this scaled sum is not even $\mathcal{G}_\infty$ measurable. Yet, a qualitatively similar result holds pointwise along the given sequence of events. In particular, the two conditional limit laws, which are what an applied analyst uses, carry over unchanged. See Appendix E for a complete treatment.

[33] The within-cluster variance $\hat{w}^2_{K,N}$ is both even and feasible, so it requires no adjustment.

allows exactly even policies to pick up information about the error signs, breaking, once again, the sign–magnitude factorization. Section 7 makes the boundary concrete by assessing Theorem 6.8 in a grid which interacts policy evenness and error symmetry.

6.4 Beyond Clustered Sampling

Nothing in the preceding argument is specific to clusters. Stripped to its essentials, the mechanism only requires sequential sampling of units according to even functionals of the data and symmetric errors. As a result, the theory admits a much more general representation that I collect in a standalone technical note in Appendix D. The same boundary, however, travels with the principle: adaptation through odd functionals of the data or error asymmetry are not covered by the general theory.

The leading application of this theory is the oldest design rule in the book: Neyman allocations with plug-in variances. At a high level, a designer would like to estimate a set of populations means, where each population has a different variance. The designer could then estimate the variance of each arm through a uniform pilot, and then allocate the remaining observations proportionally to the estimated standard deviations. The second-stage of the design is driven by sample variances, so the policy is asymptotically even and the general theory applies off the shelf. Other examples include the two-stage propensity score allocations of Hahn et al. (2011), which are a function of the ratio of pilot-estimated conditional standard deviations, hence asymptotically even.

Corollary 6.10. *Two-Stage Neyman Allocations (Informal). Imagine a population with K arms, each containing symmetric iid errors, a strictly positive variance, and finite fourth moments. Consider the two-stage implementable design that samples all arms uniformly over a fixed share of the horizon and then sets the terminal counts proportional to the pilot standard deviations. Conditional on any sufficiently coarse event—quantile cells of the realized counts included—the studentized arm means are jointly asymptotically normal: computed on the full sample, with the usual standard errors, and with no correction. Proposition D.35 and Corollary D.37 contain the formal statements.*

7 Simulations

This Section confronts the theoretical prescriptions with simulation evidence: Section 7.1 for performance, Section 7.2 for inference. Data are generated from the survey model in (1) with $\mu = 0$, $\omega = (\tau^2, \lambda_0) = (1, 5)$ and parameter space $\Omega = [\tau_m^2, \tau_M^2] \times [\lambda_m, \lambda_M] = [0.25, 4] \times [1, 100]$. The cost structure is kept fixed at $(F, V) = (20, 2)$, and the baseline budget is $B = 2{,}000$, which places the oracle design at $\big(K^*(\lambda_0), N^*(\lambda_0)\big) = (58.6, 7.1)$ and the minimal pilot at $(K_M, N_m) = (24, 3)$. Assumption 3.4 holds across all instances considered below. Errors are Gaussian in the performance exercises, and take a range of symmetric and asymmetric shapes in the inference exercises. All statistics are computed from 10^5 independent replications of the complete design-and-estimation cycle, so that all the randomness entering the selection of the terminal design is accounted for.

7.1 Performance

For each algorithm, Figure 3 reports the distribution of the the pilot estimate $\hat{\lambda}_{pB}$ that enters the optimal map and the resulting terminal design (K_B^π, N_B^π). Both algorithms concentrate around the oracle design—as prescribed by Corollary 4.7—yet `ACS` bunches more aggressively. This is to be expected as `ACS` features a weakly larger set of clusters entering the pilot compared to `EtC-CS`, what

improves the precision of the λ_0 estimator. The distribution of $\hat{\lambda}_{pB}$ itself is sharp around the truth but visibly right-skewed. The reason is that $\hat{\tau}^2$ is recovered as a difference of two random objects, i.e. $\hat{\tau}^2 = \hat{b}^2 - \hat{w}^2/N_p$, so when τ^2 is small relative to σ^2, $\hat{\tau}^2$ is estimated as a noisy residual between two large and separately noisy quantities. Furthermore, $\hat{\tau}^2$ enters the variance ratio through the convex map $\hat{\tau}^2 \mapsto 1/\hat{\tau}^2$, so any symmetric noise in the denominator manifests as a heavy right tail in $\hat{\lambda}_{pB}$. The skew in $\hat{\lambda}_{pB}$ is, nevertheless, partially absorbed by the design because $\hat{\lambda}_{pB}$ enters the optimal map through a square root which compresses the upper tail, what makes the excess variance flat to the right of λ_0 and steep to its left. Overall, Figure 14 in Appendix F shows that both algorithms track the oracle design relative well across λ_0 with minor depth oversampling and width undersampling.

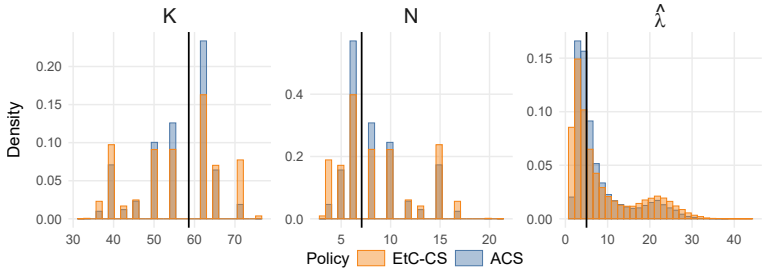


Figure 3: Design (K_B^π, N_B^π) and pilot estimate $\hat{\lambda}_{pB}$ for `EtC-CS` and `ACS`

Notes: Solid black lines mark the oracle values $K_{\underline{0}}$, $N_{\underline{0}}$ and λ_0. $\omega = (1, 5)$, $B = 2{,}000$, $F = 20$, $V = 2$. 10^5 replications.

Figure 4 compares the excess variance of the adaptive algorithms and five static policies along the λ_0 space. Static policies are indexed by the variance ratio they input. In this setting, the minimax policy is given by $\lambda_{mM} = 30.25$, so the static policy with $\lambda = 31.6$ is nearly identical to the maximin design in van Breukelen and Candel (2015). Static policies do not incur sampling risk, so their excess variance is deterministic. As previously discussed, the excess variance of fixed designs vanishes whenever $\lambda \approx \lambda_0$ and rises as they get away from the true ratio. By contrast, the excess variance of the adaptive algorithms stays uniformly close to the oracle. The median performance of the algorithms (right) is even better than its mean performance (left) as the adaptive designs occasionally draw an unlucky pilot and pay for it. Performance is also best at small λ_0, where τ^2 is large relative to σ^2 and the pilot estimate is well-conditioned. The red markers in the endpoints specify the worst excess variance of each static policy. The message is clear: in finite samples, static policies can beat the algorithms in a neighborhood of they value they input, but they remain asymptotically worst case dominated by either adaptive algorithm. Adaptation is the robust alternative to ICC misspecification.

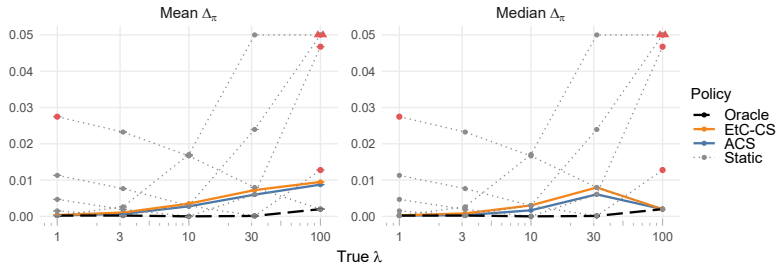


Figure 4: Excess variance across λ_0

Notes: Static policies imputed at $\lambda \in \{1, 3.16, 10, 31.6, 100\}$. $\mu = 0$, $\tau^2 = 1$, $B = 2{,}000$, $F = 20$, $V = 2$. 10^5 replications. Red triangles signal that worst case excess was capped at $\Delta_\pi = .05$ for display purposes.

Finally, Figure 5 plots the excess variance as a function of budget in absolute terms (left) and logs (right). The left panel shows that, in this environment, the excess variance of `EtC-CS` and `ACS` is already small at modest budgets and falls sharply until converging with oracle variance. The right panel shows slopes approximately of -1 for the static policies and of -2 for the adaptive algorithms, what is consistent with the results in Lemma 5.2 and Theorem 5.4, i.e. $\Theta(1/B)$ and $O(1/B^2)$, respectively. `ACS` dominates `EtC-CS` uniformly, with the gap being widest at small budgets. Recall that the excess variance of both algorithms is of the same order, and so `ACS` late-commitment advantage may only manifest through a constant and not a rate, which is more salient at small budgets.

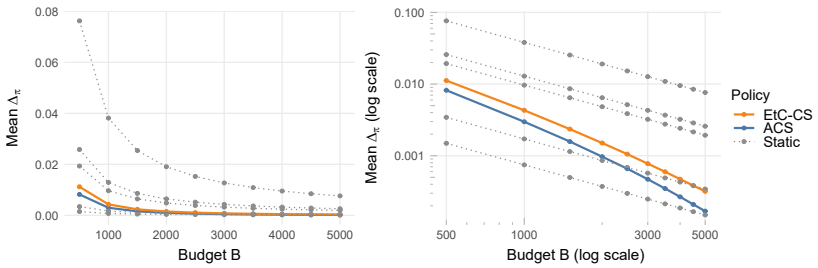


Figure 5: Mean excess variance across budget

Notes: Static policies imputed at $\lambda \in \{1, 3.16, 10, 31.6, 100\}$. 10^5 replications. $\mu = 0$, $\tau^2 = 1$, $F = 20$, $V = 2$.

7.2 Conditional Inference

I now turn to inference with an emphasis on `ACS`.[34] For a grid of symmetric error distributions—including bimodal and Laplace (heavy-tailed) distributions—, Figure 6 reports the density of the scaled sum $Z_B^\pi/\theta_{N_B^\pi}$ both unconditionally and conditional on the modal design, i.e. the (K, N) design with the highest frequency across simulation draws. The standard normal density is plotted alongside for comparison.[35] The evidence supports the results of Theorem 6.8: the limiting law remains normal after conditioning on any realized design.

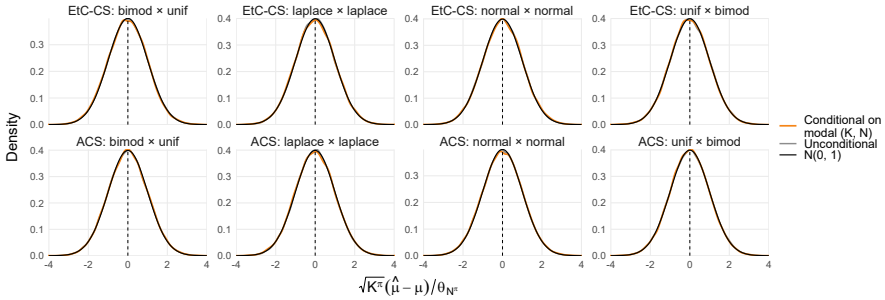


Figure 6: Empirical density of $Z_B^\pi/\theta_{N_B^\pi}$ at the modal design and unconditionally under symmetry

Notes: Distribution of u_k (left) and ε_{ki} (right). Distributions considered: Bimodal-Uniform, Laplace-Laplace, Normal-Normal, Uniform-Bimodal. $\mu = 0$, $\omega = (1, 5)$, $B = 2{,}000$, $F = 20$, $V = 2$. 10^5 replications.

For each realized K, Figure 7 reports the share of replications the interval $\hat{\mu}_{K^\pi,N^\pi} \pm 1.96\,\hat{b}_{K^\pi,N^\pi}/\sqrt{K^\pi}$ covers the true μ. Coverage is close to 95% uniformly across designs, as predicted by Theorem 6.9.

[34] `ACS` interacts with the data far more than `EtC-CS`. It conducts nearly all cluster exploration inside the pilot, so the design not only contains more data, but depends on the whole sequence $(\hat{\lambda}_{k,N_p})_{k\le s}$ rather than on a single statistic $\hat{\lambda}_p$. It is therefore the more demanding test of the results in Section 6.

[35] Figure 15 in the Appendix displays the full grid of combinations.

Distortions from normality are a finite sample artifact linked to studentization.[36]

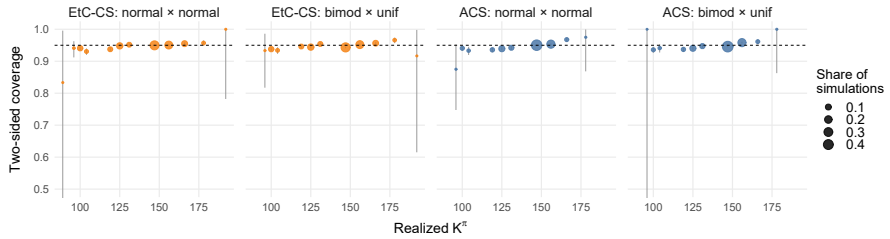


Figure 7: Design-conditional coverage of $\hat{\mu}_{K_B^\pi, N_B^\pi} \pm 1.96\, \hat{b}_{K_B^\pi, N_B^\pi} / \sqrt{K_B^\pi}$

Notes: Normal–normal and bimodal–uniform errors. Grey lines are Clopper–Pearson 95% intervals at each realized K_B^π. $\mu = 0$, $\omega = (1, 5)$, $B = 5{,}000$, $F = 20$, $V = 2$. 10^5 replications.

I next document the role of evenness in my results. Theorem 6.8 implies that for any asymptotically even policy $\mathbb{E}[Z_B^\pi \mid A_B] \to \mathbb{E}[Z_B^\pi] = 0$. I show that the symmetry requirement bites by comparing three asymptotically even policies—a naive-even policiy, `EtC-CS` and `ACS`—against a naive-odd policy that reads levels (and signs). The naive-odd and naive-even policies follow the same structure. They first run a minimal pilot (K_M, N_m), just like `EtC-CS`, and in the second stage they set

$$\text{(Odd)}\quad K^o = \begin{cases} K_M & \text{if } \bar{M}_{K_p, N_p} \geq 0 \\ K_m & \text{if } \bar{M}_{K_p, N_p} < 0 \end{cases} \qquad \text{(Even)}\quad K^e = \begin{cases} K_M & \text{if } K_p^{-1} \sum_k |M_{k,N_p}| \geq \mathbb{E}|M_{k,N_p}| \\ K_m & \text{if } K_p^{-1} \sum_k |M_{k,N_p}| < \mathbb{E}|M_{k,N_p}| \end{cases}$$

In both cases, the terminal depth is chosen to exhaust the budget.[37] The naive-odd policy reads the sign of the pilot mean, so conditioning on the realized design carries information about the signs $(\chi_{k,N})_k$ and breaks the sign-magnitude factorization of Lemma 6.3. Conversely, the naive-even policy only reads magnitudes, hence it is covered by the inference theory. Let $(C_j)_j$ be the j-th quintile of the realized width. For each C_j, Figure 8 plots the conditional average scaled sum, i.e. $\approx \mathbb{E}[Z_B^\pi \mid C_j]$. Conditional averages are indistinguishable from zero for all three even policies. By contrast, the conditional averages of the odd policy live visibly away from zero and change sign as predicted.[38]

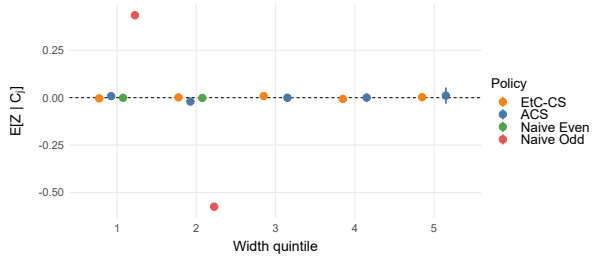


Figure 8: Design-conditional expectation $\mathbb{E}[Z_B^\pi \mid C_j]$ over width quintiles

Notes: Gaussian errors. The naive-even and naive-odd policies are two-valued, so only two quintiles carry positive mass. $\mu = 0$, $\omega = (1, 5)$, $B = 2{,}000$, $F = 20$, $V = 2$. 10^5 replications.

[36] The studentization tilt is driven by a selection effect. The feasible variance $\hat{b}^2_{K^\pi, N^\pi}$ depends on the error signs, and therefore is correlated with the conditioning event. The bias disappears approximately at a rate of $B^{-1/2}$ as $\hat{b}^2_{K^\pi, N^\pi} \to \theta^2_{N_0}$, as described in the Appendix. Observe as well that the tilt balances across cells, as unconditional coverage can be established without symmetry through the Anscombe route outlined in the text. The bias disappears when standardizing by the true variance $\theta^2_{N_B^\pi}$, as in Figure 6.

[37] $\mathbb{E}|M_{k,N}| = \theta_{N_p} \sqrt{2/\pi}$ is the exact mean of the half-normal under Gaussian errors. Hence, by a CLT argument, the designer should expect roughly 1/2 of the designs below the cutoff and 1/2 above.

[38] Observe that the odd policy does not only read the signs but also the levels. As a result, the policy is not measurable with respect to the filtration of residuals, and so it is not covered by Proposition 6.1 even under Gaussian errors.

Let $D_B := \sup_{t\in\mathcal{T}} \max_j \left|\mathbb{E}\left[e^{itZ_B^\pi}\mathbb{1}_{C_j}\right] - \phi(t)\,\mathbb{P}(C_j)\right|$ be the maximum discrepancy in conditional c.f. across width quintiles. According to Theorem 6.8, $D_B \to 0$ for any asymptotically even policy under error symmetry. Figure 9 plots D_B as a function of budget for four policy–error combinations: the naive-odd policy with Gaussian errors, `ACS` with Gaussian errors, `ACS` with Laplace errors, and `ACS` with asymmetric errors, i.e. $\text{Skew}(u_k) = 2$.[39]

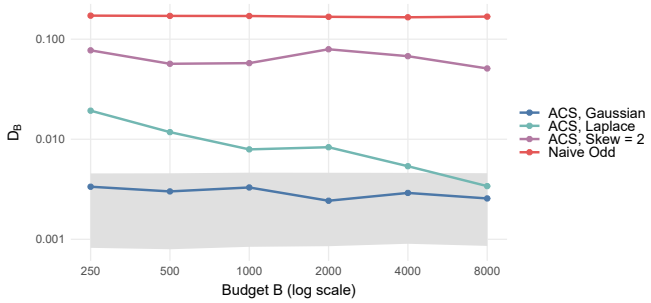


Figure 9: Maximal discrepancy in characteristic functions across width quintiles

Notes: Grey band represents the 95th percentile of D_B Monte Carlo distribution under exact conditional normality. Symmetric errors are $N(0,1)$, asymmetric errors are centered $\Gamma(1,1)$. $\mu = 0$, $\omega = (1,5)$, $F = 20$, $V = 2$. 10^5 replications.

The figure delineates the boundaries of the theory: either dropping evenness or symmetry compromises inference. D_B is uniformly bounded away from zero for both the naive-odd policy with Gaussian errors and `ACS` with asymmetric errors. This result suggest that, in general, neither condition can be dispensed with. By contrast, where both hold, the theory cannot be rejected. For `ACS` with Laplace errors, D_B declines steadily in B, exactly as predicted by Theorem 6.9. Moreover, for `ACS` with Gaussian errors, D_B sits uniformly at the simulation noise floor as predicted by Proposition 6.1.[40]

Finally, I illustrate the monotonic dependence between asymmetry and conditional bias. Figure 10 plots the conditional average scaled sum for each width quintile against the skew of the cluster level error. In the symmetric case (0 skew), all averages are conditionally unbiased; however, as skewness rises, conditional averages fan out. The mechanism is well-understood: under asymmetry, the realized design becomes informative about the error signs. Extreme designs reveal more about the signs, which is why the departures are largest in quintiles 1 and 5. Similarly, departures are larger under `ACS` because their induced designs depend on a weakly larger share of the total data.

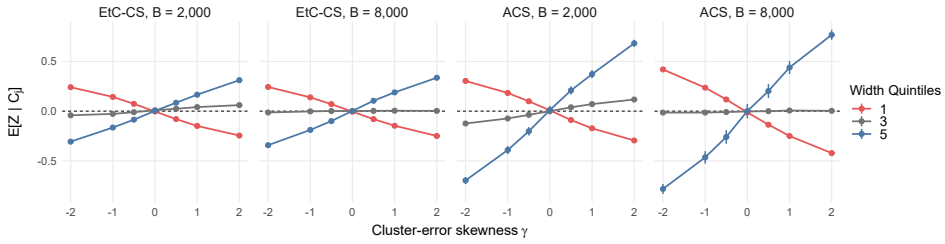


Figure 10: Design-conditional expectation $\mathbb{E}[Z_B^\pi \mid C_j]$ as a function of $\text{Skew}(u_k)$

Notes: $u_k \sim \Gamma(\alpha,\beta)$, with $\alpha = 4/\text{Skew}(u_k)^2$, $\beta = 1$. $\mu = 0$, $\omega = (1,5)$, $F = 20$, $V = 2$. 10^5 replications.

[39] $u_k \sim \Gamma(\alpha,\beta)$ with $\alpha = \beta = 1$ so that $\text{Skew}(u_k) = 2/\sqrt{\alpha} = 2$

[40] The grey band is the 95% Monte Carlo bound on D_B under exact conditional normality across 200 synthetic samples, each comprising 10^5 independent replications of $Z \sim N(0,1)$. Within each sample, replications are assigned to a design with the same frequency as the empirical distribution of designs induced by `ACS`. Finally, D_B is computed independently for each dataset.

8 Empirical Application: Fink et al. (2020b)

8.1 Setting and Design Problem

FJM study whether relaxing liquidity constraints during the agricultural lean season affects harvest outcomes and labor allocation among smallholder farmers in Chipata District, Zambia. To do so, they offered farmers favorable loans, in cash or in maize, repayable after the harvest season. The experiment was implemented through a cluster design with $K = 175$ villages and a target of $N = 22$ households per village. The treatment was assigned at the village level in three equally-sized arms (cash loan, maize loan and control) after a baseline survey.[41] The experiment is a natural setting for adaptive sampling: clear two-tier clustered setting, large number of equally-sized clusters, ex-ante fixed budget, and relevant fixed costs.

Importantly, unlike many other studies, FJM were not inattentive to the ICC. The authors run a small pilot and estimated the ICC, which they later used to power the study appropriately but not in a cost-aware fashion.[42] Conventional power calculations fix a detectable effect and solve for a sample size, but leave aside the cost structure.[43] In fact, the authors' design appears to be driven by logistic concerns. For instance, the decision to sample 22 households per village responds to the fact that that was the size of the smallest village on their initial sample.[44] These practices are not unique to FJM, but the *status quo* in field experiments that this manuscript seeks to replace.

There are two caveats in the counterfactual exercise below. First, I do not observe the cost parameters (F, V), so every result below is stated for a grid of cost ratios $F/V \in [1, 100]$.[45] In that sense, none of my claims should suggest that the realized design was suboptimal at the ratio FJM actually faced, even if the design was not intentionally selected in a cost-aware fashion. Second, the exercise uses baseline data alone to identify the design as the post-treatment outcomes are collected after one year, what leaves no room for adaptation. I therefore assume the variance ratio is stable across survey rounds t for every arm d, so that $\tau^2_{td} = \tau^2_d$ and $\lambda_{td} = \lambda_{0d}$. This assumption is prevalent in power analyses with fixed designs. Time homogeneity together with random assignment at baseline further implies that $\tau^2_d = \tau^2$ and $\lambda_{0d} = \lambda_0$. The assumptions are tested in Appendix G.2 and cannot be rejected at standard confidence levels.

The designer's problem. With three arms of equal size, the intention-to-treat (ITT) estimator for arm d specializes the cluster-level treatment estimator of Section 3 and Appendix A.2.1 to

$$\hat{\text{ITT}}^{(d)}_{K,N} \;=\; \hat{\mu}_d - \hat{\mu}_0 \;=\; \frac{3}{K}\left(\sum_{k\,:\,D_k=d} \frac{1}{N}\sum_{i=1}^{N} y_{ki} \;-\; \sum_{k\,:\,D_k=0} \frac{1}{N}\sum_{i=1}^{N} y_{ki}\right)$$

[41] Realized depth effectively ranged from 5 to 22 households, averaging 17.9 ≈ 18 households per village. This leads to a baseline sample of 3,139 households, of which 3,028 appear in the year-one survey. Appendix G.1 describes the sampling frame, the outcome variables, and the residualization process entering the analysis.

[42] The experiment register notes that the study was powered to detect a 10% increase in farm production value, see Fink et al. (2024). More details can be found in the Online Appendix (Fink et al., 2020a) and PAP (Fink et al., 2014).

[43] There is also no evidence that the size of the pilot was selected according to a statistical or cost-relevant criterion.

[44] Other features of the design point in the same direction. The authors initially considered sampling a list of 250 villages composed by a round five villages per administrative block. Villages were excluded from this list for reasons such as the impossibility of reaching them by vehicle during the rainy season. Fixed costs and logistics are then plausibly first-order. Yet, none of these concerns on their own imply that the realized design was necessarily cost-suboptimal.

[45] In practice, the distinction between fixed and variable costs may be blurry. Yet, in conversations with the authors and practitioners in the region, they are sympathetic to the idea that the realized F/V ratio could have been between 1 and 2. In that sense, I take a (very) conservative approach by always qualifying my assertions over the entire grid of cost ratios $[1, 100]$.

Under the homogeneity assumptions above, the design problem reduces to[46]

$$\min_{K,N} \max_{d\in\{1,2\}} \mathrm{Var}\big(\mathrm{I\hat{T}T}^{(d)}_{K,N}\big) \;=\; \min_{K,N} \frac{6\tau^2}{K}\left(1+\frac{\lambda}{N}\right) \qquad \text{s.t.} \quad K(F+NV)\le B \;\text{ a.s.}$$

This is the objective of Section 3 scaled by a constant, and so the optimal map and oracle allocation remain unchanged. Both treatment arms cost approximately the same per household—about 60 ZMW excluding administration costs—so V should be read as the average per-household cost across arms.

Design inputs. I estimate λ_0 from the full baseline sample and calibrate the λ interval $[\lambda_m, \lambda_M]$ through a cluster bootstrap of $\hat{\lambda}$, taking λ_m just below the 2.5th percentile and λ_M just above the 97.5th.[47] Table 4 reports the results for the two outcomes of interest: log(1 + harvest value), the register's primary outcome, and log(1 + ganyu mandays), the informal-labor measure the published paper emphasizes. Outcomes are residualized using the covariates included in the paper's specification.

Outcome	λ_0 $(=\hat{\lambda}_{K,N})$	ICC	$Q_{0.025}(\hat{\lambda}^{(b)}_{K,N})$	$Q_{0.975}(\hat{\lambda}^{(b)}_{K,N})$	λ_m	λ_M
log(harvest value + 1)	9.658	0.094	7.072	14.295	7	15
log(ganyu mandays + 1)	27.683	0.035	18.337	53.187	18	54

Table 4: Design inputs

Notes: Full baseline sample. Percentiles computed from a 2,000-replication bootstrap while holding village sizes fixed. Covariate-residualized data.

Is inference valid here? The results of Section 6 rest on Assumption 6.2, which requires the whole error block $C_k = (u_k, \varepsilon_{k1}, \varepsilon_{k2}, \dots)$ to satisfy symmetry. That restriction is not directly testable, since the analyst only observes the residual cluster averages $\hat{M}_{k,N}$ and not the residual blocks $\hat{C}_k$ themselves. The assumption is then only partially testable because symmetry of $M_{k,N}$ at every fixed N is necessary but not sufficient for block symmetry. Nevertheless, in the absence of nested error structures, as in the general theory of Appendix D, symmetry can be tested using standard methods. In this case, I use the symmetry test of Miao et al. (2006) on the residual averages $\hat{M}_{k,N}$ at $N \in \{4, 8, 12, 16, 20\}$. The results are calibrated against a Rademacher sign-flip null and aggregated across subsampling draws of size N by the dependence-robust rule of Meinshausen et al. (2009). Symmetry cannot be rejected for either outcome at any depth, at any conventional level. Although the test remains underpowered, the evidence is consistent with block symmetry. Appendix G.3 explains the symmetry testing methodology in greater detail and reports the full set of results. Under the symmetry null, standard errors and significance levels can be computed in the ordinary way: on the full sample, with no correction, and using Gaussian asymptotics.

8.2 Was the Realized Design Cost-Optimal?

I first inspect if the realized design can be rationalized by some cost structure, and how much is lost across the grid. Normalizing $V = 1$ and varying F, each ratio implies a budget $B = 175 \times (F + 17.9V)$ and an oracle design $(K^*_B(\lambda_0), N^*(\lambda_0))$. Figure 11 plots the potential excess variance of the realized design for each cost ratio.

[46] The current manuscript does not take any stance on how to trade across different outcome variables.

[47] This λ box is possibly too wide in a way that hurts the implementation of the adaptive algorithms. In practice, the authors probably had access to prior information that could narrow the interval. The intervals for both variables satisfy Assumption 3.4 over the full F/V grid.

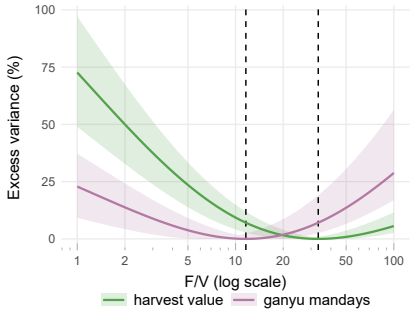


Figure 11: Excess variance of the realized FJM design across F/V

Notes: Excess variance relative to the variance of the oracle design (%). V set to 1. Rationalizing cost-ratios in dashed. $B = 175 \times (F + 17.9V)$. Ribbons are 95% cluster-bootstrap intervals, obtained by plugging-in the bootstrap percentiles of $\hat{\lambda}$ from Table 4. Covariate-residualized data.

The realized design is rationalized at a cost ratio of 33.2 for harvest value and 11.6 for ganyu mandays. Away from the rationalized costs ratios the excess variance grows quickly. For harvest value, the design carries up to 70% excess variance at the low-end of the cost grid, the likely scenario. For ganyu mandays, the design carries up to 30% excess variance at the low-end of the cost grid. Although one cannot categorically assert that the realized FJM design was not cost optimal, the broader point prevails: without proper accounting of the ICC and the cost structure, no design is safe from incurring large excess variance.[48]

8.3 What an Adaptive Design Would Have Delivered

Counterfactual designs cannot be run on the realized data, since any design at the same budget requires either more villages or more households per village than those surveyed by the authors. I therefore build a semi-synthetic population under the additive model $y_{ki} - \boldsymbol{x}_{ki}\boldsymbol{\beta} = \mu + u_k + \varepsilon_{ki}$, resampling the rescaled empirical cluster- and household-level residuals to match the first and second moments $(\hat{\mu}, \hat{\tau}^2, \hat{\sigma}^2)$ exactly. Appendix G.4 documents the semi-synthetic construction and validates it against the realized distribution. Figure 12 compares the excess variance of the realized design against the excess variance achieved by `EtC-CS` and `ACS` on the semi-synthetic data across the cost grid.

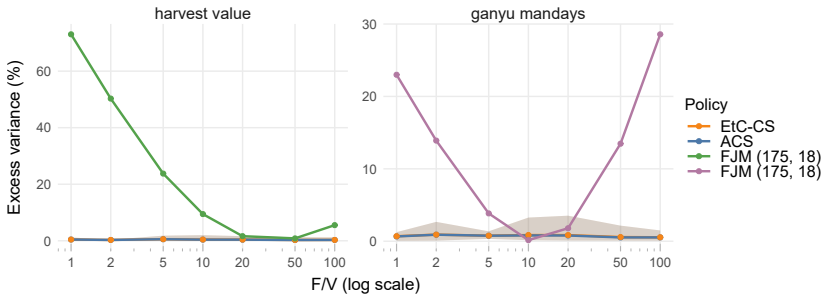


Figure 12: Excess variance across cost ratios

Notes: Excess variance relative to the variance of the oracle design (%). 10^4 replications. 95% confidence intervals across simulations.

[48] Cluster availability is not a binding constraint at any point on the grid. The village list from which FJM drew their initial sample contained 733 villages with ten or more smallholder households, 853 with five or more, and 973 with three or more. Additional districts around Chipata District within Eastern Province were also available had they been needed.

Three results stand out from the figure. First, both algorithms are essentially optimal at every cost ratio, with mean excess variance close to zero and narrow confidence bands, especially for the harvest value outcome. In the case of harvest value, entirely removing an excess variance of up to 70% corresponds to a variance reduction of roughly 41% percent, the figure invoked in the abstract and introduction. Second, both algorithms behave in a similar way. In this application, the budget supports several hundred villages, so even the worst case conservative pilot (K_M, N_m) entering `EtC-CS` identifies λ_0 with high precision, lowering the informational value of the extra clusters in `ACS`. Third, both algorithms dominate the realized design across the grid except in a neighborhood of the rationalizing ratio—which is unknown to the designer, hence the rationale for the exercise.

Cost savings at equal precision. Intuitively, minimizing the design variance subject to a budget is dual to minimizing the cost subject to attaining some precision level. In fact, in the deterministic continuous version of the problem the equivalence is exact. As a result, one may wonder how much money could have been saved by implementing an adaptive design which attained the same precision as the fixed design in FJM. Figure 13 shows that an adaptive design could have saved up to 42% of the total budget at equal precision for harvest value, and up to 22% for ganyu mandays. To produce this graph, each of the algorithms are run on a budget grid until the average precision of the induced designs across simulations matches the precision of FJM. Confidence intervals represent the 95% variation at the selected budget. The same counterfactual exercise is repeated for each cost ratio.

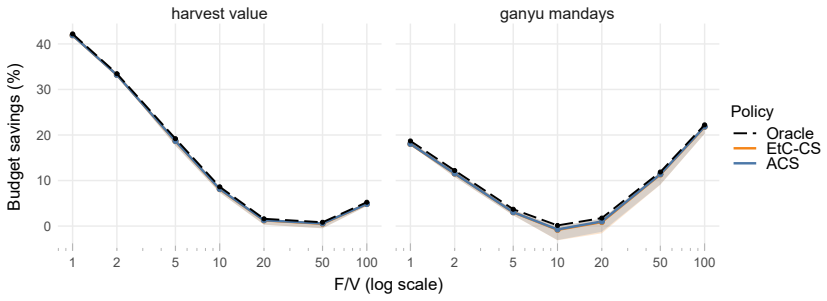


Figure 13: Budget savings across cost ratios

Notes: Budget savings (%) relative to the imputed budget $B = 175 \times (F + 17.9V)$. 10^4 replications. 95% confidence intervals across simulations at selected budget.

Consequences for the reported estimates. Finally, I ask what the design would have meant for the results in the paper. Applying the same construction to year-one outcomes, I re-estimate the treatment-effect specification in Table 4 Panel B on data generated under each counterfactual design. Tables 5 and 6 report the results. The synthetic replication of the realized design (column 3) tracks the real data (column 2) closely. The gap in standard errors comes from the fact that the synthetic design is balanced while the realized design is not, a phenomenon that is pronounced for ganyu mandays.[49] Point estimates are stable across designs—as they should be because the design affects precision, not identification—while standard errors fall by up to 17 percent for harvest value and 19 percent for ganyu mandays relative to the balanced realized design (column 3). The reduction is even more salient when compared to the implemented unbalanced design (column 2).

[49] My estimates differ slightly from those in Fink et al. (2020a) (column 1) because the published table pools year one and year two, whereas I use year one alone. Generating year-two data would have required modeling treatment rotation, differential attrition, and post-default exclusions, none of which the additive model accommodates cleanly.

	Paper	FJM	FJM	F/V						
	(Real)	(Real)	(Synthetic)	1	2	5	10	20	50	100
K/N	175/17.9	175/17.9	175/18	774/3	543/4	332/7	245/10	194/14	165/22	157/32
Cash	0.105*	0.112**	0.112***	0.109***	0.111***	0.114***	0.113***	0.112***	0.111***	0.111***
	(0.041)	(0.046)	(0.042)	(0.035)	(0.038)	(0.039)	(0.041)	(0.042)	(0.042)	(0.040)
Maize	0.056	0.037	0.036	0.034	0.037	0.037	0.036	0.037	0.035	0.038
	(0.054)	(0.047)	(0.042)	(0.035)	(0.038)	(0.039)	(0.041)	(0.042)	(0.041)	(0.040)

Table 5: Treatment-effect estimates for log(harvest value + 1), year-one outcomes.

Notes: Column 1 is reproduced from Fink et al. (2020a), Table 4 Panel B (years one and two pooled). Counterfactual designs are generated by `ACS`; coefficients and standard errors are means over 1,000 simulations, and K/N reports the mean realized design. CR1 cluster standard errors, as in the original. Stars from the normal approximation: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	FJM	FJM	F/V						
	(Real)	(Synthetic)	1	2	5	10	20	50	100
K/N	175/17.9	175/18	521/6	357/8	236/12	181/17	151/24	135/38	135/53
Cash	-0.182*	-0.180**	-0.186***	-0.180**	-0.184**	-0.186**	-0.182**	-0.182**	-0.178**
	(0.096)	(0.080)	(0.065)	(0.071)	(0.076)	(0.079)	(0.080)	(0.078)	(0.074)
Maize	-0.210**	-0.209***	-0.211***	-0.209***	-0.213***	-0.212***	-0.208***	-0.209***	-0.209***
	(0.101)	(0.080)	(0.065)	(0.071)	(0.077)	(0.079)	(0.080)	(0.078)	(0.075)

Table 6: Treatment-effect estimates for log(ganyu mandays + 1), year-one outcomes.

Notes: This specification is not reported in Fink et al. (2020b). Counterfactual designs are generated by `ACS`; coefficients and standard errors are means over 1,000 simulations, and K/N reports the mean realized design. CR1 cluster standard errors. Stars from the normal approximation: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

9 Extension: Heterogeneous Cluster Variances

9.1 The Problem of the Oracle Designer

The baseline model explicitly imposes variance homogeneity across clusters. This section relaxes this assumption by allowing clusters to carry observable covariates that shift the dispersion of the cluster-level errors,

$$y_{ki} = \mu + u_k + \varepsilon_{ki}, \qquad \mathbb{E}[u_k \mid X_k] = 0, \;\; \mathbb{E}\big[u_k^2 \mid X_k\big] = \tau_k^2 = \ell(X_k, \gamma), \qquad (X_k)_k \overset{iid}{\sim} F_X \tag{11}$$

where the link ℓ is known up to a finite-dimensional parameter $\gamma \in \Gamma$, $F_X \in \mathscr{F}_X$ is supported in $[x_m, x_M]$ and the remaining error structure is unchanged: cross-cluster independence, homogeneous unit-level variance σ^2 with $\varepsilon_{ki} \perp\!\!\!\perp X_k$, and independence between the error components $u_k \perp\!\!\!\perp \varepsilon_{ki}$. The environment is defined via $\omega_{\text{het}} = (\sigma^2, \gamma, F_X) \in [\sigma_m^2, \sigma_M^2] \times \Gamma \times \mathscr{F}_X := \Omega_{\text{het}}$. [50] For some fixed K, define as well $X^K := X_1, \ldots, X_K$. I further introduce two new population objects, namely the arithmetic and harmonic cluster variance means and the corresponding variance ratios. Expectations are taken

[50] Appendix H collects the formal assumptions (H1)–(H4): error structure and variance link, τ_k^2 compactness, interiority, and subgaussianity. The cost structure remains unchanged.

with respect to the distribution of covariates F_X.

$$\bar{\tau}^2 := \mathbb{E}\big[\tau_k^2\big], \qquad \tau_{\mathrm{h}}^2 := \frac{1}{\mathbb{E}\big[1/\tau_k^2\big]}, \qquad \bar{\lambda}_0 := \frac{\sigma^2}{\bar{\tau}^2}, \qquad \lambda_{0\mathrm{h}} := \frac{\sigma^2}{\tau_{\mathrm{h}}^2}$$

Timeline. The designer first commits to an estimator by choosing weights $(\vartheta_k)_k$ entering a weighted average of cluster means $\hat{\mu}_\vartheta = \sum_k \vartheta_k \hat{\mu}^k_{N^k} / \sum_k \vartheta_k$. The estimator class is selected before any sampling, but weights themselves are measurable with respect to the terminal information set of the designer, and so they may depend on the realized sequence of outcomes and covariates. The decision to recruit a new cluster may depend on all the information available to the designer at that time, but not on the covariate vector of the cluster itself. In particular, when recruiting a new cluster, the designer first pays the fixed cost F, and only then observes the cluster covariates X_k. By contrast, depth can be chosen at will, and may depend on the covariates of the current cluster as well as on the realized covariates of clusters recruited subsequently. Definition 9.1 formalizes the policy class.

Definition 9.1. *Admissible Policies under Heterogeneity. Let $\mathcal{I}_k$ be the designer's information when deciding whether to recruit cluster $k+1$, i.e. the set of covariates and outcomes of the first k clusters, plus independent randomization. For the oracle designer, all information sets contain, additionally, the environment ω_{het}. A policy is admissible if*

(i) Random cluster recruitment. $\{K \geq k+1\} \in \mathcal{I}_k$, with $(X_{k+1}, u_{k+1}, (\varepsilon_{k+1,i})_i) \perp\!\!\!\perp \mathcal{I}_k$.

(ii) Free revisits. Additional units from any recruited cluster can be added at any time at cost V. Depths may depend on all information available when they are chosen, including covariates of clusters recruited subsequently.

(iii) Budget. $KF + V\sum_{k\leq K} N^k \leq B$ almost surely.

Let π be a policy which induces a design $\big(K, (N^k, \vartheta_k)_k\big)$ with $N^k \geq 2$. Let $\theta_k^2 := \tau_k^2 + \sigma^2/N^k$ denote the variance of the k-th cluster mean for some realized design and covariate sequence. The problem of the designer is to select π to minimize the expected variance of the induced estimator subject to a budget constraint, i.e.

$$\min_{\pi} \ \mathbb{E}\left[W_\pi^{\mathrm{het}}\right] \qquad \text{s.t. } KF + \sum_{k=1}^{K} VN^k \leq B \ \text{ a.s.} \qquad \text{where } W_\pi^{\mathrm{het}} := \frac{\sum_{k\leq K} \vartheta_k^2\, \theta_k^2}{\big(\sum_{k\leq K} \vartheta_k\big)^2} \tag{12}$$

Unlike the benchmark model, the design variance depends on the sequence of realized covariates, and so even the oracle designer with access to the environment ω_{het} minimizes the variance in expectation. I study (12) for two leading classes of estimators: the unweighted average with weights $\vartheta_k = 1/K$, and the GLS average with weights $\vartheta_k = 1/(\tau_k^2 + \sigma^2/N^k)$.[51]

Heterogeneity is irrelevant for the unweighted average. Consider an oracle policy which fixes K before any sampling. When setting $\vartheta_k = 1/K$ in (12), the cluster variances $(\tau_k^2)_k$ do not interact with the depth choices. Intuitively, once cluster k is recruited, τ_k^2 is sunk and enters the variance additively, so no choice of N^k touches it. Depth only buys a reduction of σ^2/N^k, which does not depend on X_k. In the restricted class, the variance-minimizing allocation is therefore balanced and covariate-free, and the problem reduces to the original formulation. Furthermore, the fixed variance-minimizing width

[51] GLS weights are given by the inverse of the conditional variance of the cluster mean given the realized X_k.

does not depend on the covariate realizations, and so the optimal allocation is trivially admissible. When considering the unrestricted admissible class, the designer may improve on the policy above by sampling clusters strategically in an adapted manner. While there is no closed-form solution to the adapted oracle policy, Proposition 9.2 shows that adapted sampling of new clusters may only deliver a rate-negligible improvement. As a result, the policy which fixes width ex-ante, ignores covariates and balances depth across clusters is near-optimal in the unweighted problem.

Proposition 9.2. *Oracle Value under Equal Weights. Let $\vartheta_k = 1/K$ for every recruited cluste. Under (H1)–(H3) and the usual cost structure, the policy π that fixes $\left(K^*(\bar{\lambda}_0), N^*(\bar{\lambda}_0)\right)$ ex ante, ignoring the covariates X^K, is admissible and attains*

$$\mathbb{E}[W_\pi^{het}] = W_{UNW}^* = \frac{\left(\sqrt{F\bar{\tau}^2} + \sqrt{V\sigma^2}\right)^2}{B}$$

Moreover, there exists a constant $C < \infty$ not depending on B such that every admissible policy satisfies

$$\mathbb{E}\left[W_\pi^{het}\right] \geq W_{UNW}^* - \frac{C}{B^2}$$

Corollary 9.3. *`EtC-CS` is Robust to Heterogeneity. Let Π_{UNW} be the set of admissible policies with weights $\vartheta_k = 1/K$. Under (H1)–(H4), there exist constants C_1, C_2 not depending on B such that*

$$\max_{\omega_{het} \in \Omega_{het}} \left(\mathbb{E}\left[W_{\texttt{EtC-CS}}^{het}\right] - W_{UNW}^*\right) \leq \frac{C_1}{B^2}, \qquad \max_{\omega_{het} \in \Omega_{het}} \sup_{\pi \in \Pi_{UNW}} \left(\mathbb{E}\left[W_{\texttt{EtC-CS}}^{het}\right] - \mathbb{E}\left[W_\pi^{het}\right]\right) \leq \frac{C_2}{B^2}$$

The takeaway is that heterogeneity does not induce unbalanced sampling on its own. Unbalanced sampling only emerges from the interaction between heterogeneity and weighted estimation.

GLS weights. Consider again a policy which fixes K ex-ante. GLS weights $\vartheta_k = (\tau_k^2 + \sigma^2/N^k)^{-1}$ attain variance $1/\sum_k \vartheta_k$, thus depth choices now interact with the heterogeneous variances τ_k^2. In this context, a policy that optimally allocates depth after observing the covariate sequence X^K will equalize $\tau_k^2 N^k$ across clusters, making it inversely proportional to the cluster variance, the opposite of the Neyman logic. The reason is that, from the perspective of depth, τ_k^2 operates as an irreducible noise floor. The GLS factors will downweight noisy clusters anyway, and so depth can be used to lower the variance σ^2/N^k of quiet clusters. For these policies (with fixed (K, X^K)), (12) reduces to

$$\min_\pi \mathbb{E}\left[\frac{1}{K}\left(\tau_{\mathrm{h},K}^2 + \frac{\sigma^2}{\bar{N}}\right)\right] \qquad \text{s.t. } FK + V\sum_{k=1}^K N^k \leq B \text{ a.s.}, \qquad \text{where } \tau_{\mathrm{h},K}^2 := \frac{K}{\sum_k 1/\tau_k^2} \tag{13}$$

Equation (13) is analogous to the benchmark problem with the realized harmonic mean $\tau_{\mathrm{h},K}^2$ in place of τ^2 and the average depth $\bar{N}$ in place of N. Nevertheless, unlike the unweighted average case, the K-conditional minimization of (13) is inadmissible: the minimizing width depends on the realized sequence of covariates X^K via $\tau_{\mathrm{h},K}^2$. I circumvent this issue in two steps. First, I characterize a Wald-Jensen lower bound W_{GLS}^* to the general problem in (12). Then, I show that the feasible version of the minimizer to equation (13) with λ_{h} in place of $\lambda_{\mathrm{h},K}$ attains the lower bound up to an $O(1/B^2)$ term.

Proposition 9.4. *Oracle Value under GLS Weights. Define*

$$K^*_{GLS} = \frac{B}{F + \sqrt{\lambda_{0\mathrm{h}} F V}}, \qquad N^{k*}_{GLS} = \frac{\tau^2_{\mathrm{h},K}}{\tau^2_k}\sqrt{\lambda_{0\mathrm{h}}\frac{F}{V}}, \qquad W^*_{GLS} = \frac{\left(\sqrt{F\tau^2_{\mathrm{h}}} + \sqrt{V\sigma^2}\right)^2}{B}$$

Under (H1)–(H3), every admissible policy, whatever its weights, satisfies

$$\mathbb{E}[W^{het}_\pi] - W^*_{GLS} \geq 0, \qquad \forall\ \omega_{het}$$

Thus, W^*_{GLS} is a uniform lower bound to (12). Next, consider an oracle designer who fixes $K = K^*_{GLS}$ ex ante, recruits K clusters, and then allocates depths by $N^k = N^{k*}_{GLS}$. The oracle policy is admissible, satisfies the budget constraint a.s., has X^K-measurable depth, and attains $\mathbb{E}[W^{het}_\pi] \leq W^*_{GLS} + C/B^2$ for some finite C not depending on B.

The near-optimal allocations mimic the oracle allocations in the benchmark problem with $\lambda_{0\mathrm{h}}$ in place of λ_0 and the additional depth scaling term $\tau^2_{\mathrm{h},K}/\tau^2_k$. I conclude this subsection by arguing that heterogeneity should not be ignored. A designer who overlooks heterogeneity incurs $\Theta(1/B)$ excess variance, of the same order as that arising from λ-misspecification in the homogeneous setting.

Proposition 9.5. *Ignoring Heterogeneity Incurs a First-Order Loss. Fix some environment $\omega_{het} \in \Omega_{het}$. Under (H1)–(H2),*

$$W^*_{UNW} - W^*_{GLS} = \frac{F\left(\bar{\tau}^2 - \tau^2_{\mathrm{h}}\right) + 2\sqrt{VF\sigma^2}\left(\bar{\tau} - \tau_{\mathrm{h}}\right)}{B} \ \geq\ 0 \tag{14}$$

with equality iff τ^2_k is degenerate. The gap is otherwise $\Theta(1/B)$ and non-decreasing under mean-preserving spreads of τ^2_k.

9.2 The Problem of the Learner

The identification of the design inputs is considerably more challenging than in the homogeneous setting, especially for τ^2_k. Each cluster contributes a single draw of u_k, so τ^2_k is only identified by pooling across clusters through the link. To do so, the learner may use pilot observations to construct conditionally unbiased pseudo-cluster errors $q_{k,K}$, and then, estimate γ via nonlinear least squares of $q_{k,K}$ on $\ell(X_k, \cdot)$. In a second stage, the learner commits to the width $K^*(\hat{\lambda}_{\mathrm{h},K^p})$, recruits the remaining clusters at a conservative depth and, once all covariates have been observed, tops up every cluster according to the estimated optimal depth profile. Such policy exhausts the budget exactly. This heuristic is concretized in the weighted analogue of `EtC-CS`, denoted `wEtC-CS`, which guarantees implementability by running a conservative pilot in the first stage.[52]

Conjecture 9.6. *Performance of `wEtC-CS`. Under (H1)–(H4) and the regularity conditions of Appendix H, $0 \leq \max_{\omega_{het}\in\Omega_{het}} \mathbb{E}[W^{het}_{\texttt{wEtC-CS}}] - W^*_{GLS} = O(1/B^2)$.*

The lower bound is immediate from Proposition 9.4, since `wEtC-CS` is admissible. As for `EtC-CS`, the terminal width $K^*(\hat{\lambda}_{\mathrm{h},K_M})$ depends on the pilot only, so a proof can follow the structure of Theorem 5.4 and Corollary 9.3. Three ingredients are new. First, the inputs $(\hat{w}^2, \hat{\gamma}, \hat{\tau}^2_{\mathrm{h}}, \hat{\lambda}_{\mathrm{h}})$ must concentrate around the truth at a sufficient rate. The main difficulty lies in $\hat{\gamma}$, which is estimated by

[52] Further details can be found in Appendix H.

nonlinear least squares on cross-sectionally dependent pseudo-outcomes, so the maximal Bernstein arguments of Appendix C must be adapted to the nonlinear setting and hold uniformly in Ω_{het}. Second, $\hat{\gamma}$ enters the depth profile and the weights not only through $\hat{\lambda}_{\text{h}}$ but also through $\hat{\tau}_k^2$. The resulting misallocation must then be shown to cost no more than $O(1/B^2)$. Third, the terminal design depends on the pilot composition, which generates a covariance term similar to the one appearing in Corollary 9.3. *A priori*, no individual step is incompatible with the target rate of $O(1/B^2)$, but I leave the formal argument to future work.

10 Conclusion

This is the first paper to provide optimality guarantees for the joint width-versus-depth allocation problem without any knowledge on the ICC nor any constraints on the policy class. Optimality is achieved by two algorithms—a two-stage rule and a fully sequential rule—that learn the ICC within the experiment and implement the oracle design in the limit. They achieve expected excess variance of order $1/B^2$, a full order of the budget below the cost of misspecified static designs, maximin included. By deriving a matching lower bound, I show that their rate cannot be improved upon.

Variance gains do not come at the expense of inference. Under a symmetry restriction on the errors, estimators remain asymptotically normal after conditioning on non-degenerate asymptotically even designs with no reweighting, no sample splitting, and no discarded pilot data. This principle extends beyond the cluster setting and validates standard inference for popular sampling procedures like Neyman allocations computed from pilot variances. Its boundary, however, remains equally sharp: odd policies or asymmetric errors can restore the distortions identified by the reweighting literature.

The empirical results show that the precision gains are economically meaningful. In a reconstruction of Fink et al. (2020b), the adaptive allocation delivered up to 41% lower variance at the same cost compared to the realized design. The implementation of a live survey in real time using one of the algorithms further confirms the feasibility of my methods. Some of the modeling decisions shed light on the next steps. For instance, symmetry still excludes important distributions such as binary and censored outcomes; costs are assumed to be known, linear and irreversible; adaptivity costs are not accounted for in the decision problem; and the current framework only accommodates for a single outcome variable. Despite these limitations, the results provide a foundation for cost-aware data collection in multistage surveys and clustered randomized trials.

References

Abadie, A., Athey, S., Imbens, G. W., and Wooldridge, J. M. (2023). When should you adjust standard errors for clustering? *The Quarterly Journal of Economics*, 138(1):1–35.

Adusumilli, K. (2026). How to sample and when to stop sampling: The generalised wald problem and minimax policies. *Review of Economic Studies*, 93(1):1–34.

Anscombe, F. J. (1952). Large-sample theory of sequential estimation. In *Mathematical Proceedings of the Cambridge Philosophical Society*, volume 48, pages 600–607. Cambridge University Press.

Bai, Y. (2022). Optimality of matched-pair designs in randomized controlled trials. *American Economic Review*, 112(12):3911–3940.

Ballin, M. and Barcaroli, G. (2013). Joint determination of optimal stratification and sample allocation using genetic algorithm. *Survey Methodology*, 39(2):369–393.

Barcaroli, G., Fasulo, A., Guandalini, A., and Terribili, M. D. (2022). Two-stage sampling design and sample selection with the r package r2beat. *arXiv preprint arXiv:2209.06532*.

Bethel, J. (1989). Sample allocation in multivariate surveys. *Survey methodology*, 15(1):47–57.

Biemer, P. P. and Lyberg, L. E. (2003). *Introduction to survey quality*. John Wiley & Sons.

Bruhn, M. and McKenzie, D. (2009). In pursuit of balance: Randomization in practice in development field experiments. *American economic journal: applied economics*, 1(4):200–232.

Campbell, M. K., Grimshaw, J. M., and Elbourne, D. R. (2004). Intracluster correlation coefficients in cluster randomized trials: empirical insights into how should they be reported. *BMC Medical Research Methodology*, 4(1):9.

Canay, I. A., Romano, J. P., and Shaikh, A. M. (2017). Randomization tests under an approximate symmetry assumption. *Econometrica*, 85(3):1013–1030.

Cesa-Bianchi, N., Colomboni, R., and Kasy, M. (2025). Adaptive maximization of social welfare. *Econometrica*, 93(3):1073–1104.

Chambaz, A., van Der Laan, M. J., and Zheng, W. (2015). Targeted covariate-adjusted response-adaptive lasso-based randomized controlled trials. *Modern adaptive randomized clinical trials: statistical, operational, and regulatory aspects*, pages 345–368.

Chow, S.-C. and Corey, R. (2011). Benefits, challenges and obstacles of adaptive clinical trial designs. *Orphanet journal of rare diseases*, 6(1):79.

Chow, Y. S. and Robbins, H. (1965). On the asymptotic theory of fixed-width sequential confidence intervals for the mean. *The Annals of Mathematical Statistics*, 36(2):457–462.

Cicchitelli, G., Herzel, A., and Montanari, G. E. (1992). *Il campionamento statistico*. il Mulino.

Cochran, W. G. (1977). *Sampling techniques*, volume 3. wiley New York.

Cytrynbaum, M. (2021). Optimal stratification of survey experiments. *arXiv:2111.08157*.

Cytrynbaum, M. (2024). Covariate adjustment in stratified experiments. *Quantitative Economics*, 15(4):971–998.

Dai, J., Gradu, P., and Harshaw, C. (2023). Clip-OGD: An experimental design for adaptive Neyman allocation in sequential experiments. In *Advances in Neural Information Processing Systems*, volume 36, pages 32235–32269.

Dudewicz, E. J. and Dalal, S. R. (1975). Allocation of observations in ranking and selection with unequal variances. *Sankhyā: The Indian Journal of Statistics, Series B*, pages 28–78.

Fink, G., Jack, B. K., and Masiye, F. (2014). Food constraints, yield uncertainty and "ganyu" labour: Pre-analysis plan. Pre-analysis plan. AEA RCT Registry ID: AEARCTR-0000130.

Fink, G., Jack, B. K., and Masiye, F. (2020a). Online appendix to seasonal liquidity, rural labor markets, and agricultural production. Online appendix. Appendix to American Economic Review, 110(11): 3351–3392.

Fink, G., Jack, B. K., and Masiye, F. (2020b). Seasonal liquidity, rural labor markets, and agricultural production. *American Economic Review*, 110(11):3351–3392.

Fink, G., Jack, B. K., and Masiye, F. (2024). Food constraints, yield uncertainty and "ganyu" labour. *AEA RCT Registry*.

Fithian, W., Sun, D., and Taylor, J. (2014). Optimal inference after model selection. *arXiv preprint arXiv:1410.2597*.

Gould, A. L. and Shih, W. J. (1992). Sample size re-estimation without unblinding for normally distributed outcomes with unknown variance. *Communications in Statistics – Theory and Methods*, 21(10):2833–2853.

Hadad, V., Hirshberg, D. A., Zhan, R., Wager, S., and Athey, S. (2021). Confidence intervals for policy evaluation in adaptive experiments. *Proceedings of the national academy of sciences*, 118(15):e2014602118.

Hahn, J., Hirano, K., and Karlan, D. (2011). Adaptive experimental design using the propensity score. *Journal of Business & Economic Statistics*, 29(1):96–108.

Hansen, M. H. and Hurwitz, W. N. (1942). Relative efficiencies of various sampling units in population inquiries. *Journal of the American Statistical Association*, 37(217):89–94.

Kasy, M. and Sautmann, A. (2021). Adaptive treatment assignment in experiments for policy choice. *Econometrica*, 89(1):113–132.

Kiefer, J. and Wolfowitz, J. (1960). The equivalence of two extremum problems. *Canadian Journal of Mathematics*, 12:363–366.

Kieser, M. and Friede, T. (2003). Simple procedures for blinded sample size adjustment that do not affect the type I error rate. *Statistics in Medicine*, 22(23):3571–3581.

Kish, L. (1965). Survey sampling. new york: John wesley & sons. *Am Polit Sci Rev*, 59(4):1025.

Knox, S. A. and Chondros, P. (2004). Observed intra-cluster correlation coefficients in a cluster survey sample of patient encounters in general practice in australia. *BMC Medical Research Methodology*, 4(1):30.

Lake, S., Kammann, E., Klar, N., and Betensky, R. (2002). Sample size re-estimation in cluster randomization trials. *Statistics in Medicine*, 21(10):1337–1350.

Lattimore, T. and Szepesvári, C. (2020). *Bandit algorithms*. Cambridge University Press.

Li, D., Raymond, L., and Bergman, P. (2026). Hiring as exploration. *Review of Economic Studies*, 93(2):1200–1240.

Lu, J. (2025). Big data analysis. *Springer Books*.

Meinshausen, N., Meier, L., and Bühlmann, P. (2009). P-values for high-dimensional regression. *Journal of the American Statistical Association*, 104(488):1671–1681.

Miao, W., Gel, Y. R., and Gastwirth, J. L. (2006). A new test of symmetry about an unknown median. In *Random walk, sequential analysis and related topics: A festschrift in honor of Yuan-Shih Chow*, pages 199–214. World Scientific.

Moerbeek, M., van Breukelen, G. J. P., and Berger, M. P. F. (2000). Design issues for experiments in multilevel populations. *Journal of Educational and Behavioral Statistics*, 25(3):271–284.

Neopane, O., Ramdas, A., and Singh, A. (2024). Logarithmic neyman regret for adaptive estimation of the average treatment effect. *arXiv preprint arXiv:2411.14341*.

Neopane, O., Ramdas, A., and Singh, A. (2025). Optimistic algorithms for adaptive estimation of the average treatment effect. *arXiv preprint arXiv:2502.04673*.

Nie, X., Tian, X., Taylor, J., and Zou, J. (2018). Why adaptively collected data have negative bias and how to correct for it. In *International Conference on Artificial Intelligence and Statistics*, pages 1261–1269. PMLR.

Raudenbush, S. W. (1997). Statistical analysis and optimal design for cluster randomized trials. *Psychological Methods*, 2(2):173–185.

Rinott, Y. (1978). On two-stage selection procedures and related probability-inequalities. *Communications in Statistics-Theory and methods*, 7(8):799–811.

Rudelson, M. and Vershynin, R. (2013). Hanson-wright inequality and sub-gaussian concentration.

Sedgwick, P. (2015). Multistage sampling. *Bmj*, 351.

Shapiro, A. (1990). On concepts of directional differentiability. *Journal of optimization theory and applications*, 66(3):477–487.

Shapiro, A. (1991). Asymptotic analysis of stochastic programs. *Annals of Operations Research*, 30(1):169–186.

Shen, Y. and Fisher, L. (1999). Statistical inference for self-designing clinical trials with a one-sided hypothesis. *Biometrics*, 55(1):190–197.

Spiegelhalter, D. J. (2001). Bayesian methods for cluster randomized trials with continuous responses. *Statistics in Medicine*, 20(3):435–452.

Stein, C. (1945). A two-sample test for a linear hypothesis whose power is independent of the variance. *Annals of Mathematical Statistics*, 16(3):243–258.

Tabord-Meehan, M. (2023). Stratification trees for adaptive randomisation in randomised controlled trials. *Review of Economic Studies*, 90(5):2646–2673.

Tam, B. (2005). Household sample surveys in developing and transition countries.

Tschuprow, A. (1923). On the two different aspects of the representative method: the method of stratified son the mathematical expectation of the moments of frequency distributions in the case of correlated observation sampling and the method of purposive selection. *Metron*, 2(646-683):192.

Turner, R. M., Prevost, A. T., and Thompson, S. G. (2005). Prior distributions for the intracluster correlation coefficient, based on multiple previous estimates, and their application in cluster randomized trials. *Clinical Trials*, 2(2):108–118.

van Breukelen, G. J. P. and Candel, M. J. J. M. (2015). Efficient design of cluster randomized and multicentre trials with unknown intraclass correlation. *Statistical Methods in Medical Research*, 24(5):540–556.

van Breukelen, G. J. P. and Candel, M. J. J. M. (2018). Efficient design of cluster randomized trials with treatment-dependent costs and treatment-dependent unknown variances. *Statistics in Medicine*, 37:3027–3046.

van Breukelen, G. J. P. and Candel, M. J. J. M. (2021). Maximin design of cluster randomized trials with heterogeneous costs and variances. *Biometrical Journal*, 63(7):1444–1463.

van Schie, S. and Moerbeek, M. (2014). Re-estimating sample size in cluster randomised trials with active recruitment within clusters. *Statistics in Medicine*, 33(19):3253–3268.

Viviano, D. (2022). Experimental design under network interference. *arXiv preprint arXiv:2003.08421*.

Wager, S. (2024). Causal inference: A statistical learning approach. *URL https://web. stanford. edu/swager/causal_inf_book. pdf*.

Wittes, J. and Brittain, E. (1990). The role of internal pilot studies in increasing the efficiency of clinical trials. *Statistics in Medicine*, 9(1-2):65–72.

Wittes, J., Schabenberger, O., Zucker, D. M., Brittain, E., and Proschan, M. (1999). Internal pilot studies I: Type I error rate of the naive t-test. *Statistics in Medicine*, 18(24):3481–3491.

Wooldridge, J. M. (2003). Cluster-sample methods in applied econometrics. *American Economic Review*, 93(2):133–138.

Yin, G. and Shen, Y. (2005). Adaptive design and estimation in randomized clinical trials with correlated observations. *Biometrics*, 61(2):362–369.

Zhang, K., Janson, L., and Murphy, S. (2021). Statistical inference with m-estimators on adaptively collected data. *Advances in neural information processing systems*, 34:7460–7471.

A Proofs and Appendix to Section 3

A.1 Generalized Problem

Let Assumptions 3.1, 3.2, 3.3 and 3.4 hold throughout. Consider the general problem in which the designer samples an arbitrary number of units $N^k \geq 0$ from each cluster k subject to the budget constraint. Let $\boldsymbol{N} = (N^k)_k$, and define $\mathcal{K} = \{k : N^k > 0\}$ with $K = |\mathcal{K}|$. The variance of the average-of-averages estimator is given by

$$W(\boldsymbol{N}) = \text{Var}(\hat{\mu}_{\boldsymbol{N}}) = \text{Var}\left(\frac{1}{K}\sum_{k\in\mathcal{K}}\frac{1}{N^k}\sum_{i=1}^{N^k} y_{ki}\right) = \text{Var}\left(\mu + \frac{1}{K}\sum_{k\in\mathcal{K}} u_k + \frac{1}{K}\sum_{k\in\mathcal{K}}\frac{1}{N^k}\sum_{i=1}^{N^k}\varepsilon_{ki}\right)$$
$$= \frac{K}{K^2}\tau^2 + \frac{1}{K^2}\sum_{k\in\mathcal{K}}\frac{\sigma^2}{N^k} = \frac{\tau^2}{K}\left(1 + \frac{\lambda_0}{K}\sum_{k\in\mathcal{K}}\frac{1}{N^k}\right) = \frac{\tau^2}{K}\left(1 + \frac{\lambda_0}{N^H(\boldsymbol{N})}\right)$$

where the fourth equality uses cross-cluster independence and $u_k \perp\!\!\!\perp \varepsilon_{ki}$ (Assumption 3.1), and

$$N^H(\boldsymbol{N}) = \left(\frac{1}{K}\sum_{k\in\mathcal{K}}\frac{1}{N^k}\right)^{-1}$$

is the harmonic mean of the sampled depths. As expected, when $N^k = N$ for all $k \in \mathcal{K}$, the expression reduces to (2). The generalized problem can be written as

$$\min_{\boldsymbol{N}} W(\boldsymbol{N}) \qquad \text{s.t.} \quad KF + V\sum_{k\in\mathcal{K}} N^k \leq B \tag{15}$$

Conditional on $\mathcal{K}$, $W(\boldsymbol{N})$ is strictly decreasing in each N^k and so the budget constraint binds. Moreover, by conditioning on K and solving from the budget constraint, the total number of units $S = \sum_{k\in\mathcal{K}} N^k$ remains fixed at the optimum, i.e. $S = (B - KF)/V$. The variance minimization problem then reduces to minimizing $\sum_{k\in\mathcal{K}} 1/N^k$ subject to a constant total number of units S. Since $x \mapsto 1/x$ is strictly convex on $(0, \infty)$, Jensen's inequality gives

$$\frac{1}{K}\sum_{k\in\mathcal{K}}\frac{1}{N^k} \;\geq\; \frac{1}{\frac{1}{K}\sum_{k\in\mathcal{K}} N^k} \;=\; \frac{K}{S}$$

Equivalently $N^H(\boldsymbol{N}) \leq S/K$, with equality if and only if $N^k = S/K = N$ for all $k \in \mathcal{K}$. The variance minimizing allocation is therefore balanced, conditional on K. Next, impose a balanced allocation $N^k = N$ and solve for K from the budget constraint such that $K(N) = B/(F + VN)$. Finally,

$$W(K(N), N) = \frac{\tau^2(F + VN)}{B}\left(1 + \frac{\lambda_0}{N}\right) = \frac{\tau^2}{B}\left[F + V\lambda_0 + VN + \frac{F\lambda_0}{N}\right]$$

The expanded form makes both optimality conditions (FOC and SOC) immediate,

$$W_N = \frac{\tau^2}{B}\left[V - \frac{F\lambda_0}{N^2}\right] = 0 \implies N^*(\lambda_0) = \sqrt{\lambda_0 F/V}, \qquad W_{NN} = \frac{2F\lambda_0\tau^2}{BN^3} > 0 \;\text{ for } N > 0$$

The stationary point is a global minimum and interior in the feasible region due to strict convexity on $(0, \infty)$ and Assumption 3.4, respectively. Evaluating the variance object at $N^*(\lambda_0)$ returns the oracle

variance

$$W_0 = \frac{\tau^2}{B}\left[F + V\lambda_0 + 2\sqrt{FV\lambda_0}\right] \;=\; \frac{\tau^2\left(\sqrt{F}+\sqrt{V\lambda_0}\right)^2}{B}$$

The oracle allocation, optimal map and oracle variance of the generalized problem all coincide with those of the balanced problem in (3).

A.2 Experimental Design

Assumption A.1. *Linear Additive Model and Error Distributions under Potential Outcomes. For $d \in \{0,1\}$, let the potential outcomes y_{ki}^d follow an additive model with nested errors*

$$y_{ki}^d = \mu_d + u_k^d + \varepsilon_{ki}^d$$

where μ_d is the expected potential outcome under treatment $D_{ki} = d$, u_k^d is the cluster-level error for potential outcome d and ε_{ki}^d is the unit-level error for potential outcome d. $(u_k^0, u_k^1)_k$ is an iid sequence with $\mathbb{E}[u_k^d] = 0$, $\mathbb{E}[(u_k^d)^2] = \tau_d^2 > 0$, and $Cov(u_k^0, u_k^1) = \tau_{01}$; $(\varepsilon_{ki}^0, \varepsilon_{ki}^1)_{k,i}$ is an iid sequence with $\mathbb{E}[\varepsilon_{ki}^d] = 0$ and $\mathbb{E}[(\varepsilon_{ki}^d)^2] = \sigma_d^2 > 0$; $(u_k^d)_{k,d} \perp\!\!\!\perp (\varepsilon_{ki}^d)_{k,i,d}$. Treatment assignment is also assumed independent of the full collection of potential outcomes.

Assumption A.2. *Arm-Specific Linear Costs. Let treated (control) clusters carry fixed costs F_1 (F_0) and units carry variable cost V_1 (V_0), then*

(i) Clusters as Treatment Level: a design $(K_d, N_d)_d$ costs $\sum_d K_d(F_d + V_d N_d)$.

(ii) Clusters as blocks: a design (K, N_1, N_0) costs $K(F + V_1 N_1 + V_0 N_0)$.

A.2.1 Clusters as Treatment Level

The designer samples K clusters and N units per cluster, and randomly allocates $K/2$ clusters to treatment and $K/2$ to control. Let $D_{ki} \in \{0,1\}$ denote the treatment status of unit i in cluster k. Let $y_{ki}(d, \boldsymbol{d})$ denote the potential outcome of unit i in cluster k when $D_{ki} = d$ and $(D_{kj})_{j\neq i} = \boldsymbol{d}$. Under cluster-level assignment $\boldsymbol{d} = (d, \ldots, d)$, so the potential outcome admits the simplified representation $y_{ki}(d, \boldsymbol{d}) = y_{ki}(d)$.

The estimand is the total average treatment effect $\text{TATE} = \mathbb{E}[y_{ki}(1, \mathbf{1}) - y_{ki}(0, \mathbf{0})] = \mathbb{E}[y_{ki}(1) - y_{ki}(0)] = \mu_1 - \mu_0$, and the natural estimator is the difference in cluster means

$$\hat{\text{TATE}}_{K,N} = \frac{2}{K}\left(\sum_{k:D_k=1} \hat{\mu}_N^k - \sum_{k:D_k=0} \hat{\mu}_N^k\right)$$

where $D_k \in \{0,1\}$ is the treatment status of cluster k. The variance of the TATE estimator is then

$$\text{Var}\big(\hat{\text{TATE}}_{K,N}\big) = \frac{4}{K^2}\cdot\frac{K}{2}\left(\tau_1^2 + \frac{\sigma_1^2}{N}\right) + \frac{4}{K^2}\cdot\frac{K}{2}\left(\tau_0^2 + \frac{\sigma_0^2}{N}\right) = \frac{2}{K}\left(\tau_1^2 + \tau_0^2 + \frac{\sigma_1^2 + \sigma_0^2}{N}\right)$$

where no $\text{Cov}(u_k^1, u_k^0)$ term arises because the treatment arms draw on disjoint sets of clusters. Equivalently, using the notation of Table 1,

$$W(K,N) = \frac{2\tau^2}{K}\left(1 + \frac{\lambda}{N}\right), \qquad \tau^2 = \tau_1^2 + \tau_0^2, \qquad \sigma^2 = \sigma_0^2 + \sigma_1^2, \qquad \lambda = \sigma^2/\tau^2$$

A.2.2 Clusters as Blocks under No Interference

The designer samples K clusters and N units per cluster, and randomly allocates treatment to $N/2$ units within each cluster. Assuming no interference $y_{ki}(d, \boldsymbol{d}) = y_{ki}(d) = y_{ki}^d$. Let y_{ki}^d follow the model in Assumption A.1. The relevant estimand is the average treatment effect $\text{ATE} = \mathbb{E}[y_{ki}(1) - y_{ki}(0)] = \mu_1 - \mu_0$, and the natural estimator is the average of within-cluster differences in means,

$$\hat{\text{ATE}}_{K,N} = \frac{2}{KN} \sum_{k=1}^{K} \left(\sum_{i:D_{ki}=1} y_{ki} - \sum_{i:D_{ki}=0} y_{ki} \right)$$

The term τ_{01} now appears in the variance of the estimator as both potential cluster errors (u_k^0, u_k^1) are active within the same cluster. In particular, $\text{Var}(\hat{\text{ATE}}_{K,N}) = \frac{(\tau_1^2+\tau_0^2-2\tau_{01})}{K} + \frac{2(\sigma_1^2+\sigma_0^2)}{KN}$. Re-writing into the notation of Table 1,

$$W(K,N) = \frac{\tau^2}{K}\left(1 + \frac{\lambda}{N}\right), \qquad \tau^2 = \tau_1^2 + \tau_0^2 - 2\tau_{01}, \qquad \sigma^2 = 2(\sigma_0^2 + \sigma_1^2), \qquad \lambda = \sigma^2/\tau^2$$

Remark A.3. *Degenerate Corner. When $u_k^1 = u_k^0$ almost surely, $\tau_{01} = \tau_1^2 = \tau_0^2$ and hence $\tau^2 = 0$. In other words, within-cluster differencing removes the cluster-level noise entirely, so the variance is minimized by sampling a single cluster as deeply as the budget allows. Assumption 3.2 excludes this possibility by imposing $\tau_m^2 > 0$.*

Remark A.4. *Balanced Design and Optimality. Within the respective balanced classes, $K_d = K/2, N_d = N$ for clusters as treatment level and $N_d = N/2$ for clusters as blocks, the cost structure can be re-written as $K(\bar{F} + \bar{V}N)$ and $K(F + \bar{V}N)$, respectively. Where $\bar{F} = (F_0 + F_1)/2$ and $\bar{V} = (V_0 + V_1)/2$. The results in the paper then hold verbatim with $\bar{F}$ in place of F and $\bar{V}$ in place of V. Nevertheless, the balanced design needs not be optimal in the policy unconstrained problem. In general, the unconstrained optimum is balanced iff the variance to cost ratios match, i.e. $\tau_1^2/F_1 = \tau_0^2/F_0$ and $\sigma_1^2/V_1 = \sigma_0^2/V_0$. A sufficient condition is cost and variance homogeneity across treatment arms.*

B Proofs of Section 4

Proof of Lemma 4.1. Recall the definition $M_{k,N} = u_k + \bar{\varepsilon}_{k,N} = \hat{\mu}_N^k - \mu$ where $\bar{\varepsilon}_{k,N} = \frac{1}{N}\sum_{i=1}^N \varepsilon_{ki}$. Let $\bar{M}_{K,N} = \frac{1}{K}\sum_{k=1}^K M_{k,N}$. Under Assumption 3.1, $(M_{k,N})_k$ is *iid*, mean zero, with $\mathbb{E}[M_{k,N}^2] = \tau^2 + \sigma^2/N = \theta_N^2 < \infty$.

Unbiasedness. Since each $y_{ki} - \hat{\mu}_N^k = \varepsilon_{ki} - \bar{\varepsilon}_{k,N}$ has mean zero and variance $\frac{N-1}{N}\sigma^2$,

$$\mathbb{E}\big[S_{k,N}^2\big] = \frac{1}{N-1}\sum_{i=1}^{N} \text{Var}\big(\varepsilon_{ki} - \bar{\varepsilon}_{k,N}\big) = \frac{1}{N-1} \cdot N \cdot \frac{N-1}{N}\,\sigma^2 = \sigma^2 \implies \mathbb{E}\big[\hat{w}_{K,N}^2\big] = \sigma^2$$

Moreover, by the sample-variance identity applied to the *iid* sequence $(\hat{\mu}_N^k)_k$,

$$\mathbb{E}\big[\hat{b}_{K,N}^2\big] = \frac{1}{K-1}\,\mathbb{E}\left[\sum_{k=1}^{K}\big(M_{k,N} - \bar{M}_{K,N}\big)^2\right] = \text{Var}\big(M_{k,N}\big) = \tau^2 + \frac{\sigma^2}{N} = \theta_N^2$$

(i) Fix $N \geq 2$. Observe that $\hat{w}^2_{K,N} = \frac{1}{K}\sum_{k=1}^{K} S^2_{k,N}$ is an average of *iid* summands, because of cross-cluster independence as in Assumption 3.1, and $\mathbb{E}|S^2_{k,N}| = \sigma^2 < \infty$. The SLLN along the K dimension then yields $\hat{w}^2_{K,N} \overset{\text{a.s.}}{\underset{K}{\to}} \sigma^2$.

Fix $K \geq 1$. Exchanging the order of summation and re-writing yields

$$\hat{w}^2_{K,N} = \frac{1}{K}\sum_{k=1}^{K} \frac{1}{N-1}\sum_{i=1}^{N} \left(\varepsilon_{ki} - \bar{\varepsilon}_{k,N}\right)^2 = \frac{1}{K}\sum_{k=1}^{K} \frac{N}{N-1}\left[\frac{1}{N}\sum_{i=1}^{N} \varepsilon^2_{ki} - \bar{\varepsilon}^2_{k,N}\right]$$

The SLLN gives $\frac{1}{N}\sum_i \varepsilon^2_{ki} \overset{\text{a.s.}}{\underset{N}{\to}} \sigma^2$ and $\bar{\varepsilon}_{k,N} \overset{\text{a.s.}}{\underset{N}{\to}} 0$, for each cluster k, and so the term in brackets converges a.s. to σ^2 under the CMT. Since $N/(N-1) \to 1$ and the outer average is taken over finitely many k, $\hat{w}^2_{K,N} \overset{\text{a.s.}}{\underset{N}{\to}} \sigma^2$.

(ii) Fix $N \geq 2$. Using the same decomposition and re-writing,

$$\hat{b}^2_{K,N} = \frac{1}{K-1}\sum_{k=1}^{K} \left(M_{k,N} - \bar{M}_{K,N}\right)^2 = \frac{K}{K-1}\left[\frac{1}{K}\sum_{k=1}^{K} M^2_{k,N} - \bar{M}^2_{K,N}\right]$$

The SLLN applied to the *iid* sequences $(M^2_{k,N})_k$ and $(M_{k,N})_k$ gives $\frac{1}{K}\sum_k M^2_{k,N} \overset{\text{a.s.}}{\underset{K}{\to}} \tau^2 + \sigma^2/N$ and $\bar{M}_{K,N} \overset{\text{a.s.}}{\underset{K}{\to}} 0$, respectively. Since $K/(K-1) \to 1$, $\hat{b}^2_{K,N} \overset{\text{a.s.}}{\underset{K}{\to}} \tau^2 + \sigma^2/N$ under the CMT.

(iii) Combining parts (i) and (ii) under the CMT, the unclipped cluster-variance estimator $\tilde{\tau}^2_{K,N}$ satisfies

$$\tilde{\tau}^2_{K,N} = \hat{b}^2_{K,N} - \frac{1}{N}\hat{w}^2_{K,N} \overset{\text{a.s.}}{\underset{K}{\to}} \left(\tau^2 + \frac{\sigma^2}{N}\right) - \frac{\sigma^2}{N} = \tau^2$$

The projection $\Pi_{[a,b]}$ onto a closed interval is 1-Lipschitz, hence continuous, so $\hat{\tau}^2_{K,N} = \Pi_{[\tau^2_m,\tau^2_M]}(\tilde{\tau}^2_{K,N}) \overset{\text{a.s.}}{\underset{K}{\to}} \Pi_{[\tau^2_m,\tau^2_M]}(\tau^2) = \tau^2$, where the final equality uses that $\tau^2 \in [\tau^2_m, \tau^2_M]$ by Assumption 3.2. Finally, under the CMT, $\hat{w}^2_{K,N}/\hat{\tau}^2_{K,N} \overset{\text{a.s.}}{\underset{K}{\to}} \sigma^2/\tau^2 = \lambda_0$, provided the denominator is bounded away from zero because $\tau^2_m > 0$. Invoking the projection argument again and using $\lambda_0 \in [\lambda_m, \lambda_M]$,

$$\hat{\lambda}_{K,N} = \Pi_{[\lambda_m,\lambda_M]}\left(\frac{\hat{w}^2_{K,N}}{\hat{\tau}^2_{K,N}}\right) \overset{\text{a.s.}}{\underset{K}{\to}} \Pi_{[\lambda_m,\lambda_M]}(\lambda_0) = \lambda_0 \qquad \square$$

Lemma B.1. *Random-Index Composition. Let $(X_M)_{M\geq 1}$ be a sequence of random variables with $X_M \overset{a.s.}{\underset{M}{\to}} X$, and let $(M_B)_B$ be a sequence of $\mathbb{N}$-valued random variables where $\mathbb{P}(M_B \geq m) \to 1$ for every fixed $m \in \mathbb{N}$. Then $X_{M_B} \overset{p}{\underset{B}{\to}} X$.*

Proof of Lemma B.1. Fix $\epsilon, \delta > 0$ and set $T_M = \sup_{j\geq M}|X_j - X|$, which is measurable and non-increasing in M. Almost sure convergence of X_M implies $T_M \downarrow 0$ a.s., and so there exists a finite M_0 such that $\mathbb{P}(T_{M_0} > \epsilon) < \delta/2$. Since $M_B \overset{p}{\underset{B}{\to}} \infty$, there exists a finite (budget) B_0 such that $\mathbb{P}(M_B < M_0) < \delta/2$ for all $B \geq B_0$. Observe that on the event $\{M_B \geq M_0\}$, $|X_{M_B} - X| \leq T_{M_0}$, hence for every $B \geq B_0$,

$$\mathbb{P}\left(|X_{M_B} - X| > \epsilon\right) \leq \mathbb{P}(M_B < M_0) + \mathbb{P}(T_{M_0} > \epsilon) < \delta \qquad \square$$

Remark B.2. *Lemma B.1 is invoked repeatedly to jump between a.s. convergence over a fixed divergent sequence and convergence in probability over a random divergent sequence, both in performance and inference results. Furthermore, observe that the lemma operates without any restriction on the dependence between the random index $(M_B)_B$ and the sequence $(X_M)_M$.*

Proof of Lemma 4.4. By Lemma 4.1.iii, $\hat{\lambda}_{K,N_p} \xrightarrow[K]{\text{a.s.}} \lambda_0$ along any non-random diverging sequence of widths with $N_p \geq 2$ fixed. Since $K_{pB} \xrightarrow[p]{} \infty$, Lemma B.1 applies with $X_K = \hat{\lambda}_{K,N_p}$ and $M_B = K_{pB}$, therefore $\hat{\lambda}_{pB} \xrightarrow[K]{p} \lambda_0$. Define the map

$$h(\lambda) = (F + \sqrt{FV\lambda})^{-1}$$

which I will invoke repeatedly in the derivations below. The map $h(\cdot)$ is strictly decreasing, continuously differentiable and L_h-Lipschitz on $[\lambda_m, \lambda_M]$, provided $\lambda_m > 0$. L_h is defined implicitly via $L_h = \sup_{\lambda \in [\lambda_m, \lambda_M]} |h'(\lambda)| < \infty$. Applying the CMT for any $\lambda \in [\lambda_m, \lambda_M]$ returns $\nu_B^\pi = h(\hat{\lambda}_{pB}) \xrightarrow[p]{} h(\lambda_0) = \nu_0$ and $N_B^\pi = N^*(\hat{\lambda}_{pB}) \xrightarrow[p]{} N_0$.

For the integer statement in the footnote, the flooring operator $\lfloor \cdot \rfloor$ is continuous at every non-integer point, so when $N_0 \notin \mathbb{N}$, $\lfloor N_B^\pi \rfloor \xrightarrow[p]{} N_{\underline{0}}$ by the CMT. Trivially, convergence in probability of an integer-valued sequence to an integer constant is equivalent to $\mathbb{P}(\lfloor N_B^\pi \rfloor = N_{\underline{0}}) \to 1$. □

C Proofs of Section 5

Proof of Equation (7). Let Assumptions 3.1, 3.2, 3.3 and 3.4 hold throughout. Starting from the results in Appendix A,

$$W(K(N), N) = \frac{\tau^2}{B}\left[F + V\lambda_0 + VN + \frac{F\lambda_0}{N}\right]$$

Recall $N^*(\lambda) = \sqrt{\lambda F/V}$, hence $VN^*(\lambda) = \sqrt{FV\lambda}$ and $F\lambda_0/N^*(\lambda) = \sqrt{FV}\ \lambda_0/\sqrt{\lambda}$. Therefore,

$$W(K^*(\lambda), N^*(\lambda); (\tau^2, \lambda_0)) = \frac{\tau^2}{B}\left[F + V\lambda_0 + \sqrt{FV}\left(\sqrt{\lambda} + \frac{\lambda_0}{\sqrt{\lambda}}\right)\right]$$

Extracting $\sqrt{FV}$ as a common factor and substracting W_0 returns,

$$\Delta = \frac{\tau^2\sqrt{FV}}{B}\left(\sqrt{\lambda} - 2\sqrt{\lambda_0} + \frac{\lambda_0}{\sqrt{\lambda}}\right) = \frac{\tau^2\sqrt{FV}}{B} \cdot \frac{\lambda - 2\sqrt{\lambda\lambda_0} + \lambda_0}{\sqrt{\lambda}} = \frac{\tau^2\sqrt{FV}}{B} \cdot \frac{\left(\sqrt{\lambda} - \sqrt{\lambda_0}\right)^2}{\sqrt{\lambda}}$$

□

C.1 Static Policies

Proof of Lemma 5.2. For part *(i)*, observe that

$$\frac{\left(\sqrt{\lambda_b} - \sqrt{\lambda_0}\right)^2}{\sqrt{\lambda_b}} = \frac{(\lambda_b - \lambda_0)^2}{\left(\sqrt{\lambda_b} + \sqrt{\lambda_0}\right)^2\sqrt{\lambda_b}} \in \left[\frac{(\lambda_b - \lambda_0)^2}{4\lambda_M^{3/2}}, \frac{(\lambda_b - \lambda_0)^2}{4\lambda_m^{3/2}}\right]$$

where I used $\lambda_b, \lambda_0 \in [\lambda_m, \lambda_M]$. Bring this result into (7) and substitute $(\lambda_b - \lambda_0)^2 = b^2$. The resulting bounds are of order $\Theta(1/B)$ for $b \neq 0$ as desired. For part *(ii)*, note that $\frac{\tau^2\sqrt{FV}}{B} \cdot \frac{\left(\sqrt{\lambda} - \sqrt{\lambda_0}\right)^2}{\sqrt{\lambda}}$

is monotonically increasing in $\tau^2 > 0$, and so the maximum is achieved at $\tau^2 = \tau_M^2$. Additionally, for some fixed λ, the map $\sqrt{\lambda_0} \mapsto (\sqrt{\lambda} - \sqrt{\lambda_0})^2$ is the square of an affine function, hence convex, and attains its maximum at the end points, i.e. the excess variance maximizer λ is in $\{\lambda_m, \lambda_M\}$, whichever yields a higher loss. The minimax policy is then given by

$$\min_{\lambda} \max \left\{ \frac{\left(\sqrt{\lambda} - \sqrt{\lambda_M}\right)^2}{\sqrt{\lambda}}, \frac{\left(\sqrt{\lambda} - \sqrt{\lambda_m}\right)^2}{\sqrt{\lambda}} \right\}$$

Solving recovers $\lambda_{mM} = \left(\frac{\sqrt{\lambda_m}+\sqrt{\lambda_M}}{2}\right)^2$. Finally, evaluate the worst case excess variance at $\lambda = \lambda_{mM}$,[53]

$$\max_{\omega \in \Omega} \Delta_{\pi_{mM}} = \frac{\tau_M^2 \sqrt{FV}}{B} \cdot \frac{\left(\frac{\sqrt{\lambda_m}+\sqrt{\lambda_M}}{2} - \sqrt{\lambda_M}\right)^2}{\frac{\sqrt{\lambda_m}+\sqrt{\lambda_M}}{2}} = \frac{\tau_M^2 \sqrt{FV}}{2B} \cdot \frac{\left(\sqrt{\lambda_m} - \sqrt{\lambda_M}\right)^2}{\sqrt{\lambda_m} + \sqrt{\lambda_M}} \quad \square$$

C.2 Upper Bounds

This section is organized as follows. I first introduce the notion of centered squared errors, that will be invoked repeatedly in performance and inference proofs. I then discuss some of their concentration properties under subgaussianity. Finally, I leverage these results in the derivation of an intermediate lemma that upper bounds the expected square deviation $\mathbb{E}[(\hat{\lambda}_p - \lambda_0)^2]$ for any $\lambda_0 \in [\lambda_m, \lambda_M]$, as hinted in the main text. The proof of Theorem 5.4 will then follow as a simple corollary.

Definition C.1. *Centered Squared Errors. Let $\varepsilon_k = (\varepsilon_{k1}, \varepsilon_{k2}, \dots)$. Recall the definitions of $\bar{\varepsilon}_{k,N}$, $S^2_{k,N}$, $\hat{w}^2_{K,N}$, $M_{k,N}$ and $\bar{M}_{K,N}$ from Section 3 and Appendix B. Define*

$$\alpha_{k,N} = S^2_{k,N} - \sigma^2, \qquad \beta_{k,N} = M^2_{k,N} - \theta^2_N, \qquad \text{where } \mathbb{E}[\alpha_{k,N}] = \mathbb{E}[\beta_{k,N}] = 0$$

Remark C.2. *Between-Variance Decomposition. The proofs below will exploit consistently the fact that the within- and between-cluster variances are near-martingales. The claim is exact for $\hat{w}^2_{K,N}$ as it is a sum of iid zero-mean errors, i.e. $\hat{w}^2_{K,N} - \sigma^2 = K^{-1}\sum_{k=1}^K \alpha_{k,N}$. $\hat{b}^2_{K,N}$, however, cannot be written as the sum of iid zero-mean errors because individual terms are correlated through the de-meaning object $\bar{M}_{K,N}$ (the grand error mean). An approximate decomposition remains possible*

$$\begin{aligned}
\hat{b}^2_{K,N} - \theta^2_N &= \frac{1}{K-1}\sum_{k=1}^K \left(M_{k,N} - \bar{M}_{K,N}\right)^2 - \theta^2_N \\
&= \frac{1}{K-1}\left(\sum_{k=1}^K M^2_{k,N} - K\bar{M}^2_{K,N}\right) - \theta^2_N \\
&= \frac{1}{K-1}\left(K\theta^2_N + \sum_{k=1}^K (M^2_{k,N} - \theta^2_N) - K\bar{M}^2_{K,N}\right) - \theta^2_N \\
&= \frac{1}{K-1}\sum_{k=1}^K \beta_{k,N} + \frac{\theta^2_N}{K-1} - \frac{K}{K-1}\bar{M}^2_{K,N} \approx \frac{1}{K-1}\sum_{k=1}^K \beta_{k,N}
\end{aligned}$$

[53] I am inputting $\lambda_0 = \lambda_M$. This is without loss as at $\lambda = \lambda_m$, the excess variance is equalized across endpoints $\{\lambda_m, \lambda_M\}$ and so any of the two endpoints would deliver the same excess variance.

Lemma C.3. *Concentration of Centered Square Errors. Under Assumptions 3.1, 3.2, 3.4 and 5.3, $M_{k,N}, \alpha_{k,N}, \beta_{k,N}$ are subexponential with suitable parameters (κ, b).*

Proof of Lemma C.3. Consider some set of strictly positive constants $\{c_j\}_{j\leq 4}$. Define $\kappa_M = \sqrt{\kappa_u^2 + \kappa_\varepsilon^2/N}$. By independence of $u_k \perp\!\!\!\perp \bar{\varepsilon}_{k,N}$, $M_{k,N}$ is κ_M-subgaussian. Hence, it is subexponential with parameters $\kappa = \kappa_M^2, b_M = 0$. $\beta_{k,N}$ is the centered square of $M_{k,N}$—κ_M-subgaussian variable—thus it is subexponential with parameters $\kappa_B = c_1\kappa_M^4$, $b_B = c_2\kappa_M^2$. Next, consider the centering matrices

$$\boldsymbol{P}_N = I_N - \frac{1}{N}\mathbf{1}\mathbf{1}', \qquad \tilde{\boldsymbol{P}}_N = \frac{1}{N-1}\boldsymbol{P}_N$$

then

$$\alpha_{k,N} = \frac{1}{N-1}\left(\varepsilon_k'\boldsymbol{P}_N\varepsilon_k - (N-1)\sigma^2\right) = \varepsilon_k'\tilde{\boldsymbol{P}}_N\varepsilon_k - \sigma^2$$

such that $\mathbb{E}[\varepsilon_k'\boldsymbol{P}_N\varepsilon_k] = \sigma^2\mathrm{tr}(\boldsymbol{P}_N) = \sigma^2(N-1)$ and $\alpha_{k,N}$ is centered. By Hanson-Wright inequality (Rudelson and Vershynin, 2013), $\alpha_{k,N}$ is subexponential with parameters $\kappa_A = c_3\kappa_\varepsilon^4\|\tilde{\boldsymbol{P}}_N\|_F^2$ and $b_A = c_4\kappa_\varepsilon^2\|\tilde{\boldsymbol{P}}_N\|$, where $\|\cdot\|_F$ is the Frobenius norm and $\|\cdot\|$ is the operator norm. Because $\boldsymbol{P}_N$ is an orthogonal projection of rank $N-1$, $\|\boldsymbol{P}_N\|_F^2 = (N-1)$ and $\|\boldsymbol{P}_N\| = 1$. After scaling, $\alpha_{k,N}$ is subexponential with parameters $\kappa_A = c_3\frac{\kappa_\varepsilon^4}{N-1}$ and $b_A = c_4\frac{\kappa_\varepsilon^2}{N-1}$. □

Lemma C.4. *Maximal Bernstein Inequality. Let $\xi_k)_k$ be a sequence of iid, mean-zero, (κ, b)-subexponential random variables. Then, for every $K_+ \geq 1$ and $t > 0$,*

$$\mathbb{P}\left(\max_{1\leq K\leq K_+}\left|\sum_{k=1}^{K}\xi_k\right| \geq t\right) \leq 2\exp\left(-\frac{1}{2}\min\left(\frac{t^2}{\kappa\cdot K_+}, \frac{t}{b}\right)\right)$$

Proof of Lemma C.4. The proof is omitted. See, for instance, Chapter 6 in Lu (2025). □

Lemma C.5. *$\hat{\lambda}$ Concentration for K-asymptotic Pilot Policies. Let π be an implementable pilot policy with pilot size (K_{pB}^π, N_p^π) where (i) $N_p^\pi = N_p$ fixed, and $K_{pB}^\pi \in \mathcal{I}_B = [K_{B-}, K_{B+}] \cap \mathbb{N}$ a.s.; (ii) $K_{B+}/K_{B-} \leq C_0$ for some finite positive constant C_0 not depending on B; and (iii) $K_{B-} \underset{B}{\to} \infty$. Under Assumptions 3.1, 3.2, 3.3, 3.4 and 5.3, there exist two positive finite constants C_1 and B_0 not depending on ω such that for all $B \geq B_0$ and every $\varepsilon > 0$,*

$$\mathbb{P}\left(\mid \hat{\lambda}_{K_{pB}^\pi, N_p} - \lambda_0 \mid \geq \varepsilon\right) \leq 6\exp\left(-C_1K_{B-}\varepsilon^2\right)$$

Consequently, for all $B \geq B_0$

$$\mathbb{E}\left[\left(\hat{\lambda}_{K_{pB}^\pi, N_p} - \lambda_0\right)^2\right] \leq \frac{6}{C_1K_{B-}}$$

Remark C.6. *Uniform Bounds in λ_0. Observe that the bounds in Lemma C.5 do not depend on λ_0, and so they hold uniformly in the λ-space.*

Proof of Lemma C.5. Let $D_w := \sup_{K\in\mathcal{I}_B} \mid \hat{w}_{K,N_p}^2 - \sigma^2 \mid$. For some $u > 0$

$$\begin{aligned}\mathbb{P}(D_w \geq u) &\leq \mathbb{P}\left(\frac{1}{K_{B-}}\max_{K_{B-}\leq K\leq K_{B+}}\left|\sum_{k=1}^{K}\alpha_{k,N_p}\right| \geq u\right) \leq 2\exp\left(-\frac{1}{2}\min\left(\frac{u^2K_{B-}^2}{K_{B+}\kappa_A}, \frac{uK_{B-}}{b_A}\right)\right)\\ &\leq 2\exp\left(-\frac{K_{B-}}{2}\min\left(\frac{u^2}{C_0\kappa_A}, \frac{u}{b_A}\right)\right)\end{aligned}$$

where the second inequality uses that α_{k,N_p} is sub-exponential with constants (κ_A, b_A) by Lemma C.3 and applies Lemma C.4 over α_{k,N_p} and $t = uK_{B-}$. Similarly, define $D_b := \sup_{K\in\mathcal{J}_B} \mid \hat{b}^2_{K,N_p} - \theta^2_{N_p} \mid$, and recall the decomposition in Remark C.2

$$\hat{b}^2_{K,N_p} - \theta^2_{N_p} = \frac{K}{K-1}\cdot\frac{1}{K}\sum_{k=1}^{K}\beta_{k,N_p} + \frac{\theta^2_{N_p}}{K-1} - \frac{K}{K-1}\left(\frac{1}{K}\sum_{k=1}^{K}M_{k,N_p}\right)^2$$

Define $D_B := \sup_{K\in\mathcal{J}_B} \mid \frac{1}{K}\sum_{k\leq K}\beta_{k,N_p} \mid$, $D_M := \sup_{K\in\mathcal{J}_B} \mid \frac{1}{K}\sum_{k\leq K} M_{k,N_p} \mid$, and $r_B := \frac{\tau^2_M(1+\lambda_M)}{K_{B-}-1} < \infty$ (by Assumption 3.2). Then, $D_b \leq 2D_B + 2D^2_M + r_B$, where I used that $K/(K-1) \leq 2$ for $K \geq 2$. Next, for every $u \geq 3r_B$,

$$\begin{aligned}\mathbb{P}(D_b \geq u) &\leq \mathbb{P}\left(D_B \geq \frac{u}{6}\right) + \mathbb{P}\left(D_M \geq \sqrt{\frac{u}{6}}\right) + \mathbb{P}\left(r_B \geq \frac{u}{3}\right)\\ &\leq 2\exp\left(-\frac{K_{B-}}{2}\min\left(\frac{u^2}{36C_0\kappa_B}, \frac{u}{6b_B}\right)\right) + 2\exp\left(-\frac{K_{B-}}{2}\frac{u}{6C_0\kappa_M}\right)\end{aligned}$$

where the third term is zero by selection of u and the main inequality follows from Lemma C.4 over $(\beta_{k,N_p}, M_{k,N_p})$. Recall that M_{k,N_p} is not only subexponential but subgaussian and so its bound does not include the subexponential refinement. Now I translate the bounds over $\hat{w}^2_{K,N_p}$ and $\hat{b}^2_{K,N_p}$ into bounds over $\hat{\tau}^2_{K,N_p}$ and $\hat{\lambda}_{K,N_p}$, by a double application of the triangle inequality. Specifically, $\tilde{\tau}^2_{K,N_p} - \tau^2 = (\hat{b}^2_{K,N_p} - \theta^2_{N_p}) - \frac{1}{N_p}(\hat{w}^2_{K,N_p} - \sigma^2)$, thus $\sup_{K\in\mathcal{J}_B} \mid \tilde{\tau}^2_{K,N_p} - \tau^2 \mid \leq D_b + D_w/N_p$. The same bound is mechanically preserved for the contraction $\hat{\tau}^2_{K,N_p} = \Pi_{[\tau^2_m,\tau^2_M]}(\tilde{\tau}^2_{K,N_p})$. Similarly,

$$\sup_{K\in\mathcal{J}_B} \mid \tilde{\lambda}_{K,N_p} - \lambda_0 \mid = \sup_{K\in\mathcal{J}_B}\left|\frac{\tau^2(\hat{w}^2_{K,N_p} - \sigma^2) - \sigma^2(\hat{\tau}^2_{K,N_p} - \tau^2)}{\tau^2\ \hat{\tau}^2_{K,N_p}}\right| \leq \frac{1}{\tau^2_m}\left[\left(1 + \frac{\lambda_M}{N_p}\right)D_w + \lambda_M D_b\right] \leq L(D_w + D_b)$$

with $L = (1+\lambda_M)/\tau^2_m$. The first inequality follows from $\hat{\tau}^2 \geq \tau^2_m$ by clipping, and $\sigma^2/\tau^2 = \lambda_0 \leq \lambda_M$ by compactness. The second inequality follows by noting that $N_p \geq 1$. The bound obtained for $\tilde{\lambda}$ is preserved after clipping. Therefore,

$$\mathbb{P}\left(\mid \hat{\lambda}_{K^{\pi}_{pB},N_p} - \lambda_0 \mid \geq \varepsilon\right) \leq \mathbb{P}\left(\sup_{K\in\mathcal{J}_B} \mid \hat{\lambda}_{K,N_p} - \lambda_0 \mid \geq \varepsilon\right) \leq \mathbb{P}\left(D_w \geq \frac{\varepsilon}{2L}\right) + \mathbb{P}\left(D_b \geq \frac{\varepsilon}{2L}\right)$$

Observe that the probability on the LHS is zero for $\varepsilon > \Lambda := \lambda_M - \lambda_m$ because $\hat{\lambda}_{K^{\pi}_{pB},N_p}$ and λ_0 both lie in $[\lambda_m, \lambda_M]$. Hence, it suffices to establish the inequality for $\varepsilon \in (0, \Lambda]$. To do so, I evaluate the bounds above on $u = \varepsilon/2L \in (0, u_M) := (0, \Lambda/2L)$. On a bounded range, the Bernstein exponent can be uniformly lower-bounded by a quadratic function of u, i.e.

$$\min\{au^2, bu\} \geq u^2 \cdot \min\left\{a, \frac{b}{u_M}\right\}$$

Therefore, probability bounds above can be further upper bounded by expressions of the form $2\exp(-CK_{B-}\varepsilon^2)$, for some finite positive constants (C_w, C_B, C_M). The constants only depend on the sub-exponential parameters, the window width C_0, Λ and L, but not the particular ω realization. Setting $C_1 := \min\{C_w, C_B, C_M\}$ delivers the statement in the Lemma with a prefactor of 6 for all $\varepsilon \in (\varepsilon_B, \Lambda] = (6Lr_B, \Lambda]$. Small deviations of the form $\varepsilon \in (0, \varepsilon_B)$ are inconsequential for large B because $\varepsilon_B = O(1/K_{B-})$ and $K_{B-} \underset{B}{\to} \infty$, so B_0 can be selected such that $C_1 K_{B-}\varepsilon^2_B \leq \ln 6$ for all $B \geq B_0$. Then,

for every $\varepsilon < \varepsilon_B$,

$$6\exp\left(-C_1 K_{B-}\varepsilon^2\right) \geq 6\exp\left(-C_1 K_{B-}\varepsilon_B^2\right) \geq 6e^{-\ln 6} = 1$$

and the probability statement holds trivially. Integrating the tail delivers the expectation bound

$$\mathbb{E}\left[\left(\hat{\lambda}_{K^\pi_{pB},N_p} - \lambda_0\right)^2\right] = \int_0^\infty \mathbb{P}\left(\mid \hat{\lambda}_{K^\pi_{pB},N_p} - \lambda_0 \mid > \sqrt{t}\right) dt \leq \int_0^\infty 6\exp\left(-C_1 K_{B-} t\right) dt = \frac{6}{C_1 K_{B-}}$$

□

Remark C.7. *Role of Clipping. Clipping plays two distinct roles in Lemma C.5. First, it bounds the estimation error by Λ, so that only deviations $\varepsilon \leq \Lambda$ carry probability and all Bernstein regimes collapse into the quadratic one, i.e. the clipped estimator is sub-gaussian on the effective range. Second, it bounds the denominator $\hat{\tau}^2$ away from zero. Both roles require the parameter space to be bounded away from degeneracy, i.e. $\tau_m^2 > 0$, as per Assumption 3.2. By contrast, the unclipped ratio $\tilde{\lambda}$ need not admit sub-exponential tails as it features extreme deviations coming from small-denominator issues.*

Proof of Theorem 5.4. I first argue that both algorithms satisfy the assumptions entering Lemma C.5. Both algorithms set $N^\pi_{pB} = N_m$ fixed. Moreover, $K^{\texttt{EtC-CS}}_{pB} = K_M$ deterministically, hence the window is degenerate, i.e. $K_{B-} = K_{B+}$ and $C_0 = 1$. Similarly, $K^{\texttt{ACS}}_{pB} = \max\{s-1, K_M\} \in [K_M, K_m]$ a.s., therefore, the assumption is satisfied for $C_0 = K_m/K_M \leq 2h(\lambda_m)/h(\lambda_M)$ for B large enough. Finally, both `EtC-CS` and `ACS` trivially sample $K^\pi_{pB} \geq K_M \geq Bh(\lambda_M) - 1 \underset{B}{\to} \infty$. In addition, both policies are implementable, hence they both implement an optimal-map design with $\lambda = \hat{\lambda}_{K^\pi_{pB},N_m}$. Using the representation of the excess variance in (8) and the upper bound in the proof of Lemma 5.2

$$\mathbb{E}\left[\Delta_\pi\right] \leq \frac{\tau_M^2\sqrt{FV}}{4\lambda_m^{3/2}} \cdot \frac{1}{B} \cdot \mathbb{E}\left[\left(\hat{\lambda}_{K^\pi_{pB},N_p} - \lambda_0\right)^2\right] \leq \frac{\tau_M^2\sqrt{FV}}{4\lambda_m^{3/2}} \cdot \frac{6}{C_1\ B\ K_M}$$

where the second inequality follows from Lemma C.5. Recall $K_M = \Theta(B)$, so the right hand side is $O(B^{-2})$. Absorbing constants, $\mathbb{E}_\pi[\Delta_\pi] \leq C_\pi/B^2$ with C_π depending on $(\tau_M^2, \lambda_m, \lambda_M, F, V, N_m, C_0, C_1)$ but not B. Setting $C_\pi = C_{\texttt{EtC-CS}}$ for `EtC-CS` and $C_\pi = C_{\texttt{ACS}}$ for `ACS` completes the proof with constants differing only in the window width C_0. The upper bounds hold uniformly in Ω and so upper bound $\max_{\omega\in\Omega} \mathbb{E}[\Delta_\pi]$ □

C.3 Lower Bounds – Proof of Theorem 5.5

Policy Class. I first define the relevant comparison class Π. The policy class is fairly general and contains most reasonable policies as special cases. In particular, I consider policies which, in every period $t \in \{1, \ldots, T\}$, select a cluster $k_t = k$ and, budget permitting, sample at least one unit $N_t^{k_t} \geq 1$ from that cluster. As a result, the game contains at most $\lfloor B/V \rfloor$ rounds. If no additional observation can be sampled without violating the remaining budget, the algorithms must terminate. The same cluster k can be revisited several times. Let N^k be the terminal number of samples of cluster k and K the number of clusters with at least one sampled unit. Policies must satisfy the budget constraint a.s..

Data Generating Process. Fix an instance $(\tau_0^2, \lambda_0) \in \text{int}(\Omega)$ with $\sigma^2 = \lambda_0\tau_0^2$. Consider two DGPs (y_-, y_+) indexed by $\omega_- = (\tau_-^2, \lambda_-)$ and $\omega_+ = (\tau_+^2, \lambda_+)$, where $\lambda_\pm = \lambda_0 \pm \delta_B$ for some small positive $\delta_B \underset{B}{\to} 0$. $\tau_\pm^2 = \sigma^2/\lambda_\pm$, so that the individual-level variance σ^2 is common to both environments and only the cluster-level variance differs. For sufficiently large B, $(\tau_\pm^2, \lambda_\pm) \in \Omega$. The additive errors model

parallels (1),[54]

$$y_{\pm ki} = \mu + u_{\pm k} + \varepsilon_{ki} \qquad u_{\pm k} \sim N(0, \tau_{\pm}^2), \quad \varepsilon_{ki} \sim N(0, \sigma^2)$$

Separating Events. Standard lower bound derivations characterize a separating event as a function of counts. Following this principle, I characterize a separating event E a function of the terminal harmonic mean N^H. By linearity of the cost structure $KF + \sum_{k \le K} VN^k = K(F + V\bar{N})$, where $\bar{N} = K^{-1} \sum_{k \le K} N^k$ is the arithmetic mean. Recall that $N^H \le \bar{N}$, hence $\frac{1}{K} \ge \frac{F + VN^H}{B}$. It follows

$$W(\boldsymbol{N}) = \frac{\tau^2}{K(\boldsymbol{N})} \left(1 + \frac{\lambda}{N^H(\boldsymbol{N})}\right) \ge \frac{\tau^2 (F + VN^H(\boldsymbol{N}))}{B} \left(1 + \frac{\lambda}{N^H(\boldsymbol{N})}\right) = W^H(N^H)$$

Next, let $N_{\pm} = N^*(\lambda_{\pm}) = \sqrt{\lambda_{\pm} F/V}$, so $N_- < N_0 < N_+$. Define the separating event $E = \{N^H \le N_0\}$. In environment ω_+, $W^H(\cdot; \omega_+)$ is strictly convex with minimum at $N_+ > N_0$, hence decreasing on $(0, N_+]$. It follows that

$$W(\boldsymbol{N}; \omega_+) - W(N_+; \omega_+) \ge W^H(N^H; \omega_+) - W^H(N_+; \omega_+) \underset{\text{on } E}{\ge} W^H(N_0; \omega_+) - W^H(N_+; \omega_+)$$

where the first inequality uses the harmonic mean relaxation and the fact that $W(N_+; \omega_+) = W^H(N_+; \omega_+)$, as per the balancedness argument in Appendix A. On E^c, $W^H(\cdot; \omega_-)$ is increasing on $[N_-, \infty)$ and $N^H > N_0 > N_-$, therefore the same inequality runs for ω_-. The same computation as in Equation (7)—with truth $\lambda_{\pm}$ and input λ_0—yields

$$W^H(N_0; \omega_{\pm}) - W^H(N_{\pm}; \omega_{\pm}) = \frac{\tau_{\pm}^2 \sqrt{FV}}{B} \cdot \frac{\left(\sqrt{\lambda_0} - \sqrt{\lambda_{\pm}}\right)^2}{\sqrt{\lambda_0}}$$

Finally, both branches can be uniformly bounded in terms of δ_B and common parameters by noting: (i) $\left(\sqrt{\lambda_0} - \sqrt{\lambda_{\pm}}\right)^2 = \delta_B^2 / (\sqrt{\lambda_0} + \sqrt{\lambda_{\pm}})^2 \ge \delta_B^2 / (4\lambda_M)$, (ii) $\lambda_0^{-1/2} \ge \lambda_M^{-1/2}$, and (iii) $\sigma^2 / \lambda_M \le \min\{\tau_-^2, \tau_+^2\}$

$$W^H(N^H; \omega_{\pm}) - W^H(N_{\pm}; \omega_{\pm}) \ge \frac{\sigma^2 \sqrt{FV}}{4 \lambda_M^{5/2} B} \, \delta_B^2 \qquad \text{on } E \text{ (for } \omega_+) \text{ and } E^c \text{ (for } \omega_-)$$

The result is symmetric to the excess variance of static policies in Lemma 5.2 with $\tau_{\pm}^2 \ge \sigma^2 / \lambda_M$.

Augmented Experiment and Divergence Decomposition. For some policy $\pi \in \Pi$, let $(\tilde{\mathbb{P}}_{\pi-}, \tilde{\mathbb{P}}_{\pi+})$ be the probability measures induced by the interaction between the policy π and the environments y_- and y_+, respectively. Now, consider the augmented experiment where, every time the learner samples cluster k, she observes both $(y_{ki})_{i \le N^k}$ *and* u_k. Let $(\mathbb{P}_{\pi-}, \mathbb{P}_{\pi+})$ be the probability measures induced by the interaction between the policy π and the augmented environments. By data processing,

$$D_{\mathrm{KL}}(\tilde{\mathbb{P}}_{\pi-}, \tilde{\mathbb{P}}_{\pi+}) \le D_{\mathrm{KL}}(\mathbb{P}_{\pi-}, \mathbb{P}_{\pi+})$$

The augmented game simplifies the analysis of the KL-divergence significantly because outcomes remain *iid* conditional on the history and revisiting clusters does not further increase the divergence.[55] In addition, unit-level errors share the same distribution across environments, so the KL-divergence

[54] Gaussian errors are exactly subgaussian, so the lower bound is established over a strict subclass of the environment considered in Theorem 5.5. The lower bound then carries over to the general environment class given in Assumption 3.1.

[55] For a complete derivation of the chain rule under history-conditional *iid* outcomes see Lemma 15.1 in Lattimore and Szepesvári (2020).

analysis is limited to the cluster-level, i.e.

$$D_{\mathrm{KL}}(\mathbb{P}_{\pi-}, \mathbb{P}_{\pi+}) = \mathbb{E}_{\pi-}[K]\, D_{\mathrm{KL}}(N(0,\tau_-^2), N(0,\tau_+^2))$$

Solving for D_{KL} explicitly,

$$\begin{aligned} D_{\mathrm{KL}}(N(0,\tau_-^2), N(0,\tau_+^2)) &= \frac{1}{2}\left[\frac{\tau_-^2}{\tau_+^2} - 1 - \ln\left(\frac{\tau_-^2}{\tau_+^2}\right)\right] = \frac{1}{2}\left[\frac{\lambda_+}{\lambda_-} - 1 - \ln\left(\frac{\lambda_+}{\lambda_-}\right)\right] \\ &= \frac{1}{2}\phi\left(\frac{\lambda_+}{\lambda_-}\right) = \frac{1}{2}\phi\left(1 + \frac{2\delta_B}{\lambda_0 - \delta_B}\right) \end{aligned}$$

where the third equality serves as definition of $\phi(x) = x - 1 - \ln(x)$. $\phi(x)$ is monotonically increasing in $x \geq 1$ because $\phi'(x) = 1 - 1/x$. Using the facts that for all sufficiently large B $\delta_B \leq \lambda_0/2$, and that $\phi(1+x) \leq x^2/2$

$$\frac{1}{2}\phi\left(\frac{\lambda_+}{\lambda_-}\right) \leq \frac{1}{2}\phi\left(1 + \frac{4\delta_B}{\lambda_0}\right) \leq \frac{4\delta_B^2}{\lambda_0^2}$$

Finally, observe that for all policies $\pi \in \Pi$, $FK + \sum_k VN^k \leq B$ a.s., thus $K \leq B/F$ a.s.. Bringing all pieces together

$$D_{\mathrm{KL}}(\tilde{\mathbb{P}}_{\pi-}, \tilde{\mathbb{P}}_{\pi+}) \leq D_{\mathrm{KL}}(\mathbb{P}_{\pi-}, \mathbb{P}_{\pi+}) \leq \frac{B}{F}\frac{4\delta_B^2}{\lambda_0^2} = C_{\mathrm{KL}} B \delta_B^2$$

Collecting Terms and BH Inequality. The argument is now completed through a standard Bretagnolle-Huber argument[56]

$$\begin{aligned} &\tilde{\mathbb{E}}_{\pi-}[W(\boldsymbol{N};\omega_-)] - W(K^*(\lambda_-), N^*(\lambda_-);\omega_-) + \tilde{\mathbb{E}}_{\pi+}[W(\boldsymbol{N};\omega_+)] - W(K^*(\lambda_+), N^*(\lambda_+);\omega_+) \geq \\ &\tilde{\mathbb{E}}_{\pi-}\left[W^H(N^H;\omega_-)\right] - W^H(N_-;\omega_-) + \tilde{\mathbb{E}}_{\pi+}\left[W^H(N^H;\omega_+)\right] - W^H(N_+;\omega_+) \geq \\ &\tilde{\mathbb{P}}_{\pi-}(E^c)\cdot\left(\tilde{\mathbb{E}}_{\pi-}\left[W^H(N^H;\omega_-) \mid E^c\right] - W^H(N_-;\omega_-)\right) + \tilde{\mathbb{P}}_{\pi+}(E)\cdot\left(\tilde{\mathbb{E}}_{\pi+}\left[W^H(N^H;\omega_+) \mid E\right] - W^H(N_+;\omega_+)\right) \geq \\ &\frac{\sigma^2\sqrt{FV}}{4B\lambda_M^{5/2}}\delta_B^2\left(\tilde{\mathbb{P}}_{\pi-}(E^c) + \tilde{\mathbb{P}}_{\pi+}(E)\right) \geq \frac{\sigma^2\sqrt{FV}}{4B\lambda_M^{5/2}}\delta_B^2\frac{1}{2}\exp\left(-D_{\mathrm{KL}}(\tilde{\mathbb{P}}_{\pi-}, \tilde{\mathbb{P}}_{\pi+})\right) \geq \\ &\frac{\sigma^2\sqrt{FV}}{4B\lambda_M^{5/2}}\delta_B^2\frac{1}{2}\exp\left(-D_{\mathrm{KL}}(\mathbb{P}_{\pi-}, \mathbb{P}_{\pi+})\right) \geq \frac{\sigma^2\sqrt{FV}}{8B\lambda_M^{5/2}}\delta_B^2\exp\left(-C_{\mathrm{KL}}\delta_B^2 B\right) \end{aligned}$$

Where the first inequality follows from the variance relaxation under the harmonic mean, the second inequality is a simple event partition, the third inequality follows from the uniform lower bound on excess variance, the fourth inequality follows from Bretagnolle-Huber inequality, the fifth inequality follows from a data processing argument, and the final inequality follows from the derivation above. Finally, by selecting $\delta_B = \sqrt{1/(C_{\mathrm{KL}}B)} \underset{B}{\to} 0$,

$$\frac{\sigma^2\sqrt{FV}}{8B\lambda_M^{5/2}}\frac{1}{C_{\mathrm{KL}}B}\exp(-1) \geq C_2/B^2$$

for some finite and strictly positive constant C_2 not depending on (π, B). The statement of the theorem is completed by applying the inequality $\max\{a,b\} \geq (a+b)/2$ and absorbing it into a constant C. □

[56] See Theorem 14.2 in Lattimore and Szepesvári (2020).

D Design-Robust Inference for Asymptotically Even Policies

This section is a self-contained technical note on inference for adaptive designs under symmetry. In a nutshell, the thesis of this document is that, under a symmetry assumption on the errors, inference on sample averages remains invariant after conditioning on designs induced by even or asymptotically even policies. This principle extends well beyond the adaptive algorithms and the clustered sampling framework presented in the main text. The notation follows the literature on multi-armed adaptive designs: N is the horizon (indexed by i) and K the number of arms (indexed by k). It should not be conflated with the K versus N notation of the main text.

	Main text (Section 6)	**This Appendix**
Asymptotic index	Budget B	Horizon N
Groups	One population	K arms
Design Choices	(K, N)	$(N^k)_{k\leq K}$
Summands	Clusters $k \leq K_B$	Pulls $s \leq N_N^k$

D.1 Setup and Definitions

Definition D.1. *Sampling Protocol. In each period $i = 1, 2, \ldots, N$: (i) the learner selects an arm $k_i \in [K] = \{1, \ldots, K\}$ according to a history-adapted policy π; (ii) the pull counts update, $N^k(i) = N^k(i-1) + \mathbb{1}(k_i = k)$ with $N^k(0) = 0$; and (iii) the learner observes*

$$X_i = \mu_{k_i} + \varepsilon^{k_i}_{N^{k_i}(i)}$$

Write $N_N^k := N^k(N)$ for the terminal count, also referred to as terminal design. It is convenient to index arm-level statistics by pulls rather than periods, so that $X_s^k = \mu_k + \varepsilon_s^k$ denotes the s-th observation from arm k. Moreover, for $M \geq 1$ set

$$\hat{\mu}_M^k = \frac{1}{M}\sum_{s=1}^{M} X_s^k, \qquad \bar{\varepsilon}_M^k = \frac{1}{M}\sum_{s=1}^{M} \varepsilon_s^k, \qquad Z_M^k = \sqrt{M}\big(\hat{\mu}_M^k - \mu_k\big) = \frac{1}{\sqrt{M}}\sum_{s=1}^{M} \varepsilon_s^k$$

with $Z_0^k = 0$. The terminal average of arm k is $\hat{\mu}^k_{N_N^k}$, defined on $\{N_N^k > 0\}$.

Definition D.2. *Random Sequences. For each arm $k \in [K]$ define an infinite iid error sequence along the pull count dimension $(\varepsilon_s^k)_{s\geq 1}$. Let $(\zeta_s^k)_{k\geq 1, s\geq 1}$ be an iid Rademacher array, independent from the errors, used to break ties in the sign decomposition of Definition D.9, and let $(U_i)_{i\geq 1}$ be an iid* Unif$[0,1]$ *sequence, also independent from the errors, used as an independent randomization device.*[57]

Assumption D.3. *Errors.*

i) ***Moments.*** *$\mathbb{E}[\varepsilon_s^k] = 0$ and $\mathbb{E}[(\varepsilon_s^k)^2] = \sigma_k^2 \in (0, \infty)$ for all k, s.*

ii) ***Independence.*** *The array $(\varepsilon_s^k)_{k,s}$ has mutually independent entries, and the three collections $(\varepsilon_s^k)_{k,s}$, $(\zeta_s^k)_{k,s}$, $(U_i)_i$ are mutually independent.*

iii) ***Symmetry.*** *$\varepsilon_s^k \overset{d}{=} -\varepsilon_s^k$ for all k, s.*

Remark D.4. *Independence across Both Indices. Assumption D.3.ii) emphasizes independence be-*

[57] $(U_i)_i$ is absent in every argument below. It is carried so that policies employing external randomization, like a randomized tie-breaking or random arm selection, remain inside the even class of Definition D.17, and so that an even surrogate could share the randomization of the asymptotically even policy it couples to.

tween arms and pulls. Every draw therefore carries its own sign, which makes possible the coordinate-wise definition of symmetry in Assumption D.3.iii) and the array factorization of Lemma D.11. Do not conflate the double indexing in the current appendix (arms and counts) with the double indexing in the main text (clusters and units). In particular, the sequence of cluster error averages $M_{k,N}$ are certainly not independent along the N dimension.

Definition D.5. *Even σ-algebra. The even σ-algebra is defined via* $\mathcal{E}_\infty := \sigma\big(((\varepsilon_s^k)^2)_{k,s},\ (U_i)_i\big)$.

D.2 The Gaussian Benchmark

Before turning to symmetry, I recall what exact Gaussianity buys compared to symmetry. Under Gaussian errors, the sample average of each arm is independent of its residuals, so any sampling policy driven by residuals—sample variances included—can be conditioned on at no cost. The inference-invariant policy class can then be extended from measurable wrt the even σ-algebra (squared errors) to measurable wrt the residual σ-algebra.

Definition D.6. *Residual σ-algebra and Nested Designs. Fix some deterministic pilot lengths $(m^k)_{k\in[K]}$ with $m^k \geq 2$. The residual σ-algebra is defined via*

$$\mathcal{R} := \sigma\Big(\big(\varepsilon_s^k - \bar{\varepsilon}_{m^k}^k\big)_{s\leq m^k,\ k\in[K]},\ (U_i)_i\Big)$$

Moreover, a pilot design $(m_N^k)_{k\in[K]}$ is nested at $(N^k)_k$ if $m_N^k \leq N_N^k$ almost surely for every k.

Remark D.7. *By considering deterministic pilot lengths $(m^k)_{k\in[K]}$, the exposition is kept tidy. Random commitment stages are accommodated by considering the stopped residual filtration, as it is the case of `ACS` in the cluster specialization of Appendix E.1.*

Proposition D.8. *Exact Conditional Normality under Gaussian Errors. Let $\varepsilon_s^k \sim N(0,\sigma_k^2)$, independent across (k,s), and let π induce a design $(N_N^k)_{k\in[K]}$ that is $\mathcal{R}$-measurable and nested at $(m^k)_k$. Then, for every horizon N and every event $A \in \mathcal{R}$ with $\mathbb{P}(A) > 0$,*

$$\mathcal{L}\left(\left(Z_{N_N^k}^k/\sigma_k\right)_{k\in[K]} \,\middle|\, A\right) \;=\; N(\mathbf{0}, I) \qquad \textit{exactly}$$

Proof of Proposition D.8. *Step 1. Orthogonality.* Fix an arm k and any $M^k \geq m^k$. For every $s \leq m^k$,

$$\mathrm{Cov}\big(\bar{\varepsilon}_{M^k}^k,\ \varepsilon_s^k - \bar{\varepsilon}_{m^k}^k\big) \;=\; \frac{\sigma_k^2}{M^k} - \frac{1}{m^k}\sum_{j\leq m^k}\mathrm{Cov}\big(\bar{\varepsilon}_{M^k}^k, \varepsilon_j^k\big) \;=\; \frac{\sigma_k^2}{M^k} - \frac{1}{m^k}\cdot m^k \cdot \frac{\sigma_k^2}{M^k} \;=\; 0$$

where I used $m^k \leq M^k$. All the variables involved are jointly Gaussian, so zero covariance implies independence, $\bar{\varepsilon}_{M^k}^k \perp\!\!\!\perp \big(\varepsilon_s^k - \bar{\varepsilon}_{m^k}^k\big)_{s\leq m^k}$. Independence across arms holds by assumption, and $(U_i)_i$ is independent of the errors. Since $Z_{M^k}^k = \sqrt{M^k}\,\bar{\varepsilon}_{M^k}^k$, it follows that for every deterministic vector $(M^k)_k$ with $M^k \geq m^k$,

$$\big(Z_{M^k}^k\big)_{k\in[K]} \perp\!\!\!\perp \mathcal{R}, \qquad \big(Z_{M^k}^k/\sigma_k\big)_{k\in[K]} \sim N(\mathbf{0}, I)$$

Step 2. Cell-wise factorization. The terminal counts take values in the finite grid $\mathcal{M} = \{(M^k)_k \in \mathbb{N}^K : \sum_k M^k = N,\ M^k \geq m^k\}$, and $\{(N_N^k)_k = (M^k)_k\} \in \mathcal{R}$ by $\mathcal{R}$-measurability of the

design. For any design event $A \in \mathcal{R}$, define $A_{(M^k)_k} = A \cap \{(N_N^k)_k = (M^k)_k\} \in \mathcal{R}$, then, for any fixed $(t_k)_k \in \mathbb{R}^K$,

$$\mathbb{E}\left[\exp\left(i\sum_k t_k \frac{Z^k_{N^k_N}}{\sigma_k}\right)\mathbb{1}_A\right] = \sum_{(M^k)_k \in \mathcal{M}} \mathbb{E}\left[\exp\left(i\sum_k t_k \frac{Z^k_{M^k}}{\sigma_k}\right)\mathbb{1}_{A_{(M^k)_k}}\right]$$
$$= \sum_{(M^k)_k \in \mathcal{M}} \exp\big(-\|t\|^2/2\big)\,\mathbb{P}\big(A_{(M^k)_k}\big) \;=\; \exp\big(-\|t\|^2/2\big)\,\mathbb{P}(A)$$

where the second equality applies Step 1 cell by cell, and the nesting condition guarantees $M^k \geq m^k$ throughout. The proof concludes by dividing by $\mathbb{P}(A) > 0$ and applying Lévy's continuity theorem. □

Four Observations on Gaussianity vs Symmetry. First, strictly speaking $\mathcal{R}$ and $\mathcal{E}_\infty$ are incomparable, but in practice the set of $\mathcal{R}$-measurable policies is much bigger than the set of $\mathcal{E}_\infty$-measurable policies.[58] In that way, the two frameworks trade distributional assumptions against policy generality. Under Gaussianity, (i) the design may depend arbitrarily on the residuals—including through odd functionals of the data—and (ii) normality is preserved exactly at every horizon (Proposition D.8). Under symmetry, a substantial weakening of the error distribution class, the conditioning σ-algebra *de facto* contracts from the residuals to their magnitudes; thus, the invariant policy class contracts accordingly from any mapping of the residuals to (asymptotically) even policies.

Second, neither framework covers policies that condition on the *level* of the data. The sample average is precisely the direction demeaning removes, so a design that reads a running mean like futility rules, threshold rules, or many bandit algorithms induce designs outside both $\mathcal{R}$ and $\mathcal{E}_\infty$. Conditional normality fails even under Gaussian errors and the adjustment tradition reviewed in the main text becomes necessary. Third, nesting is the general-theory analogue of *implementability*: the observations used to choose the design must be a subset of the observations entering the terminal average. Finally, exactness is confined to standardization under the true variance σ_k. Any variance estimate will be a function of residuals themselves and so the distribution of the standardized object will be affected after conditioning. Exact studentized statements require the specific two-stage weighting of Stein (1945), and alternative studentized results remain purely asymptotic.

D.3 Signs and Magnitudes

For the remainder of this appendix, I move away from Gaussianity and focus on symmetry.

Definition D.9. *Signs and Magnitudes. For each (k,s), write $\varepsilon_s^k = \chi_s^k R_s^k$ with*

$$\chi_s^k = \mathbb{1}(\varepsilon_s^k \neq 0)\,\mathrm{sign}(\varepsilon_s^k) + \mathbb{1}(\varepsilon_s^k = 0)\,\zeta_s^k, \qquad R_s^k = |\varepsilon_s^k|$$

Lemma D.10. *Signs are Independent of Magnitudes. Under Assumption D.3, for each (k,s), $\chi_s^k \sim$ Rad and $\chi_s^k \perp\!\!\!\perp R_s^k$.*

Proof of Lemma D.10. Fix (k,s) and drop indices. For any Borel $A \subseteq [0,\infty)$,

$$\mathbb{P}(\chi = 1, R \in C) = \mathbb{P}\big(\varepsilon > 0, |\varepsilon| \in C\big) + \tfrac{1}{2}\,\mathbb{P}(\varepsilon = 0)\,\mathbb{1}(0 \in C)$$

[58] Residuals themselves change sign under $\varepsilon \mapsto -\varepsilon$, hence $\mathcal{R} \not\subseteq \mathcal{E}_\infty$. However, the σ-algebras remain incomparable because while $\mathcal{E}_\infty$ records $(\varepsilon_s^k)^2$ at every s, $\mathcal{R}$ only sees the first m^k draws of each arm, so neither contains the other.

where the second term uses $\zeta \sim \text{Rad}$, independent of ε. By symmetry, $\mathbb{P}(\varepsilon > 0, |\varepsilon| \in C) = \mathbb{P}(-\varepsilon > 0, |-\varepsilon| \in C) = \mathbb{P}(\varepsilon < 0, |\varepsilon| \in C)$. Since $\{\varepsilon < 0, |\varepsilon| \in C\}$ and $\{\varepsilon > 0, |\varepsilon| \in C\}$ partition $\{|\varepsilon| \in C \setminus \{0\}\}$, each has probability $\frac{1}{2}\mathbb{P}(|\varepsilon| \in C\setminus\{0\})$. Adding the tie-break term gives $\mathbb{P}(\chi = 1, R \in C) = \frac{1}{2}\mathbb{P}(R \in C)$. The same argument holds for $\chi = -1$. The joint law factorizes, delivering both claims. □

Lemma D.11. *Signs are Independent of the Even σ-algebra. Under Assumption D.3, $(\chi_s^k)_{k\geq 1, s\geq 1} \perp\!\!\!\perp \mathcal{E}_\infty$.*

Proof of Lemma D.11. The pairs (χ_s^k, R_s^k) are functions of $(\varepsilon_s^k, \zeta_s^k)$, which by Assumption D.3.ii) are independent of $(U_i)_i$ and independent across both k and s (Remark D.4). Each pair factorizes by Lemma D.10, so the joint law of the array also factorizes with $(\chi_s^k)_{k,s} \perp\!\!\!\perp (R_s^k)_{k,s}$. Since

$$\mathcal{E}_\infty = \sigma\big(((\varepsilon_s^k)^2)_{k,s}, (U_i)_i\big) = \sigma\big(((R_s^k)^2)_{k,s}, (U_i)_i\big) = \sigma\big((R_s^k)_{k,s}, (U_i)_i\big)$$

and $(\zeta_s^k)_{k,s} \perp\!\!\!\perp \mathcal{E}_\infty$, it follows that $(\chi_s^k)_{k,s} \perp\!\!\!\perp \mathcal{E}_\infty$. □

D.4 Conditional CLT for a Fixed Design

Throughout this subsection I fix a single arm k and drop it from subscripts and superscripts. I further replace the possibly random count N_N^k by a deterministic M. Assumption D.3 is imposed throughout.

Lemma D.12. *Frozen Conditions. Let $(R_s)_s$ be iid with $\mathbb{E}[R_s^2] = \sigma^2 < \infty$. Then, there is an almost sure event $\Omega_0 \in \mathcal{E}_\infty$ on which, for every $\delta > 0$,*

$$\textit{(i)} \quad \frac{1}{M}\sum_{s=1}^{M} R_s^2 \overset{a.s.}{\underset{M}{\to}} \sigma^2, \qquad \textit{(ii)} \quad \frac{1}{M}\sum_{s=1}^{M} R_s^2 \, \mathbb{1}\{R_s > \delta\sqrt{M}\} \overset{a.s.}{\underset{M}{\to}} 0$$

Remark D.13. *Why "Frozen"? The squared errors become frozen once the even σ-algebra is conditioned upon, i.e. $\mathbb{E}[\varepsilon_s^2 \mid \mathcal{E}_\infty] = R_s^2$. Lemma D.12 verifies that the two inputs entering a Lindeberg–Feller Theorem hold pathwise on the frozen array.*

Proof of Lemma D.12. Part (i) follows directly from the SLLN, applicable since $\mathbb{E}[R_s^2] < \infty$. For the Lindeberg condition in part (ii), fix $C > 0$. Then, for every $M \geq C^2/\delta^2 \implies \delta\sqrt{M} \geq C$, hence $\mathbb{1}\{R_s > \delta\sqrt{M}\} \leq \mathbb{1}\{R_s > C\}$ and

$$\limsup_{M\to\infty} \frac{1}{M}\sum_{s=1}^{M} R_s^2 \mathbb{1}\{R_s > \delta\sqrt{M}\} \;\leq\; \limsup_{M\to\infty} \frac{1}{M}\sum_{s=1}^{M} R_s^2 \mathbb{1}\{R_s > C\} \;=\; \mathbb{E}\big[R_1^2 \mathbb{1}\{R_1 > C\}\big]$$

where the equality follows from the SLLN applied to the integrable *iid* summands $R_s^2\mathbb{1}\{R_s > C\}$. Dominated convergence sends the bound to zero as $C \to \infty$. Finally, define the almost sure event Ω_0 by intersecting the almost sure event of part (i) with an intersection over rational $\delta > 0$ and integer C—which can be extended to all $\delta > 0$ by monotonicity. □

Theorem D.14. *L^1 Convergence of the Conditional Characteristic Function. Let $\phi_M(t) := \mathbb{E}\big[\exp(itZ_M) \mid \mathcal{E}_\infty\big]$ denote the* conditional *characteristic function and $\phi(t) = \exp(-\sigma^2 t^2/2)$ the characteristic function of $N(0, \sigma^2)$. Then*

$$\mathbb{E}\big|\phi_M(t) - \phi(t)\big| \to 0 \qquad \forall t \in \mathbb{R}$$

Proof of Theorem D.14. *Step 1. Pathwise Lindeberg–Feller.* Fix ω in the almost sure

event Ω_0 and $t \in \mathbb{R}$. Write $Y_{s,M} = \varepsilon_s/\sqrt{M} = \chi_s R_s/\sqrt{M}$, so $Z_M = \sum_{s\le M} Y_{s,M}$. By Lemma D.11, when conditioning on $\mathcal{E}_\infty$, the magnitudes R_s are fixed and the signs χ_s remain *iid* Rademacher; hence, conditional on $\mathcal{E}_\infty$, the $Y_{s,M}$ are independent, mean zero, and have variance R_s^2/M, i.e.

$$\operatorname{Var}\big(Z_M \mid \mathcal{E}_\infty\big) = \sum_{s\le M} \operatorname{Var}\big(Y_{s,M} \mid \mathcal{E}_\infty\big) = \frac{1}{M}\sum_{s\le M} R_s(\omega)^2 \overset{\text{a.s.}}{\to} \sigma^2$$

by Lemma D.12.i), with conditional Lindeberg condition given by by Lemma D.12.ii),

$$\sum_{s\le M} \mathbb{E}\big[Y_{s,M}^2 \mathbb{1}\{|Y_{s,M}| > \delta\} \,\big|\, \mathcal{E}_\infty\big] = \frac{1}{M}\sum_{s\le M} R_s(\omega)^2\, \mathbb{1}\{R_s(\omega) > \delta\sqrt{M}\} \overset{\text{a.s.}}{\to} 0$$

The Lindeberg–Feller CLT for independent, non-identically distributed triangular arrays therefore applies along $M \to \infty$, giving $\phi_M(t)(\omega) \to \exp(-\sigma^2 t^2/2)$. As this holds for every $\omega \in \Omega_0$, convergence is almost sure. *Step 2. L^1 convergence.* Characteristic functions are bounded by one, so $|\phi_M(t) - \phi(t)| \le 2$. a.s. convergence plus uniform boundedness gives $\mathbb{E}|\phi_M(t) - \phi(t)| \to 0$ through a dominated convergence argument. □

Corollary D.15. *Uniform Mixing. Under the conclusion of Theorem D.14,*

$$\sup_{A\in\mathcal{E}_\infty,\ t\in\mathbb{R}} \big|\mathbb{E}[\phi_M(t)\mathbb{1}_A] - \phi(t)\mathbb{P}(A)\big| \to 0, \quad \iff \quad Z_M \implies N(0,\sigma^2) \ \ \mathcal{E}_\infty\text{-uniformly mixing}$$

Proof of Corollary D.15. For any $A \in \mathcal{E}_\infty$ and $t \in \mathbb{R}$,

$$\begin{aligned}\big|\mathbb{E}[\exp(itZ_M)\mathbb{1}_A] - \phi(t)\mathbb{P}(A)\big| &= \big|\mathbb{E}\big[\mathbb{E}[\exp(itZ_M)\mathbb{1}_A \mid \mathcal{E}_\infty]\big] - \mathbb{E}[\phi(t)\mathbb{1}_A]\big| \\ &= \big|\mathbb{E}\big[\big(\phi_M(t) - \phi(t)\big)\mathbb{1}_A\big]\big| \ \le\ \mathbb{E}\big|\phi_M(t) - \phi(t)\big| \ \to\ 0,\end{aligned}$$

where the second equality uses the $\mathcal{E}_\infty$-measurability of $\mathbb{1}_A$. The convergence is uniform over $A \in \mathcal{E}_\infty$ because the final bound does not depend on A. □

Remark D.16. *Is Uniform Mixing Necessary? Uniformity allows the conditioning event to move with the index in a way that standard pointwise mixing cannot accommodate. This is essential below, where the conditional design events of interest will move along the horizon.*

D.5 Even Policies and Terminal Events

Definition D.17. *Policy Class. Let $\nu_N^{\mathcal{K}} = (N_N^k/N)_{k\in\mathcal{K}}$ for some subset $\mathcal{K} \subseteq [K]$. A policy π is*

(i) ***Even*** *if the terminal design $(N_N^k)_{k\in[K]}$ is $\mathcal{E}_\infty$-measurable;*

(ii) ***Non-$\mathcal{K}$-starving*** *if $\mathbb{P}\big(\nu_N^{\mathcal{K}} \ge c\,\mathbf{1}\big) \to 1$ for some $c > 0$.*

Remark D.18. *Reading Definition D.17. Of the two, evenness is the substantive restriction. Were the policy able to select the next arm k_i on the basis of the raw errors $(\varepsilon_s^k)_{k,s<i}$, the terminal design and its associated events would not be $\mathcal{E}_\infty$-measurable. Non-starvation is required because asymptotic statements can only concern arms played infinitely often, and nothing in evenness prevents a policy from ceasing to play an arm. Weaker divergence conditions, not necessarily linear in N, would suffice for most results in this note, but linear non-starvation becomes indispensable when transitioning towards asymptotically even policies. Divergence is then stated in linear terms throughout for consistency.*

D.6 Conditional CLT for Random Even Designs

Theorem D.19. *Mixing CLT Conditional on the Design. Write* $\phi^k(t) = \exp(-\sigma_k^2 t^2/2)$. *For a subset* $\mathcal{K} \subseteq [K]$, *let* $(N_N^k)_{k\in[K]}$ *be a terminal design induced by an even, non-*$\mathcal{K}$*-starving policy, and let* $\big(A_N^{\mathcal{K}}\big)_N$ *with* $A_N^{\mathcal{K}} = \{(N_N^k)_{k\in\mathcal{K}} \in C\} \in \mathcal{E}_\infty$ *be a sequence of design events. Then*

$$\left|\mathbb{E}\left[\exp\left(i\sum_{k\in\mathcal{K}} t_k Z^k_{N_N^k}\right)\mathbb{1}_{A_N^{\mathcal{K}}}\right] - \mathbb{P}\big(A_N^{\mathcal{K}}\big)\prod_{k\in\mathcal{K}}\phi^k(t_k)\right| \to 0 \qquad \forall (t_k)_k \in \mathbb{R}^{|\mathcal{K}|}$$

Moreover, for any such sequence with $\liminf_N \mathbb{P}(A_N^{\mathcal{K}}) > 0$,

$$\mathcal{L}\left(\left(Z^k_{N_N^k}\right)_{k\in\mathcal{K}} \;\middle|\; A_N^{\mathcal{K}}\right) \implies N\big(\mathbf{0}, \mathrm{diag}(\sigma_k^2)_{k\in\mathcal{K}}\big)$$

Proof of Theorem D.19. *Step 1: Decomposition of the joint conditional characteristic function.* For any $(t_k)_k \in \mathbb{R}^{|\mathcal{K}|}$,

$$\begin{aligned}
\mathbb{E}\left[\exp\left(i\sum_{k\in\mathcal{K}} t_k Z^k_{N_N^k}\right)\mathbb{1}_{A_N^{\mathcal{K}}}\right] &= \mathbb{E}\left[\mathbb{E}\left[\exp\left(i\sum_{k\in\mathcal{K}} t_k Z^k_{N_N^k}\right)\mathbb{1}_{A_N^{\mathcal{K}}}\middle|\mathcal{E}_\infty\right]\right] \\
&= \mathbb{E}\left[\mathbb{E}\left[\sum_{(M^k)_{k\in\mathcal{K}}} \mathbb{1}\big\{(N_N^k)_{k\in\mathcal{K}} = (M^k)_{k\in\mathcal{K}}\big\}\exp\left(i\sum_{k\in\mathcal{K}} t_k Z^k_{M^k}\right)\middle|\mathcal{E}_\infty\right]\mathbb{1}_{A_N^{\mathcal{K}}}\right] \\
&= \mathbb{E}\left[\sum_{(M^k)_{k\in\mathcal{K}}} \mathbb{1}\big\{(N_N^k)_{k\in\mathcal{K}} = (M^k)_{k\in\mathcal{K}}\big\}\,\mathbb{E}\left[\exp\left(i\sum_{k\in\mathcal{K}} t_k Z^k_{M^k}\right)\middle|\mathcal{E}_\infty\right]\mathbb{1}_{A_N^{\mathcal{K}}}\right] \\
&= \mathbb{E}\left[\sum_{(M^k)_{k\in\mathcal{K}}} \mathbb{1}\big\{(N_N^k)_{k\in\mathcal{K}} = (M^k)_{k\in\mathcal{K}}\big\}\prod_{k\in\mathcal{K}}\phi^k_{M^k}(t_k)\,\mathbb{1}_{A_N^{\mathcal{K}}}\right] \;=\; \mathbb{E}\left[\prod_{k\in\mathcal{K}}\phi^k_{N_N^k}(t_k)\,\mathbb{1}_{A_N^{\mathcal{K}}}\right]
\end{aligned}$$

The second equality follows by $\mathcal{E}_\infty$-measurability of $A_N^{\mathcal{K}}$, the third by $\mathcal{E}_\infty$-measurability of the design indicators together with the finiteness of the grid of count vectors. The fourth equality is the load-bearing step: after conditioning on $\mathcal{E}_\infty$ the magnitudes entering each $Z^k_{M^k}$ are frozen and the signs remain *iid* and independent across arms, so the conditional expectation factorizes over k.

Step 2: L^1 convergence along the diverging counts. Define $\phi^k_M(t) = \mathbb{E}[\exp(itZ^k_M) \mid \mathcal{E}_\infty]$. Step 1 of Theorem D.14 established $|\phi^k_M(t) - \phi^k(t)| \xrightarrow[M]{\text{a.s.}} 0$ on Ω_0. Non-starvation gives $N_N^k \xrightarrow{p} \infty$ for each $k \in \mathcal{K}$, so Lemma B.1 yields $|\phi^k_{N_N^k}(t) - \phi^k(t)| \xrightarrow{p} 0$. A union bound over the finite set $\mathcal{K}$ delivers the vector statement. Uniform boundedness upgrades convergence in probability to L^1.

Step 3: Uniform mixing. By Corollary D.15, L^1 convergence of the conditional characteristic functions gives uniform mixing over $\mathcal{E}_\infty$, and $(A_N^{\mathcal{K}})_N$ is $\mathcal{E}_\infty$-measurable by evenness. Finally, the first claim is established by noting that $\mathbb{E}\big|\prod_k \phi^k_{N_N^k}(t_k) - \prod_k \phi^k(t_k)\big| \to 0$, given that each factor converges in L^1 and all factors are bounded by one. The second claim follows by dividing by $\mathbb{P}(A_N^{\mathcal{K}})$, which is eventually bounded away from zero by the theorem's hypothesis, and applying Lévy's theorem. □

Lemma D.20. *Variance Consistency. For a deterministic diverging index* M, *the within-arm variance estimator*

$$\hat\sigma^2_{k,M} := \frac{1}{M}\sum_{s=1}^{M}\big(X_s^k - \hat\mu_M^k\big)^2 \qquad \textit{satisfies} \qquad \hat\sigma^2_{k,M} \xrightarrow[M]{\text{a.s.}} \sigma_k^2$$

Proof of Lemma D.20. Since $X_s^k - \hat{\mu}_M^k = \varepsilon_s^k - \bar{\varepsilon}_M^k$,

$$\hat{\sigma}_{k,M}^2 = \frac{1}{M}\sum_{s=1}^{M}\left(\varepsilon_s^k\right)^2 - \left(\bar{\varepsilon}_M^k\right)^2 \xrightarrow[M]{\text{a.s.}} \sigma_k^2$$

by the CMT and the SLLN applied to the *iid* sequences $((\varepsilon_s^k)^2, (\varepsilon_s^k))_s$ with finite second moments. □

Corollary D.21. *Studentized Conditional Inference. Let $\sigma_k^2 > 0$ for all $k \in \mathcal{K}$, let $(N_N^k)_{k\in[K]}$ be a design induced by an even, non-$\mathcal{K}$-starving policy, and let $(A_N^{\mathcal{K}})_N$ be a sequence of $\mathcal{E}_\infty$-measurable design events with $\liminf_N \mathbb{P}(A_N^{\mathcal{K}}) > 0$. Then*

$$\mathcal{L}\left(\left(\frac{\sqrt{N_N^k}(\hat{\mu}_{N_N^k}^k - \mu_k)}{\hat{\sigma}_{k,N_N^k}}\right)_{k\in\mathcal{K}} \,\middle|\, A_N^{\mathcal{K}}\right) \implies N(\mathbf{0}, I)$$

Proof of Corollary D.21. Decompose

$$\left(\frac{Z_{N_N^k}^k}{\hat{\sigma}_{k,N_N^k}}\right)_{k\in\mathcal{K}} = \left(\frac{Z_{N_N^k}^k}{\sigma_k}\right)_{k\in\mathcal{K}} \odot \left(\frac{\sigma_k}{\hat{\sigma}_{k,N_N^k}}\right)_{k\in\mathcal{K}}$$

Lemma D.20 gives $\hat{\sigma}_{k,M}^2 \to \sigma_k^2$ a.s. along a deterministic M and non-starvation gives $N_N^k \xrightarrow{p} \infty$, so Lemma B.1 recovers convergence in probability $\sigma_{k,N_N^k}^2 \xrightarrow[N]{p} \sigma_k^2$. Consequently, for every $\delta > 0$,

$$\begin{aligned}\mathbb{P}\left(\left\|(\sigma_k/\hat{\sigma}_{k,N_N^k})_{k\in\mathcal{K}} - \mathbf{1}\right\| > \delta \mid A_N^{\mathcal{K}}\right) &= \frac{\mathbb{P}\left(\left\{\left\|(\sigma_k/\hat{\sigma}_{k,N_N^k})_k - \mathbf{1}\right\| > \delta\right\} \cap A_N^{\mathcal{K}}\right)}{\mathbb{P}(A_N^{\mathcal{K}})} \\ &\le \frac{\mathbb{P}\left(\left\|(\sigma_k/\hat{\sigma}_{k,N_N^k})_k - \mathbf{1}\right\| > \delta\right)}{\mathbb{P}(A_N^{\mathcal{K}})} \;\to\; 0\end{aligned}$$

The ratio is well defined because $\mathbb{P}(\min_{k\in\mathcal{K}} \hat{\sigma}_{k,N_N^k} > 0) \to 1$ and $\liminf_N \mathbb{P}(A_N^{\mathcal{K}}) > 0$. By Theorem D.19, $\mathcal{L}\left((Z_{N_N^k}^k/\sigma_k)_{k\in\mathcal{K}} \mid A_N^{\mathcal{K}}\right) \implies N(\mathbf{0}, I)$. Applying a vectorial Slutsky theorem to the conditional law completes the proof. □

Remark D.22. *Studentization Does Not Require Evenness. The variance estimators $(\hat{\sigma}_{k,N_N^k}^2)_{k\in\mathcal{K}}$ are not $\mathcal{E}_\infty$-measurable because they demean by the arm sample means $\hat{\mu}^k$ rather than by the true means μ_k, so the resulting cross terms depend on relative signs. Nonetheless, the corollary goes through because $\hat{\sigma}^2$ converges to a constant in the limit, which is absorbed by Slutsky regardless of measurability.*

D.7 Asymptotically Even Designs

Even policies are generally infeasible as the learner only observes the residuals, not the true errors. So the learner could, at best, emulate even policies by acting on functions of squared residuals, such as the within-arm sample variance $\hat{\sigma}_{k,M}^2$. However, policies depending on (sequences of) sample variances could in principle depend on the relative signs, therefore inducing designs outside the even σ-algebra. The example below describes one such policy.

Example D.23. *Sample Variances Load on Relative Signs. Fix a single arm and drop its index. The designer observes $X_s = \mu + \varepsilon_s$ with μ unknown, so neither ε_s nor ε_s^2 is observed. The sample variance*

is location invariant, hence computable from outcomes for $m \geq 2$. In particular, it can be written as

$$\hat{\sigma}_m^2 = \frac{m-1}{m^2} \sum_{s \leq m} \varepsilon_s^2 \; - \; \frac{2}{m^2} U_m, \qquad U_m := \sum_{1 \leq i < j \leq m} \varepsilon_i \varepsilon_j$$

The first term entering $\hat{\sigma}_m^2$ is $\mathcal{E}_\infty$-measurable, thus $\hat{\sigma}_m^2$ is $\mathcal{E}_\infty$-measurable if and only if $U_m \in \mathcal{E}_\infty$. For $m = 2$, $U_2 = |\varepsilon_1 \varepsilon_2| \operatorname{sign}(\varepsilon_1)\operatorname{sign}(\varepsilon_2)$, which is not $\mathcal{E}_\infty$-measurable by Lemma D.11. A policy that depends on $\hat{\sigma}_2^2$ may therefore induce a design that is not $\mathcal{E}_\infty$-measurable, and the hypothesis of Theorem D.19 fails.

Remark D.24. *Enlarging $\mathcal{E}_\infty$ to contain the relative signs does not help. Conditioning on all relative signs (and all magnitudes) pins down the error path up to a single global flip, so Z_M becomes two-valued rather than Gaussian. In what follows, asymptotic evenness is not defined through leakage, but the intuition remains helpful. The odd component of $\hat{\sigma}_m^2$ is $-2U_m/m^2 = O_p(m^{-1})$, while the even component is $O_p(1)$. Thus, provided ε^2 is non-degenerate and $m \to \infty$, designs depending on terminal true variances can only differ from designs induced by terminal sample variances on events of vanishing probability.*

Definition D.25. *Asymptotically Even Policies. Let $\Xi_N^k = (N_N^k - N_{0N}^k)/\sqrt{N}$ for a deterministic centering sequence $(N_{0N}^k)_N$, and write $\Xi_N^{\mathcal{K}} = (\Xi_N^k)_{k \in \mathcal{K}}$. A policy π with terminal design $(N_N^k)_{k \in [K]}$ is asymptotically even if there exists an $\mathcal{E}_\infty$-measurable design $(\tilde{N}_N^k)_{k \in [K]}$, with $\tilde{\Xi}_N^{\mathcal{K}} = \big((\tilde{N}_N^k - N_{0N}^k)/\sqrt{N}\big)_{k \in \mathcal{K}}$, such that*

$$\big\|\Xi_N^{\mathcal{K}} - \tilde{\Xi}_N^{\mathcal{K}}\big\| \xrightarrow[N]{p} 0$$

Remark D.26. *The definition places little restrictions on the even and asymptotically even policies. The surrogate even policy need be neither feasible nor related to π in any substantive way. Moreover, π could a priori condition on events well outside $\mathcal{E}_\infty$, raw residuals included. However, in practice, the asymptotically even policy will be some functional of squared residuals, and the even surrogate, a functionally equivalent policy with the squared errors replacing the squared residuals.*

Remark D.27. *A Reduction in the Event Conditioning Class. For exactly even policies the event conditioning class is unrestricted. Theorem D.19 guarantees uniform mixing over $\mathcal{E}_\infty$, so any $\mathcal{E}_\infty$-measurable event of any scale, terminal or not, leaves the limiting law intact. The design events induced by asymptotically even policies are no longer $\mathcal{E}_\infty$-measurable, thus asymptotic normality is preserved by pairing the asymptotically even policy with an even surrogate. The class of conditioning events is reduced accordingly to the subset of events that agree asymptotically under both policies, in the sense $\mathbb{P}(A_N^{\mathcal{K}} \triangle \tilde{A}_N^{\mathcal{K}}) \to 0$. The scale of the events that can survive this transition is at most of the same order as the scale at which the two designs are coupled.*

In the applications motivating this note, terminal counts fluctuate on the $\sqrt{N}$-scale around the deterministic optimal allocations. Therefore, in this context, the $\sqrt{N}$ normalization is the finest scale at which the feasible and surrogate designs can generally be coupled while retaining non-degeneracy of the conditioning events. Non-degeneracy is very important (i) because many events of interest live on this scale, such as the quantiles of N^k, and (ii) because it makes conditional and unconditional inference meaningfully distinct. At coarser scales like $\nu_N^{\mathcal{K}} = (N_N^k/N)_k$, coupling is usually trivial but the conditioning events are vacuous. In most cases, fractional designs converge in probability, and so every conditioning event has limiting probability in $\{0, 1\}$. In this regime, the conditional law

degenerates to the unconditional law, such that a random-index central limit theorem in the spirit of Anscombe (1952) will generally suffice without ever invoking symmetry. By contrast, at the unit-scale coupling is usually not possible and requires stronger convergence than granted by Definition D.25. This trichotomy is discussed in greater detail in the main text.

Lemma D.28. *Asymptotically Even Events. Let $A_N^{\mathcal{K}} = \{\Xi_N^{\mathcal{K}} \in C\}$ for a Borel set $C \subseteq \mathbb{R}^{|\mathcal{K}|}$, generated by an asymptotically even, non-$\mathcal{K}$-starving policy π. Let C satisfy the margin condition*

$$\lim_{\eta \downarrow 0} \limsup_N \mathbb{P}\big(\operatorname{dist}(\Xi_N^{\mathcal{K}}, \partial C) \le \eta\big) = 0 \tag{16}$$

Then, the event $\tilde{A}_N^{\mathcal{K}} = \{\tilde{\Xi}_N^{\mathcal{K}} \in C\}$ belongs to $\mathcal{E}_\infty$ and satisfies $\mathbb{P}(A_N^{\mathcal{K}} \triangle \tilde{A}_N^{\mathcal{K}}) \to 0$.

Proof of Lemma D.28. Measurability of $\tilde{A}_N^{\mathcal{K}}$ is immediate, since $(\tilde{N}_N^k)_k$ is $\mathcal{E}_\infty$-measurable and the centering and scaling are deterministic. If the two indicators disagree at a point, that means that one of $(\Xi_N^{\mathcal{K}}, \tilde{\Xi}_N^{\mathcal{K}})$ lies in C and the other does not, so the segment joining them meets ∂C and

$$A_N^{\mathcal{K}} \triangle \tilde{A}_N^{\mathcal{K}} = \big\{\mathbb{1}_C(\Xi_N^{\mathcal{K}}) \neq \mathbb{1}_C(\tilde{\Xi}_N^{\mathcal{K}})\big\} \;\subseteq\; \big\{\operatorname{dist}(\Xi_N^{\mathcal{K}}, \partial C) \le \|\Xi_N^{\mathcal{K}} - \tilde{\Xi}_N^{\mathcal{K}}\|\big\}$$

Introducing a separating $\eta > 0$,

$$\mathbb{P}\big(A_N^{\mathcal{K}} \triangle \tilde{A}_N^{\mathcal{K}}\big) \;\le\; \mathbb{P}\big(\|\Xi_N^{\mathcal{K}} - \tilde{\Xi}_N^{\mathcal{K}}\| > \eta\big) + \mathbb{P}\big(\operatorname{dist}(\Xi_N^{\mathcal{K}}, \partial C) \le \eta\big)$$

The first term vanishes for every fixed η by asymptotic evenness, so $\limsup_N \mathbb{P}(A_N^{\mathcal{K}} \triangle \tilde{A}_N^{\mathcal{K}}) \le \limsup_N \mathbb{P}(\operatorname{dist}(\Xi_N^{\mathcal{K}}, \partial C) \le \eta)$ for every $\eta > 0$. The left-hand side does not depend on η, so letting $\eta \downarrow 0$ and invoking (16) sends this term to zero. □

Remark D.29. *Reading the Margin Condition. Condition (16) is an anti-concentration requirement on the scaled counts near ∂C: it forbids the design from piling up on the boundary, which is exactly where the feasible and surrogate designs can disagree. In applications it is usually verified by analyzing the limiting law of the scaled design. For example, assume that $\Xi_N^{\mathcal{K}} \implies \Xi_\infty^{\mathcal{K}}$ for a well-defined limit law $\Xi_\infty^{\mathcal{K}}$ with $\mathbb{P}(\Xi_\infty^{\mathcal{K}} \in \partial C) = 0$, then, since the set $\{x : \operatorname{dist}(x, \partial C) \le \eta\}$ is closed, the Portmanteau Theorem gives*

$$\limsup_N \mathbb{P}\big(\operatorname{dist}(\Xi_N^{\mathcal{K}}, \partial C) \le \eta\big) \;\le\; \mathbb{P}\big(\operatorname{dist}(\Xi_\infty^{\mathcal{K}}, \partial C) \le \eta\big)$$

where the right-hand side decreases to $\mathbb{P}(\Xi_\infty^{\mathcal{K}} \in \partial C) = 0$ as $\eta \downarrow 0$. The same weak convergence argument implies that $\liminf_N \mathbb{P}(A_N^{\mathcal{K}}) > 0$ for any C whose interior receives positive limit mass.

Remark D.30. *CLT and Fourth Moments in Asymptotically Even Policies. As hinted in Remark D.29, Condition (16) is usually verified by inspecting the limiting law of the scaled design. This procedure can be implemented by establishing a central limit theorem over the underlying statistic governing the design as, whenever the mass of the limiting law spreads smoothly, the anticoncentration requirement is immediately satisfied. In particular, whenever the design is driven by second moments, the margin condition can be verified by proving a CLT over the pilot variances, whose own variance is $\mu_{4,k} - \sigma_k^4$. Such CLT can usually be established under an additional finite-fourth-moments assumption. In other words, finite fourth moments will be, in general, a clean sufficient alternative to the margin condition. On a final note, the design statistic entering the cluster problem is clipped, so the limit law may contain an atom when λ_0 sits at the boundary, and a especial treatment is required. See Lemma E.15.*

Lemma D.31. *Increment Vanishes at Ratio-close Indices. Fix a single arm and drop its index. Let $(M_N)_N$ and $(\tilde{M}_N)_N$ be $\{0,1,\dots,N\}$-valued random indices with $|M_N/N - \tilde{M}_N/N| \xrightarrow[N]{p} 0$ and $\mathbb{P}\big(M_N/N \wedge \tilde{M}_N/N \geq c\big) \to 1$ for some $c > 0$. Then*

$$\big|Z_{M_N} - Z_{\tilde{M}_N}\big| \xrightarrow{p} 0$$

Proof of Lemma D.31. Work on the event $\{\tilde{M}_N \leq M_N\}$, and write $\Sigma_M = \sum_{s\leq M} \varepsilon_s$, so that $Z_M = \Sigma_M/\sqrt{M}$,

$$\big|Z_{M_N} - Z_{\tilde{M}_N}\big| \;\leq\; \left|\sqrt{\tfrac{\tilde{M}_N}{M_N}} - 1\right| \cdot \big|Z_{\tilde{M}_N}\big| + \frac{\big|\Sigma_{M_N} - \Sigma_{\tilde{M}_N}\big|}{\sqrt{M_N}}$$

On $\{\tilde{M}_N \geq cN\}$, $|Z_{\tilde{M}_N}| \leq \max_{cN\leq m\leq N}|\Sigma_m|/\sqrt{cN} = O_p(1)$ by Kolmogorov's maximal inequality applied over the *iid*, mean-zero, finite-variance error sequence. Convergence in probability of the scaled indices together with linear non-starvation gives $|\sqrt{\tilde{M}_N/M_N} - 1| \xrightarrow{p} 0$, and so the first product is $o_p(1)$. For the second term, fix $\delta \in (0,1)$ and intersect the working event with $\{|M_N - \tilde{M}_N| \leq \delta N\} \cap \{\tilde{M}_N \geq cN\}$, whose probability tends to one under the hypotheses. Therefore,

$$\frac{\big|\Sigma_{M_N} - \Sigma_{\tilde{M}_N}\big|}{\sqrt{M_N}} \;\leq\; \max_{\substack{cN\leq \tilde{m}\leq m\leq N \\ |m-\tilde{m}|\leq \delta N}} \frac{|\Sigma_m - \Sigma_{\tilde{m}}|}{\sqrt{m}} \;\leq\; \frac{1}{\sqrt{c}} \max_{\substack{0\leq \tilde{m}\leq m\leq N \\ |m-\tilde{m}|\leq \delta N}} \frac{|\Sigma_m - \Sigma_{\tilde{m}}|}{\sqrt{N}} \;\leq\; \frac{1}{\sqrt{c}} \sup_{\substack{0\leq s\leq t\leq 1 \\ |t-s|\leq \delta}} \big|\Psi_N(t) - \Psi_N(s)\big|$$

where $\Psi_N(t) = \Sigma_{\lfloor Nt\rfloor}/\sqrt{N}$ is the rescaled partial-sum process. Under Assumption D.3, $(\Psi_N)_N$ is asymptotically uniformly equicontinuous (tight), as per Donsker's functional central limit theorem and continuity of the Brownian limit. Equivalently, $(\Psi_N)_N$ holds the local modulus-of-continuity property

$$\lim_{\delta\downarrow 0} \limsup_{N\to\infty} \mathbb{P}\left(\sup_{\substack{0\leq s\leq t\leq 1 \\ |t-s|\leq \delta}} \big|\Psi_N(t) - \Psi_N(s)\big| \geq \eta\right) = 0 \qquad \forall \eta > 0$$

Fix $\eta, \epsilon > 0$ and choose δ so that

$$\limsup_{N\to\infty} \mathbb{P}\left(\sup_{\substack{0\leq s\leq t\leq 1 \\ |t-s|\leq \delta}} \big|\Psi_N(t) - \Psi_N(s)\big| \geq \eta\sqrt{c}\right) \leq \varepsilon/2$$

Since $\mathbb{P}(|M_N - \tilde{M}_N| > \delta N) \to 0$, for all N large

$$\mathbb{P}\left(\frac{|\Sigma_{M_N} - \Sigma_{\tilde{M}_N}|}{\sqrt{M_N}} \geq \eta\right) \;\leq\; \mathbb{P}\big(|M_N - \tilde{M}_N| > \delta N\big) + \mathbb{P}\left(\sup_{\substack{0\leq s\leq t\leq 1 \\ |t-s|\leq \delta}} \big|\Psi_N(t) - \Psi_N(s)\big| \geq \eta\sqrt{c}\right) \;\leq\; \epsilon$$

As η and ϵ were arbitrary the second term is $o_p(1)$ and the claim follows. For the case where $\{\tilde{M}_N > M_N\}$ the argument is completely symmetrical and so the unconditional statement follows too. □

Remark D.32. *Lemma D.31 Is Measurability-Free. Lemma D.31 is a statement about the random walk alone. It does not condition on $\mathcal{E}_\infty$ and imposes no measurability on M_N or $\tilde{M}_N$. As a result, it will be useful to bound the distance between $Z^k_{N^k_N}$ (non-measurable) and $Z^k_{\tilde{N}^k_N}$ (measurable). Additionally, this is exactly where **linear** non-starvation enters the analysis. Tightness of the sum process*

only manifests at indices which are negligible at the scale of counts. Asymptotic evenness controls $|N_N^k - \tilde{N}_N^k|$ *only to order* $o_p(\sqrt{N})$*, so the difference between indices is negligible at the scale of counts just because the counts are assumed to be of order* N*. Slower forms of divergence will compromise convergence in probability between the scaled sums of the exactly even and asymptotically even policies.*

Theorem D.33. *Inference for Asymptotically Even Designs. Let* π *be an asymptotically even, non-*$\mathcal{K}$*-starving policy, and let* $(A_N^{\mathcal{K}})_N$*, with* $A_N^{\mathcal{K}} = \{\Xi_N^{\mathcal{K}} \in C\}$*, be a sequence of scaled design events satisfying the margin condition (16). Then*

$$\left| \mathbb{E}\left[\exp\left(i \sum_{k \in \mathcal{K}} t_k Z^k_{N^k_N} \right) \mathbb{1}_{A_N^{\mathcal{K}}} \right] - \mathbb{P}\big(A_N^{\mathcal{K}}\big) \prod_{k \in \mathcal{K}} \phi^k(t_k) \right| \to 0 \qquad \forall (t_k)_k \in \mathbb{R}^{|\mathcal{K}|}$$

If in addition $\liminf_N \mathbb{P}(A_N^{\mathcal{K}}) > 0$*, then*

$$\mathcal{L}\left((Z^k_{N^k_N})_{k \in \mathcal{K}} \;\middle|\; A_N^{\mathcal{K}} \right) \implies N\big(\mathbf{0}, \mathrm{diag}(\sigma_k^2)_{k \in \mathcal{K}}\big), \qquad \mathcal{L}\left(\left(\frac{\sqrt{N_N^k}(\hat{\mu}^k_{N^k_N} - \mu_k)}{\hat{\sigma}_{k, N^k_N}} \right)_{k \in \mathcal{K}} \;\middle|\; A_N^{\mathcal{K}} \right) \implies N(\mathbf{0}, I)$$

Remark D.34. *Not Uniform Mixing. The first statement cannot be phrased as* $\mathcal{E}_\infty$*-uniform mixing, since* $A_N^{\mathcal{K}}$ *is not even* $\mathcal{E}_\infty$*-measurable. In fact, convergence only holds pointwise along the design event sequence. For the same reason, the conditional characteristic function of the feasible design cannot be written as* $\phi^k_{N^k_N}$*. Yet, the essence of the result is precisely akin that of Theorem D.19.*

Proof of Theorem D.33. *Step 0: the surrogate inherits non-starvation.* Since $\tilde{\nu}_N^k - \nu_N^k = (\tilde{\Xi}_N^k - \Xi_N^k)/\sqrt{N}$, for any $\eta > 0$

$$\mathbb{P}\big(\tilde{\nu}_N^{\mathcal{K}} < (c - \eta)\mathbf{1}\big) \;\le\; \mathbb{P}\big(\|\Xi_N^{\mathcal{K}} - \tilde{\Xi}_N^{\mathcal{K}}\| > \sqrt{N}\,\eta\big) + \mathbb{P}\big(\nu_N^{\mathcal{K}} < c\,\mathbf{1}\big) \;\to\; 0$$

so $\tilde{\pi}$ is non-$\mathcal{K}$-starving with constant $c - \eta$. *Step 1: decomposition.* Writing $\tilde{A}_N^{\mathcal{K}}$ for the surrogate event of Lemma D.28,

$$\begin{aligned}
&\left| \mathbb{E}\Big[e^{i \sum_k t_k Z^k_{N^k_N}} \mathbb{1}_{A_N^{\mathcal{K}}} \Big] - \mathbb{P}(A_N^{\mathcal{K}}) \prod_{k \in \mathcal{K}} \phi^k(t_k) \right| \\
&\quad \le \underbrace{\left| \mathbb{E}\Big[e^{i \sum_k t_k Z^k_{N^k_N}} \mathbb{1}_{A_N^{\mathcal{K}}} \Big] - \mathbb{E}\Big[e^{i \sum_k t_k Z^k_{\tilde{N}^k_N}} \mathbb{1}_{\tilde{A}_N^{\mathcal{K}}} \Big] \right|}_{[1]} + \underbrace{\left| \mathbb{E}\Big[e^{i \sum_k t_k Z^k_{\tilde{N}^k_N}} \mathbb{1}_{\tilde{A}_N^{\mathcal{K}}} \Big] - \mathbb{P}(\tilde{A}_N^{\mathcal{K}}) \prod_{k \in \mathcal{K}} \phi^k(t_k) \right|}_{[2]} \\
&\qquad + \underbrace{\left| \prod_{k \in \mathcal{K}} \phi^k(t_k) \right| \cdot \left| \mathbb{P}(A_N^{\mathcal{K}}) - \mathbb{P}(\tilde{A}_N^{\mathcal{K}}) \right|}_{[3]}
\end{aligned}$$

Term [3] vanishes because $|\mathbb{P}(A_N^{\mathcal{K}}) - \mathbb{P}(\tilde{A}_N^{\mathcal{K}})| \le \mathbb{P}(A_N^{\mathcal{K}} \triangle \tilde{A}_N^{\mathcal{K}}) \to 0$ by Lemma D.28, and the characteristic functions are bounded by one. Term [2] vanishes by Theorem D.19 applied to $\big((\tilde{N}_N^k)_{k \in \mathcal{K}}, (\tilde{A}_N^{\mathcal{K}})_N\big)$: the surrogate design and its events are $\mathcal{E}_\infty$-measurable by construction, and the surrogate counts diverge (linearly) with probability tending to one by Step 0, so the freezing identity $\mathbb{E}\big[e^{i \sum_k t_k Z^k_{\tilde{N}^k_N}} \mid \mathcal{E}_\infty \big] =$

$\prod_k \phi^k_{\tilde{N}^k_N}(t_k)$ is available. *Step 2: the coupling term.* Split

$$[1] \;\le\; \left|\mathbb{E}\Big[\Big(e^{i\sum_k t_k Z^k_{N^k_N}} - e^{i\sum_k t_k Z^k_{\tilde{N}^k_N}}\Big)\mathbb{1}_{A^{\mathcal{K}}_N}\Big]\right| + \left|\mathbb{E}\Big[e^{i\sum_k t_k Z^k_{\tilde{N}^k_N}}\big(\mathbb{1}_{A^{\mathcal{K}}_N} - \mathbb{1}_{\tilde{A}^{\mathcal{K}}_N}\big)\Big]\right|$$

The second piece is bounded by $\mathbb{P}(A^{\mathcal{K}}_N \triangle \tilde{A}^{\mathcal{K}}_N) \to 0$, the integrand having modulus one. For the first term, using the fact that $|e^{ia} - e^{ib}| \le \min\{2, |a-b|\}$ together with subadditivity of $x \mapsto \min\{2, x\}$ yields

$$\mathbb{E}\Big|e^{i\sum_k t_k Z^k_{N^k_N}} - e^{i\sum_k t_k Z^k_{\tilde{N}^k_N}}\Big| \;\le\; \sum_{k\in\mathcal{K}} \mathbb{E}\Big[\min\big\{2, |t_k|\cdot\big|Z^k_{N^k_N} - Z^k_{\tilde{N}^k_N}\big|\big\}\Big]$$

For each $k \in \mathcal{K}$, Step 0 gives $\mathbb{P}(N^k_N/N \wedge \tilde{N}^k_N/N \ge c') \to 1$ for some $c' > 0$, and asymptotic evenness gives $|N^k_N/N - \tilde{N}^k_N/N| = o_p(\sqrt{N})/N \xrightarrow{p} 0$, so the hypotheses of Lemma D.31 hold and each increment is $o_p(1)$. Uniform boundedness of the integrand by 2 upgrades convergence in probability to L^1 convergence. Moreover, the sum runs over finitely many arms, thus $[1] \to 0$, establishing the first statement.

Step 3. The conditional limits. Dividing by $\mathbb{P}(A^{\mathcal{K}}_N)$, eventually bounded away from zero by hypothesis, and applying Lévy's continuity theorem gives the first conditional statement. Studentization only requires $\hat{\sigma}^2_{k,N^k_N} \xrightarrow{p} \sigma^2_k$, which holds under non-starvation by Lemma D.20. The argument of Corollary D.21 therefore applies verbatim. Measurability of N^k_N and $A^{\mathcal{K}}_N$ were never invoked in the derivation. □

D.8 Policy Examples

Proposition D.35. *Two-Stage Neyman Allocation Is Asymptotically Even. Let $\sigma^2 = (\sigma^2_k)_k > \mathbf{0}$ and let $0 < d < \min_k f_k(\sigma)$ be known to the designer, with $dN \in \mathbb{N}$, $f_k(\sigma) = \sigma_k/\Sigma_\sigma$ and $\Sigma_\sigma = \sum_l \sigma_l$. Consider the policy π that samples uniformly for the first dKN periods—so every arm receives dN pulls—computes the pilot variances $\hat{\sigma}^2_{k,dN} = (dN)^{-1}\sum_{s\le dN}\big(X^k_s - \hat{\mu}^k_{dN}\big)^2$, and allocates the remaining periods so that the terminal counts equal $N^k_N = \lfloor N f_k(\hat{\sigma})\rfloor$, with leftovers assigned by an arbitrary deterministic rule.*[59] *Then, π is non-$[K]$-starving and asymptotically even for $[K]$.*

Proof of Proposition D.35. *Surrogate evenness.* The pilot allocation is deterministic, hence so are the pilot counts. Define the surrogate counts $\tilde{N}^k_N = \lfloor N f_k(q)\rfloor$ under the same leftover rule, with $q^2_k = (dN)^{-1}\sum_{s\le dN}(\varepsilon^k_s)^2$. Each q^2_k is a function of $((\varepsilon^k_s)^2)_{s\le dN} \in \mathcal{E}_\infty$, and the rounding and leftover rules are deterministic, so $(\tilde{N}^k_N)_k$ is $\mathcal{E}_\infty$-measurable. *Non-starvation.* Every arm receives at least dN pulls in the pilot, which are later included within the terminal counts, so $N^k_N \ge dN$ almost surely. *Asymptotic evenness.* The pilot variance demeans by the pilot mean, so $q^2_k - \hat{\sigma}^2_{k,dN} = (\bar{\varepsilon}^k_{dN})^2 = O_p(1/N)$, provided $\bar{\varepsilon}^k_{dN} = O_p(N^{-1/2})$. Since $\sigma_k > 0$, the square root operator is Lipschitz on a neighbourhood of σ^2_k with probability tending to one, thus $|\hat{\sigma}_k - q_k| = O_p(1/N)$. Similarly, since $\Sigma_\sigma > 0$, f is Lipschitz on a neighbourhood of σ, thus $|f_k(\hat{\sigma}) - f_k(q)| = O_p(1/N)$. Finally, note that $|\lfloor Nx\rfloor - \lfloor Ny\rfloor| \le N|x-y| + 1$ and that at most K units can be allocated to arm k through the leftover rule. Combining these three elements,

$$\big|N^k_N - \tilde{N}^k_N\big| \;\le\; N\big|f_k(\hat{\sigma}) - f_k(q)\big| + 1 + K \;=\; O_p(1)$$

[59] Since $\min_k f_k(\sigma) > d$ strictly, $\mathbb{P}(\lfloor N f_k(\hat{\sigma})\rfloor \ge dN\ \forall k) \to 1$ and the target allocation is implementable with probability approaching one. The complementary event has vanishing probability, thus affects none of the statements below.

so $|N_N^k - \tilde{N}_N^k|/\sqrt{N} \xrightarrow{p} 0$ and π is asymptotically even with even surrogate $(\tilde{N}_N^k)_k$. □

Lemma D.36. *Central Limit Theorem for the Scaled Counts. Together with Assumption D.3, assume that $\mu_{4,k} := \mathbb{E}[(\varepsilon_s^k)^4] < \infty$ and $\mu_{4,k} > \sigma_k^4$ for every k. Consider the usual definitions for $(\Xi_N^{[K]}, \tilde{\Xi}_N^{[K]})$ centered at the oracle allocation $N_{\underline{0}N}^k = \lfloor N f_k(\sigma) \rfloor$. Then*

$$\Xi_N^{[K]} \implies G, \qquad \tilde{\Xi}_N^{[K]} \implies G, \qquad G = Jz$$

where $z \sim N(\mathbf{0}, \mathrm{diag}(v_k)_k)$ with $v_k = (\mu_{4,k} - \sigma_k^4)/(4d\sigma_k^2) > 0$, and J is the Jacobian of f at σ with entries $J_{kl} = (\mathbb{1}\{k = l\}\Sigma_\sigma - \sigma_k)/\Sigma_\sigma^2$.

Proof of Lemma D.36. Consider the surrogate first. The variables $((\varepsilon_s^k)^2)_s$ are *iid* with mean σ_k^2 and variance $\mu_{4,k} - \sigma_k^4 < \infty$. The pilot size dN is deterministic, so $\sqrt{N}(q_k^2 - \sigma_k^2) \implies N\big(0, (\mu_{4,k} - \sigma_k^4)/d\big)$, jointly across arms with independent arm coordinates. The delta method applied to $x \mapsto \sqrt{x}$, whose derivative at σ_k^2 is $1/(2\sigma_k)$, gives $\sqrt{N}(q_k - \sigma_k) \implies N(0, v_k)$; applied to f it gives $\sqrt{N}(f(q) - f(\sigma)) \implies Jz = G$. Rounding and the leftover rule perturb $\tilde{\Xi}_N^k$ by at most $(1+K)/\sqrt{N} \to 0$, so $\tilde{\Xi}_N^{[K]} \implies G$ by Slutsky. Finally $\|\Xi_N^{[K]} - \tilde{\Xi}_N^{[K]}\| = O_p(1)/\sqrt{N} \xrightarrow{p} 0$ by Proposition D.35, so $\Xi_N^{[K]} \implies G$ as well. □

Corollary D.37. *Design-Conditional Inference for Two-Stage Neyman. Let π be the two-stage Neyman policy of Proposition D.35, and let $A_N^{[K]} = \{\Xi_N^{[K]} \in C\}$ for a Borel set $C \subseteq \mathbb{R}^K$, with $\Xi_N^{[K]}$ centered at $N_{\underline{0}N}^k = \lfloor N f_k(\sigma) \rfloor$. (a) Under Assumption D.3, if C satisfies (16) and $\liminf_N \mathbb{P}(A_N^{[K]}) > 0$, then*

$$\mathcal{L}\left(\left(\frac{\sqrt{N_N^k}\,(\hat{\mu}_{N_N^k}^k - \mu_k)}{\hat{\sigma}_{k, N_N^k}} \right)_{k \in [K]} \,\middle|\, A_N^{[K]} \right) \implies N(\mathbf{0}, I)$$

(b) If in addition $\mu_{4,k} < \infty$ for all k, (a) holds whenever $\mathbb{P}(G \in \partial C) = 0$ and $\mathbb{P}(G \in C) > 0$, with G as in Lemma D.36.

Proof. (a) By Proposition D.35, π is asymptotically even and non-$[K]$-starving. The margin condition (16) holds by assumption, and the result follows from Theorem D.33. (b) Under fourth moments, Lemma D.36 demonstrates that $\Xi_N^{[K]} \implies G$. Assuming $\mathbb{P}(G \in \partial C) = 0$, the Portmanteau Theorem in Remark D.29 then yields (16) and $\mathbb{P}(A_N^{[K]}) \to \mathbb{P}(G \in C) > 0$. □

E Proofs of Section 6

This section concretizes the general theory of Appendix D for the clustered sampling problem and, in doing so, proves the results of Section 6. Two features of the cluster problem make the translation not mechanical. First, the cluster problem carries *two* error sequences, $(u_k)_k$ and $(\varepsilon_{ki})_{k,i}$, and *two* degrees of freedom (K, N), only one of which diverges with the budget. I handle this issue by establishing N-consistency and extending the results depth by depth. Second, establishing asymptotic evenness for `EtC-CS` follows the general recipe outlined in the two-stage Neyman allocation example, i.e. replacing residuals by errors. Nonetheless, this replacement is insufficient for `ACS`, as doing so would still leave the analyst with the difficult task of pairing two random stopping times. The problem is avoided entirely by considering an even surrogate with fixed stopping time K_{0B} and localizing the stopping time of `ACS` in a window of order $O_p(\sqrt{B})$ around K_{0B}. The definitions and lemmas below are held to a minimum

to avoid duplicities. Connections to the general theory are made throughout, so I recommend reading this section in light of the results in the standalone Appendix D.

E.1 Exact Gaussianity

Proposition 6.1 is the cluster instance of Proposition D.8 with two adaptations. Orthogonality must be established at two layers, and the stopped residual σ-algebra must be characterized appropriately, given that the commitment stage `ACS` is random.

Definition E.1. *Residual σ-algebras. For a fixed pilot depth N_p, an iid auxiliary sequence $(U_k)_k \sim \mathrm{Unif}[0,1]$ independent of the errors, and $j \geq 2$, define the residual σ-algebra for the first j clusters as*

$$\mathcal{R}_j := \sigma\Big(\big(\varepsilon_{ki} - \bar{\varepsilon}_{k,N_p}\big)_{i\leq N_p,\, k\leq j},\ \big(M_{k,N_p} - \bar{M}_{j,N_p}\big)_{k\leq j},\ (U_k)_{k\geq 1} \Big)$$

Definition E.2. *Residual Filtration and Pilot σ-algebras. Given $\mathcal{R}_j$ with $j \geq 2$, then,*

(i) Filtration. *$\mathcal{R}_k \subseteq \mathcal{R}_j$ for all $2 \leq k \leq j$, since the k-demeaned cluster residuals can be recovered from the j-demeaned ones through*

$$M_{l,N_p} - \bar{M}_{k,N_p} = \big(M_{l,N_p} - \bar{M}_{j,N_p}\big) - \frac{1}{k}\sum_{l'\leq k} \big(M_{l',N_p} - \bar{M}_{j,N_p}\big), \qquad l \leq k \leq j$$

(ii) Measurability. *For every $2 \leq k \leq j$, $(\hat{w}^2_{k,N_p}, \hat{b}^2_{k,N_p}, \hat{\lambda}_{k,N_p})$ are $\mathcal{R}_j$-measurable, where*

$$\hat{w}^2_{k,N_p} = \frac{1}{k}\sum_{l\leq k} \frac{1}{N_p - 1}\sum_{i\leq N_p} \big(\varepsilon_{li} - \bar{\varepsilon}_{l,N_p}\big)^2, \qquad \hat{b}^2_{k,N_p} = \frac{1}{k-1}\sum_{l\leq k} \big(M_{l,N_p} - \bar{M}_{k,N_p}\big)^2$$

(iii) Stopping. *The commitment stages $T^{\texttt{EtC-CS}} = K_M$ and $T^{\texttt{ACS}} = s$ are stopping times for $(\mathcal{R}_j)_j$. The first is deterministic so inclusion is trivial. Define the pilot σ-algebra as $\mathcal{P}^{\texttt{EtC-CS}} := \mathcal{R}_{K_M}$. For `ACS` see that*

$$\{s = j\} = \big\{j+1 > K^*(\hat{\lambda}_{j,N_p})\big\} \cap \bigcap_{l=K_M}^{j-1} \big\{l+1 \leq K^*(\hat{\lambda}_{l,N_p})\big\} \in \mathcal{R}_j$$

and so the stopped pilot σ-algebra is $\mathcal{P}^{\texttt{ACS}} := \mathcal{R}_s = \big\{A : A \cap \{s = j\} \in \mathcal{R}_j \ \text{ for every } j \geq 2\big\}$. For $\pi \in \{\texttt{EtC-CS}, \texttt{ACS}\}$, the terminal design (K^π_B, N^π_B) is $\mathcal{P}^\pi$-measurable by construction.

Proof of Proposition 6.1. *Step 1. Two-layer orthogonality.* Fix $j \geq 2$ and let $K' \geq j$, $N \geq N_p$. Since $\hat{\mu}_{K',N} - \mu = \bar{u}_{K'} + (K')^{-1}\sum_{k\leq K'} \bar{\varepsilon}_{k,N}$, and $u \perp\!\!\!\perp \varepsilon$ with independence across clusters, it suffices to verify orthogonality layer by layer against the generators of $\mathcal{R}_j$. At the unit layer, for $i \leq N_p$ and $k \leq j$, only the kth term contributes to the covariance, i.e.

$$\mathrm{Cov}\big(\bar{\varepsilon}_{k,N},\ \varepsilon_{ki} - \bar{\varepsilon}_{k,N_p}\big) = \frac{\sigma^2}{N} - \frac{1}{N_p}\cdot N_p \cdot \frac{\sigma^2}{N} = 0$$

both evaluations using $N_p \leq N$. At the cluster layer, for $k \leq j$,

$$\mathrm{Cov}\big(\bar{u}_{K'},\ u_k - \bar{u}_j\big) = \frac{\tau^2}{K'} - \frac{1}{j}\cdot j \cdot \frac{\tau^2}{K'} = 0$$

both evaluations using $j \le K'$. The remaining piece of $M_{k,N_p} - \bar{M}_{j,N_p}$, namely $\bar{\varepsilon}_{k,N_p} - j^{-1}\sum_{l\le j}\bar{\varepsilon}_{l,N_p}$, is also orthogonal to $(K')^{-1}\sum_k \bar{\varepsilon}_{k,N}$ by the same argument as in the unit layer. All variables are jointly Gaussian, so zero covariance implies independence: $\hat{\mu}_{K',N} \perp\!\!\!\perp \mathcal{R}_j$ with $j \le K'$ and $N_p \le N$.

Step 2. Nesting. $T^\pi \le K_B^\pi$ and $N_p \le N_B^\pi$ almost surely. For `EtC-CS` the result follows because $T^{\texttt{EtC-CS}} = K_M \le K_B^{\texttt{EtC-CS}}$, and for `ACS` because $s \le K^*(\hat{\lambda}_{\max\{s-1,K_M\},N_p}) = K_B^{\texttt{ACS}}$. The hypotheses of Step 1 then hold on every cell of the design grid. The attentive reader will note that nesting is the inference analogue of implementability.

Step 3. Cell-wise factorization. The triple $(T^\pi, K_B^\pi, N_B^\pi)$ takes finitely many values. Fix $A \in \mathcal{P}^\pi$ with $\mathbb{P}(A) > 0$ and write $A_{(j,K,N)} := A \cap \{(T^\pi, K_B^\pi, N_B^\pi) = (j,K,N)\}$. Since (K_B^π, N_B^π) is $\mathcal{P}^\pi$-measurable, $A \cap \{(K_B^\pi, N_B^\pi) = (K,N)\} \in \mathcal{P}^\pi$, and intersecting with $\{T^\pi = j\}$ places $A_{(j,K,N)}$ in $\mathcal{R}_j$ by the stopped σ-algebra construction of Lemma E.2.iii). By Steps 1 and 2, $\sqrt{K}(\hat{\mu}_{K,N} - \mu)/\theta_N \sim N(0,1)$ independently of $\mathcal{R}_j \ni A_{(j,K,N)}$ on every cell, so

$$\mathbb{E}\left[\exp\left(\frac{itZ_B^\pi}{\theta_{N_B^\pi}}\right)\mathbb{1}_A\right] = \sum_{(j,K,N)} e^{-t^2/2}\,\mathbb{P}\big(A_{(j,K,N)}\big) = e^{-t^2/2}\,\mathbb{P}(A)$$

The proof is concluded by dividing by $\mathbb{P}(A)$ and applying Lévy's continuity theorem. □

E.2 Random Sequences in the Cluster Problem

Definition E.3. *Cluster Error Sequences. For a fixed depth N, recall $M_{k,N} = u_k + N^{-1}\sum_{i\le N}\varepsilon_{ki}$, indexed by the cluster k. Let $(\zeta_k)_{k\ge1}$ be an iid Rademacher sequence, used to break ties in Lemma 6.3, and $(U_k)_{k\ge1}$ an iid* Unif$[0,1]$ *sequence available to the policy as an independent randomization device, both independent of $(C_k)_k$.*

Remark E.4. *What Carries Over? For each fixed N, Assumptions 3.1 and 6.2 make $(M_{k,N})_{k\ge1}$ an iid, mean-zero, symmetric sequence with $\mathbb{E}[M_{k,N}^2] = \theta_N^2 < \infty$. Hence, $(M_{k,N})_{k\ge1}$ with fixed N plays the role of the error sequence in Assumption D.3. Then every result of Appendix D applies over the "single-arm" sequence $(M_{k,N})_{k\ge1}$, after replacing the pull index by the cluster index, and the horizon N by the diverging cluster count K_B.*

E.3 Signs and Magnitudes

Proof of Lemma 6.3. Fix N. By Remark E.4, $(M_{k,N})_k$ is *iid*, mean zero and symmetric, so Lemma D.10 applies coordinatewise with ε_s^k replaced by $M_{k,N}$ and the tie-breaker ζ_s^k by ζ_k. Notably, independence is established only across the width dimension and not the depth dimension. □

Proof of Corollary 6.4. Fix N and let $r = (r_k)_k \in \{\pm1\}^\infty$. Define the cluster-flip map $g_r : (C_k)_k \mapsto (r_k C_k)_k$. By Assumption 6.2 and cross-cluster independence, g_r is measure preserving. In particular, it leaves every generator of $\mathcal{G}_\infty$ invariant, since

$$S^2_{k,N_p}(r_kC_k) = S^2_{k,N_p}(C_k), \qquad M^2_{k,N'}(r_kC_k) = \big(r_k M_{k,N'}(C_k)\big)^2 = M^2_{k,N'}(C_k) \quad \forall N' \ge 1$$

while sending $\chi_{k,N} \mapsto r_k\chi_{k,N}$. Hence for bounded measurable f and bounded $\mathcal{G}_\infty$-measurable h, $\mathbb{E}[f((\chi_{k,N})_k)h] = \mathbb{E}[f((r_k\chi_{k,N})_k)h]$ at every fixed r. Averaging over r drawn *iid* Rademacher independently of everything, $(r_k\chi_{k,N})_k$ is *iid* Rademacher for any $(\chi_{k,N})_k$, so the inner expectation is the

constant $\mathbb{E}[f(\mathrm{Rad})]$ and $\mathbb{E}\big[f\big((\chi_{k,N})_k\big)\,h\big] = \mathbb{E}\big[f(\mathrm{Rad})\big]\,\mathbb{E}[h]$, delivering both claims. □

E.4 Conditional CLT for a Fixed Design

This subsection establishes uniform mixing of Z_B^π over $\mathcal{G}_\infty$ at a *deterministic* design. Consider $(K_{0B}, N_{0B}) \in \mathbb{N}^2$ with $K_{0B}, N_{0B} \geq 2$, $K_{0B}/B \to c' \in (0,1]$ and $N_{0B} \to N_{\underline{0}}$. Since N_{0B} is integer valued, there is a budget B_0 with $N_{0B} = N_{\underline{0}}$ for all $B \geq B_0$.

Corollary E.5. *Frozen Conditions in the Cluster Problem. There is an almost sure event $\Omega_0 \in \mathcal{G}_\infty$ on which, for every $\delta > 0$,*

$$(i)\quad \frac{1}{K_{0B}} \sum_{k=1}^{K_{0B}} R_{k,N_{0B}}^2 \overset{a.s.}{\underset{B}{\to}} \theta_{N_{\underline{0}}}^2 = \tau^2 + \frac{\sigma^2}{N_{\underline{0}}}, \qquad (ii)\quad \frac{1}{K_{0B}} \sum_{k=1}^{K_{0B}} R_{k,N_{0B}}^2 \,\mathbb{1}\big\{R_{k,N_{0B}} > \delta\sqrt{K_{0B}}\big\} \overset{a.s.}{\underset{B}{\to}} 0$$

Proof of Corollary E.5. For $B \geq B_0$ the depth is fixed at $N_{\underline{0}}$, so both statements reduce to Lemma D.12 applied to the *iid* sequence $(R_{k,N_{\underline{0}}})_k$, which has $\mathbb{E}[R_{k,N_{\underline{0}}}^2] = \theta_{N_{\underline{0}}}^2 < \infty$ by Assumption 3.1. Take Ω_0 to be the corresponding almost sure event. □

Corollary E.6. *L^1 Convergence of the Conditional Characteristic Function. Let $Z_{0B} := K_{0B}^{-1/2} \sum_{k \leq K_{0B}} M_{k,N_{0B}}$, let $\phi_{0B}(t) := \mathbb{E}[\exp(itZ_{0B}) \mid \mathcal{G}_\infty]$, and let $\phi(t) = \exp(-\theta_{N_{\underline{0}}}^2 t^2/2)$. Then*

$$\big|\phi_{0B}(t) - \phi(t)\big| \overset{a.s.}{\underset{B}{\to}} 0 \qquad \text{and} \qquad \mathbb{E}\big|\phi_{0B}(t) - \phi(t)\big| \to 0 \qquad \forall t \in \mathbb{R}$$

Proof of Corollary E.6. Immediate from Theorem D.14 in light of Corollary E.5, the signs $(\chi_{k,N_{\underline{0}}})_k$ being *iid* Rademacher and independent of $\mathcal{G}_\infty$ by Corollary 6.4 at the fixed depth $N_{\underline{0}}$. □

E.5 Conditional CLT for Random Even Designs

Proof of Proposition 6.6. The argument mirrors Theorem D.19 and Corollary D.21. The only difference is that the depth margin must be frozen before the width asymptotics can run. *Step 1.* For fixed B, the decomposition of the conditional characteristic function in Step 1 of Theorem D.19 goes through by summing over the finite grid of budget-feasible designs $\{K_B^\pi = K_{0B}, N_B^\pi = N_{0B}\}$, which is $\mathcal{G}_\infty$-measurable by evenness. On each cell the depth is fixed, so Corollary 6.4 applies and the conditional expectation freezes.

Step 2. Let $\phi_{K_{0B},N_{0B}}(t)$ be the conditional characteristic function at a deterministic design (K_{0B}, N_{0B}), and define $T_B := \sup_{K_0 \geq cB} |\phi_{K_{0B},N_{\underline{0}}}(t) - \phi(t)|$. On the probability-one frozen event $\{N_{0B} = N_{\underline{0}}\}$, $\phi_{K_{0B},N_{\underline{0}}}(t) \underset{B}{\to} \phi(t)$. Hence, $T_B \to 0$ a.s. for any $c \geq c'$. Then

$$\mathbb{P}\big(|\phi_B(t) - \phi(t)| > \eta\big) \;\leq\; \mathbb{P}\big(N_B^\pi \neq N_{\underline{0}}\big) + \mathbb{P}\big(K_B^\pi < cB\big) + \mathbb{P}(T_B > \eta) \;\to\; 0$$

The first term vanishes by N-consistency and the second by non-starvation. Uniform boundedness upgrades convergence in probability to L^1 convergence. *Step 3.* The events $(A_B)_B$ are $\mathcal{G}_\infty$-measurable by evenness, thus Corollary D.15 delivers the uniform mixing statement over $\mathcal{G}_\infty$. The first conditional statement follows by dividing by $\mathbb{P}(A_B)$, eventually bounded away from zero, and applying Lévy's continuity theorem. For the studentized statement, Lemma 4.1.ii) gives $\hat{b}_{K,N_{\underline{0}}}^2 \to \theta_{N_{\underline{0}}}^2$ almost surely along any diverging K, so the same composition argument as in Step 2 guarantees convergence in

probability over the random index. Slutsky concludes exactly as in Corollary D.21. □

E.6 Asymptotically Even Designs

Corollary E.7. *Asymptotically Even Events in the Cluster Problem. Let π be an asymptotically even, N-consistent policy with surrogate $(\tilde{K}_B, \tilde{N}_B)$, and let $A_B = \{(\Xi_B^\pi, N_B^\pi) \in C\}$ for a Borel $C \subseteq \mathbb{R} \times \mathbb{N}$ satisfying the margin condition (10). Set $\tilde{A}_B = \{(\tilde{\Xi}_B, \tilde{N}_B) \in C\} \in \mathcal{G}_\infty$. Then $\mathbb{P}(A_B \triangle \tilde{A}_B) \to 0$.*

Proof of Corollary E.7. Because N_B^π and $\tilde{N}_B$ are integer valued, $|N_B^\pi - \tilde{N}_B| \xrightarrow{p} 0$ is equivalent to $\mathbb{P}(N_B^\pi = \tilde{N}_B) \to 1$. The depth coordinate is asymptotically fixed and can contribute to the symmetric difference only on an event of vanishing probability. Hence

$$\mathbb{P}\big(A_B \triangle \tilde{A}_B\big) \;\le\; \mathbb{P}\big(N_B^\pi \neq \tilde{N}_B\big) + \mathbb{P}\Big(\big\{\mathbb{1}_C(\Xi_B^\pi, N_B^\pi) \neq \mathbb{1}_C(\tilde{\Xi}_B, N_B^\pi)\big\}\Big)$$

where the second term is of the form of the object bounded in Lemma D.28. In particular, the margin condition and $|\Xi_B^\pi - \tilde{\Xi}_B| \xrightarrow{p} 0$ send it to zero. The depth is eventually frozen and enters only through the section $C_{N_0} = \{x : (x, N_0) \in C\}$. □

Lemma E.8. *Increment Vanishes at Ratio-close Designs. Let (K_B^π, N_B^π) and $(\tilde{K}_B, \tilde{N}_B)$ be designs with bounded depths $2 \le N_B^\pi, \tilde{N}_B \le N_M < \infty$ a.s., such that $|\nu_B^\pi - \tilde{\nu}_B| \xrightarrow[B]{p} 0$, $\mathbb{P}(\nu_B^\pi \wedge \tilde{\nu}_B \ge c) \to 1$ for some $c > 0$, and $\mathbb{P}(N_B^\pi = \tilde{N}_B) \to 1$. Writing $\tilde{Z}_B = \tilde{K}_B^{-1/2} \sum_{k \le \tilde{K}_B} M_{k,\tilde{N}_B}$,*

$$\big|Z_B^\pi - \tilde{Z}_B\big| \xrightarrow[B]{p} 0$$

Proof of Lemma E.8. Split the difference at the common depth N_B^π,

$$\big|Z_B^\pi - \tilde{Z}_B\big| \;\le\; \underbrace{\left|\frac{1}{\sqrt{K_B^\pi}} \sum_{k \le K_B^\pi} M_{k,N_B^\pi} - \frac{1}{\sqrt{\tilde{K}_B}} \sum_{k \le \tilde{K}_B} M_{k,N_B^\pi}\right|}_{\text{width}} + \underbrace{\left|\frac{1}{\sqrt{\tilde{K}_B}} \sum_{k \le \tilde{K}_B} \big(M_{k,N_B^\pi} - M_{k,\tilde{N}_B}\big)\right|}_{\text{depth}}$$

The depth term cancels exactly at the probability-1 event $\{N_B^\pi = \tilde{N}_B\}$. For the width term, fix $\eta > 0$ and decompose over the finitely many admissible depths

$$\mathbb{P}\big(\text{width} > \eta\big) \;\le\; \sum_{N=2}^{N_M} \mathbb{P}\left(\left|\frac{1}{\sqrt{K_B^\pi}} \sum_{k \le K_B^\pi} M_{k,N} - \frac{1}{\sqrt{\tilde{K}_B}} \sum_{k \le \tilde{K}_B} M_{k,N}\right| > \eta\right)$$

For each fixed N, the sequence $(M_{k,N})_k$ is *iid*, mean zero, with finite variance θ_N^2, and the indices $K_B^\pi, \tilde{K}_B$ satisfy the hypotheses of Lemma D.31 with asymptotics running over B (not N). The sum runs over finitely many terms, and each of them vanishes completing the result. □

Theorem E.9. *Inference for Asymptotically Even Designs in the Cluster Problem. Let π be an asymptotically even, non-starving, N-consistent policy, and let $(A_B)_B$ be a sequence of scaled design events as in (9) satisfying the margin condition (10). Then*

$$\big|\mathbb{E}\big[\exp(itZ_B^\pi)\,\mathbb{1}_{A_B}\big] - \phi(t)\,\mathbb{P}(A_B)\big| \to 0 \qquad \forall t \in \mathbb{R}$$

with $\phi(t) = \exp(-\theta^2_{N_0} t^2/2)$. *If in addition* $\liminf_B \mathbb{P}(A_B) > 0$, *then*

$$\mathcal{L}\Big(Z_B^\pi \;\Big|\; A_B\Big) \implies N\big(0, \theta^2_{N_0}\big), \qquad \mathcal{L}\left(\frac{\sqrt{K_B^\pi}\big(\hat{\mu}_{K_B^\pi, N_B^\pi} - \mu\big)}{\hat{b}_{K_B^\pi, N_B^\pi}} \;\middle|\; A_B\right) \implies N(0,1)$$

Proof of Theorem E.9. The argument is that of Theorem D.33 with minor changes. *Step 0: the surrogate also inherits N-consistency.* Non-starvation transfers exactly as in Step 0 of Theorem D.33. For N-consistency, $\mathbb{P}\big(\tilde{N}_B \neq N_0\big) \leq \mathbb{P}\big(\tilde{N}_B \neq N_B^\pi\big) + \mathbb{P}\big(N_B^\pi \neq N_0\big) \to 0$, by asymptotic evenness and N-consistency of π respectively. The surrogate is therefore even, non-starving and N-consistent, and so Proposition 6.6 applies.

Step 1: decomposition. Bound the target by the three terms of Step 1 in Theorem D.33, with $(\tilde{K}_B, \tilde{N}_B)$ and $\tilde{A}_B$ in place of the surrogate design and event. The third term is controlled by $|\mathbb{P}(A_B) - \mathbb{P}(\tilde{A}_B)| \leq \mathbb{P}(A_B \triangle \tilde{A}_B) \to 0$ from Corollary E.7, and the second term by Proposition 6.6 applied to the surrogate. *Step 2: the coupling term.* Decompose the first term as in Step 2 of Theorem D.33. The event-mismatch piece is bounded by $\mathbb{P}(A_B \triangle \tilde{A}_B) \to 0$. The increment piece is bounded using the same ingredients: $|e^{ia} - e^{ib}| \leq \min\{2, |a-b|\}$ and $\mathbb{E}[\min\{2, |t||Z_B^\pi - \tilde{Z}_B|\}] \to 0$, by Lemma E.8 and uniform boundedness. Lemma E.8 replaces here Lemma D.31 precisely because the cluster design carries a second, non-divergent coordinate. *Step 3: conditional limits.* Dividing by $\mathbb{P}(A_B)$ and applying Lévy's continuity theorem gives the first conditional statement. Studentization only requires $\hat{b}^2_{K_B^\pi, N_B^\pi} \xrightarrow{p} \theta^2_{N_0}$, which has been established in Step 3 of Proposition 6.6, together with $\liminf_B \mathbb{P}(A_B) > 0$. Measurability of (K_B^π, N_B^π) or A_B was never invoked. □

E.7 Application to `EtC-CS` and `ACS`

It remains to verify that both algorithms are asymptotically even, non-starving and N-consistent, and that the relevant terminal design events satisfy the margin condition. As anticipated in the main text and Appendix D, the margin condition is concretized through a CLT on the design statistic, which requires finite fourth moments. The assumption is strictly weaker than the subgaussian Assumption 5.3 needed for performance.

Assumption E.10. *Fourth Moments.* $\mathbb{E}[u_k^4] < \infty$ *and* $\mathbb{E}[\varepsilon_{ki}^4] < \infty$.

Throughout, recall $\alpha_{k,N_p} = S^2_{k,N_p} - \sigma^2$ and $\beta_{k,N_p} = M^2_{k,N_p} - \theta^2_{N_p}$, both *iid* across clusters, mean zero, and with finite variance under Assumption E.10.

Lemma E.11. *Uniform Rates for the Even Surrogate. Fix N_p and define*

$$\tilde{b}^2_{K,N_p} = \frac{1}{K}\sum_{k \leq K} M^2_{k,N_p}, \qquad \tilde{\lambda}_{K,N_p} = \Pi_{[\lambda_m, \lambda_M]}\left(\frac{\hat{w}^2_{K,N_p}}{\Pi_{[\tau_m^2, \tau_M^2]}\big(\tilde{b}^2_{K,N_p} - \hat{w}^2_{K,N_p}/N_p\big)}\right)$$

Let $K \in [K_M, K_m]$ a.s., then under Assumptions 3.1, 3.2, 3.4 and E.10,

i) $\hat{w}^2_{K,N_p}$, $\tilde{b}^2_{K,N_p}$ *and* $\tilde{\lambda}_{K,N_p}$ *are* $\mathcal{G}_\infty$*-measurable for every K,*

ii) $\sup_K |\hat{b}^2_{K,N_p} - \tilde{b}^2_{K,N_p}| = O_p(1/B)$, *and consequently* $\sup_K |\hat{\lambda}_{K,N_p} - \tilde{\lambda}_{K,N_p}| = O_p(1/B)$,

iii) $\sup_K |\hat{\lambda}_{K,N_p} - \lambda_0| = O_p(B^{-1/2})$ *and* $\sup_K |\tilde{\lambda}_{K,N_p} - \lambda_0| = O_p(B^{-1/2})$,

iv) for some fixed K_{0B}, $A < \infty$, and $J = \lceil A\sqrt{B} \rceil$, $\max_{|j|\le J} \big|\tilde\lambda_{K_{0B}+j,N_p} - \tilde\lambda_{K_{0B},N_p}\big| = O_p\big(B^{-3/4}\big)$

Proof of Lemma E.11. *(i)* $\hat w^2_{K,N_p}$ and $\tilde b^2_{K,N_p}$ are averages of the $\mathcal{C}_\infty$-generators $(S^2_{k,N_p})_k$ and $(M^2_{k,N_p})_k$, and $\tilde\lambda_{K,N_p}$ is a measurable map of the two. Result follows.

(ii) Since $\hat b^2_{K,N_p} = (K-1)^{-1}\big[\sum_{k\le K} M^2_{k,N_p} - K\bar M^2_{K,N_p}\big]$,

$$\hat b^2_{K,N_p} - \tilde b^2_{K,N_p} = \frac{1}{K(K-1)} \sum_{k\le K} M^2_{k,N_p} \;-\; \frac{K}{K-1}\,\bar M^2_{K,N_p}$$

The first term is $(K-1)^{-1}$ times an average converging to $\theta^2_{N_p}$, hence $O_p(1/B)$ uniformly since $\Theta(B) = K_M \le K \le K_m = \Theta(B)$. For the second, Kolmogorov's maximal inequality gives $\max_{K\le K_m}|\sum_{k\le K} M_{k,N_p}| = O_p(\sqrt{K_m})$, so $\sup_{K\le K_m} \bar M^2_{K,N_p} \le O_p(K_m)/K^2_M = O_p(1/B)$, using $K_m/K_M = O(1)$. Therefore, $\sup_{K\le K_m}|\hat b^2_K - \tilde b^2_K| = O_p(1/B)$. For the transfer to λ, note that $\hat\tau^2_{K,N_p}$ and $\tilde\tau^2_{K,N_p} := \Pi_{[\tau^2_m,\tau^2_M]}(\tilde b^2_{K,N_p} - \hat w^2_{K,N_p}/N_p)$ both share the term $\hat w^2_{K,N_p}/N_p$, which cancels, and so $|\tilde\tau^2 - \hat\tau^2| \le |\tilde b^2 - \hat b^2|$ by 1-Lipschitzness of the clip. The two clipped denominators exceed τ^2_m, so

$$\sup_K \big|\hat\lambda_{K,N_p} - \tilde\lambda_{K,N_p}\big| \;\le\; \sup_K \hat w^2_{K,N_p}\, \frac{|\tilde\tau^2_{K,N_p} - \hat\tau^2_{K,N_p}|}{\hat\tau^2_{K,N_p}\,\tilde\tau^2_{K,N_p}} \;\le\; \frac{\sup_K \hat w^2_{K,N_p}}{(\tau^2_m)^2}\, \sup_K \big|\hat b^2_{K,N_p} - \tilde b^2_{K,N_p}\big| = O_p(1/B)$$

using $\sup_K \hat w^2_{K,N_p} \le K_M^{-1}\sum_{k\le K_m} S^2_{k,N_p} = O_p(1)$ and the 1-Lipschitzness of the outer clip.

(iii) By Kolmogorov's maximal inequality applied to $(\alpha_{k,N_p})_k$ and $(\beta_{k,N_p})_k$, both with finite variance under Assumption E.10,

$$\sup_K \big|\hat w^2_{K,N_p} - \sigma^2\big| \le \frac{1}{K_M} \max_{K\le K_m} \Big|\sum_{k\le K} \alpha_{k,N_p}\Big| = O_p\big(B^{-1/2}\big), \qquad \sup_K \big|\tilde b^2_{K,N_p} - \theta^2_{N_p}\big| = O_p\big(B^{-1/2}\big)$$

Combining this result with part (ii), $\sup_K|\hat b^2 - \theta^2_{N_p}| = O_p(B^{-1/2})$. The clipped τ^2 estimator is Lipschitz as the clipping operator is 1-Lipschitz. Moreover, the clipped λ_0 estimator is also Lipschitz because the denominator is bounded below by τ^2_m and Lipschitzness of the clipping map. Using $\lambda_0 \in [\lambda_m, \lambda_M]$, then $\sup_K|\hat\lambda_{K,N_p} - \lambda_0| = O_p(B^{-1/2})$. The statement for $\tilde\lambda$ follows from part (ii) and the triangle inequality.

(iv) This analysis is more subtle. Using the same Lipschitz argument for the clipping and $\lambda_0 \in [\lambda_m, \lambda_M]$, $|\tilde\lambda_{K_{0B}+j} - \tilde\lambda_{K_{0B}}| \le L_g\big(|\hat w^2_{K_{0B}+j} - \hat w^2_{K_{0B}}| + |\tilde b^2_{K_{0B}+j} - \tilde b^2_{K_{0B}}|\big)$. The two terms are handled identically, so I only discuss here $\hat w^2_{K,N_p}$. Decompose $\hat w^2_{K,N_p} = \sigma^2 + T_K/K$ with $T_K = \sum_{k\le K}\alpha_{k,N_p}$. Then, for every $j : |j| \le J$,

$$\hat w^2_{K_{0B}+j} - \hat w^2_{K_{0B}} = \frac{T_{K_{0B}+j} - T_{K_{0B}}}{K_{0B}+j} \;-\; \frac{j\,T_{K_{0B}}}{K_{0B}(K_{0B}+j)}$$

The first term may be interpreted as new noise, and the latter as a re-dilution term. For new noise, Kolmogorov's maximal inequality applied forwards on $(K_{0B}, K_{0B}+J]$ and backwards on $(K_{0B}-J, K_{0B}]$ around the deterministic centre K_{0B} gives $\max_{|j|\le J}|T_{K_{0B}+j} - T_{K_{0B}}| = O_p(\sqrt{J}) = O_p(B^{1/4})$. For the re-dilution term, $|T_{K_{0B}}| = O_p(\sqrt{B})$ by Chebyshev and the fact that $|j| \le J = O(\sqrt{B})$. Since, eventually, $K_{0B} - J = \Theta(B)$,

$$\max_{|j|\le J} \big|\hat w^2_{K_{0B}+j} - \hat w^2_{K_{0B}}\big| = O_p\left(\frac{B^{1/4}}{B}\right) + O_p\left(\frac{\sqrt{B}\cdot\sqrt{B}}{B^2}\right) = O_p\big(B^{-3/4}\big)$$

The proof is completed by carrying the same computation over $\tilde{b}^2_{K,N_p} = \theta^2_{N_p} + K^{-1}\sum_{k\le K}\beta_{k,N_p}$. □

Proposition E.12. *EtC-CS Is Asymptotically Even. Assume $N^*(\lambda_0) \notin \mathbb{N}$. EtC-CS is asymptotically even, non-starving and N-consistent, with even surrogate $(\tilde{K}_B, \tilde{N}_B) = \big(K^*(\tilde{\lambda}_{K_M,N_p}), N^*(\tilde{\lambda}_{K_M,N_p})\big)$ under the rounding convention of the feasible policy. Moreover $|K_B^{\texttt{EtC-CS}} - \tilde{K}_B| = O_p(1)$.*

Proof of Proposition E.12. Non-starvation and N-consistency are given by Corollary 4.7 and Lemma 4.4. The commitment stage K_M is deterministic, so $\tilde{\lambda}_{K_M,N_p}$ is $\mathcal{G}_\infty$-measurable by Lemma E.11.i) and the surrogate design is even, provided the optimal map is trivially measurable. For the width coupling, observe that h is L_h-Lipschitz on $[\lambda_m, \lambda_M]$ and $K^*(\lambda) = Bh(\lambda)$, so by Lemma E.11.ii)

$$\big|K_B^{\texttt{EtC-CS}} - \tilde{K}_B\big| \;\le\; B\,L_h\big|\hat{\lambda}_{K_M,N_p} - \tilde{\lambda}_{K_M,N_p}\big| + 1 \;=\; O_p(1),$$

the 1 accounting for any rounding decisions. Therefore, $|K_B^{\texttt{EtC-CS}} - \tilde{K}_B|/\sqrt{B} \xrightarrow{p} 0$. For the depth, N^* is Lipschitz on $[\lambda_m, \lambda_M]$, so $|N^*(\hat{\lambda}_{K_M,N_p}) - N^*(\tilde{\lambda}_{K_M,N_p})| = O_p(1/B)$, since $N^*(\lambda_0) \notin \mathbb{N}$, the rounding (flooring) coincides with probability tending to one and $|N_B^{\texttt{EtC-CS}} - \tilde{N}_B| \xrightarrow{p} 0$. □

Proposition E.13. *ACS Is Asymptotically Even. ACS is asymptotically even, non-starving and N-consistent, with even surrogate $(\tilde{K}_B, \tilde{N}_B) = \big(K^*(\tilde{\lambda}_{K_{0B},N_p}), N^*(\tilde{\lambda}_{K_{0B},N_p})\big)$, $K_{0B} = \lfloor K^*(\lambda_0)\rfloor$. Moreover $|K_B^{\texttt{ACS}} - \tilde{K}_B| = O_p(B^{1/4})$.*

Proof of Proposition E.13. Non-starvation and N-consistency are given by Corollary 4.7 and Lemma 4.4. The surrogate is even by Lemma E.11.i) and determinism of K_{0B}. For the coupling I proceed in two stages.

Step 1: localizing the commitment stage. Let $D_B := \sup_{K\le K_m}|K^*(\hat{\lambda}_{K,N_p}) - K^*(\lambda_0)| \le BL_h \sup_K|\hat{\lambda}_{K,N_p} - \lambda_0| = O_p(\sqrt{B})$ by Lemma E.11.iii). Recall $s = \inf\{k \ge K_M : k+1 > K^*(\hat{\lambda}_{k,N_p})\}$ from (5). For any $k \ge K^*(\lambda_0) + D_B$ we have $k+1 > K^*(\lambda_0) + D_B \ge K^*(\hat{\lambda}_{k,N_p})$, so the algorithm stops; and, for any $k \le K^*(\lambda_0) - D_B - 1$ we have $k+1 \le K^*(\lambda_0) - D_B \le K^*(\hat{\lambda}_{k,N_p})$, so the stopping condition is not satisfied. Let the commitment stage be equal to $p_B = \max\{s-1, K_M\} \in [K_M, K_m]$. Then, $|p_B - K_{0B}| \le D_B + 2 = O_p(\sqrt{B})$, where the constant absorbs the rounding in K_{0B}.

Step 2: coupling. Fix some constant $A < \infty$ sufficiently large so that the event $\{|p_B - K_{0B}| \le J\}$ with $J = \lceil A\sqrt{B}\rceil$ has probability increasing to 1 (by Step 1). Then

$$\begin{aligned}
\big|K_B^{\texttt{ACS}} - \tilde{K}_B\big| &\le \big|K^*(\hat{\lambda}_{p_B,N_p}) - K^*(\tilde{\lambda}_{p_B,N_p})\big| + \big|K^*(\tilde{\lambda}_{p_B,N_p}) - K^*(\tilde{\lambda}_{K_{0B},N_p})\big| + 1\\
&\le BL_h|\hat{\lambda}_{p_B,N_p} - \tilde{\lambda}_{p_B,N_p}| + BL_h|\tilde{\lambda}_{p_B,N_p} - \tilde{\lambda}_{K_{0B},N_p}| + 1\\
&\le BL_h \sup_K\big|\hat{\lambda}_{K,N_p} - \tilde{\lambda}_{K,N_p}\big| + BL_h \max_{|j|\le J}\big|\tilde{\lambda}_{K_{0B}+j,N_p} - \tilde{\lambda}_{K_{0B},N_p}\big| + 1
\end{aligned}$$

which is $O_p(1) + O_p(B^{1/4})$ by Lemma E.11 parts (ii) and (iv). Hence $|K_B^{\texttt{ACS}} - \tilde{K}_B|/\sqrt{B} \xrightarrow{p} 0$. The depth statement follows as in Proposition E.12, since $|\hat{\lambda}_{p_B,N_p} - \tilde{\lambda}_{K_{0B},N_p}| \xrightarrow{p} 0$. □

Remark E.14. *On the Choice of ACS Surrogate. The surrogate for ACS exploits decisively the content in Remark D.26. The surrogate need be neither feasible nor the "even twin" of ACS, it only needs to be $\mathcal{G}_\infty$-measurable. The natural twin policy which runs the same stopping rule on the even sequence $(\tilde{\lambda}_{k,N_p})_k$ with its own stopping time $\tilde{s}$ would be admissible, but would further require coupling two random stopping times at the $\sqrt{B}$ scale. Instead, evaluating the even statistic at the deterministic stage*

K_{0B} reduces the problem to a single maximal inequality from a fixed center as in Lemma E.11.iv).

Lemma E.15. *Limit Law of the Scaled Cluster Counts. Let Σ be the covariance matrix of $(S^2_{k,N_p}, M^2_{k,N_p})$. Let $G = (G_w, G_b) \sim N(0, \Sigma)$. For $a < b$ and $x \in [a, b]$, define the directional derivative of the clip operator at x as*

$$\Pi'_{[a,b],x}(z) = z \cdot \mathbb{1}\{a < x < b\} + \max\{z, 0\} \cdot \mathbb{1}\{x = a\} + \min\{z, 0\} \cdot \mathbb{1}\{x = b\}$$

Finally, define

$$H := \frac{G_w - \lambda_0 \, \Pi'_{[\tau_m^2, \tau_M^2], \tau^2}\big(G_b - G_w/N_p\big)}{\tau^2}, \qquad L_\pi := \frac{h'(\lambda_0)}{\sqrt{\nu^\pi}} \, \Pi'_{[\lambda_m, \lambda_M], \lambda_0}(H)$$

with $\nu^{\texttt{EtC-CS}} = h(\lambda_M)$ and $\nu^{\texttt{ACS}} = h(\lambda_0)$. Then, for $\pi \in \{\texttt{EtC-CS}, \texttt{ACS}\}$,

$$\Xi_B^\pi \implies L_\pi, \qquad \tilde{\Xi}_B^\pi \implies L_\pi$$

L_π is absolutely continuous on $\mathbb{R} \setminus \{0\}$, and has a point mass at zero, with mass in $(0, 1)$, iff $\lambda_0 \in \{\lambda_m, \lambda_M\}$. In the case where $\lambda_0 \in (\lambda_m, \lambda_M)$ and $\tau^2 \in (\tau_m^2, \tau_M^2)$, then $L_\pi \sim N(0, \Lambda_\pi^2)$ with

$$\Lambda_\pi^2 := \frac{h'(\lambda_0)^2 \, V_\lambda}{\nu^\pi}, \qquad V_\lambda = \frac{1}{\tau^8} \operatorname{Var}\Big(\theta_{N_p}^2 S^2_{k,N_p} - \sigma^2 M^2_{k,N_p}\Big) > 0$$

Proof of Lemma E.15. First, observe that Σ is finite under Assumption E.10 and non-singular by Assumption 3.1. Write $\psi(w, b) = \Pi_{[\lambda_m, \lambda_M]}\big(w/\Pi_{[\tau_m^2, \tau_M^2]}(b - w/N_p)\big)$ such that $\hat{\lambda}_{K,N_p} = \psi(\hat{w}^2_{K,N_p}, \hat{b}^2_{K,N_p})$. The pairs $(S^2_{k,N_p}, M^2_{k,N_p})$ are *iid* across clusters with finite second moments under Assumption E.10, and $\hat{b}^2_{K,N_p}$ differs from $K^{-1}\sum_{k \le K} M^2_{k,N_p}$ by $O_p(1/K)$ (Lemma E.11.ii). Hence, along any deterministic $K \to \infty$, the bivariate central limit theorem gives $\sqrt{K}\big(\hat{w}^2_{K,N_p} - \sigma^2, \hat{b}^2_{K,N_p} - \theta^2_{N_p}\big) \implies G$. The clip $\Pi_{[a,b]}$ is Lipschitz and Hadamard directionally differentiable at every $x \in [a, b]$, with derivative $\Pi'_{[a,b],x}$. The ratio map is continuously differentiable on $\{\tau^2 \ge \tau_m^2 > 0\}$, and so, by the (directional) chain rule (Shapiro, 1990), ψ is Hadamard directionally differentiable at $(\sigma^2, \theta^2_{N_p})$ with derivative $\psi'(G) = \Pi'_{[\lambda_m, \lambda_M], \lambda_0}(H)$. The directional delta method (Shapiro, 1991) then gives $\sqrt{K}(\hat{\lambda}_{K,N_p} - \lambda_0) \implies \psi'(G)$. When $(\tau^2, \lambda_0) \in \operatorname{int}(\Omega)$, then both clips are inactive in an event with probability increasing to 1. A standard delta method then runs through. In particular, ψ' is linear, $\psi'(G) = \nabla g' G \sim N(0, V_\lambda)$ with $\nabla g(\sigma^2, \theta^2_{N_p}) = \Big(\theta^2_{N_p}/\tau^4, -\sigma^2/\tau^4\Big)$. By Lemma E.11.ii) the feasible and surrogate statistics differ by $O_p(1/B) = o_p(B^{-1/2})$, and so they share every $\sqrt{K}$-scale limit.

For `EtC-CS` the commitment stage K_M is deterministic with $K_M/B \to h(\lambda_M)$. Since ψ' is positively homogeneous of degree one, $\sqrt{B}(\hat{\lambda}_{K_M,N_p} - \lambda_0) \implies \psi'(G)/\sqrt{h(\lambda_M)}$ (under the generalized delta method theory above for non-differentiable maps). For `ACS` the commitment stage p_B is random, but $p_B/K_{0B} \xrightarrow{p} 1$ by Step 1 of Proposition E.13 and $K_{0B}/B \to h(\lambda_0)$. Normalized *iid* partial sums are uniformly continuous in probability, so Anscombe's random-index central limit theorem (Anscombe, 1952) applies to the bivariate mean (jointly, via Cramér–Wold) over the random index p_B. Slutsky then gives convergence at the deterministic rate $\sqrt{B}$ with limit $G/\sqrt{h(\lambda_0)}$, and the directional delta method gives $\sqrt{B}(\hat{\lambda}_{p_B,N_p} - \lambda_0) \implies \psi'(G)/\sqrt{h(\lambda_0)}$.

Finally, $\Xi_B^\pi = \sqrt{B}\big(h(\hat{\lambda}_{\cdot,N_p}) - h(\lambda_0)\big) + O(B^{-1/2})$, with the remainder absorbing the rounding in K_{0B}. Since h is continuously differentiable on $(0, \infty)$, one last application of the chain rule through

h delivers $\Xi_B^\pi \implies L_\pi$. The surrogate shares the limit because $|\Xi_B^\pi - \tilde{\Xi}_B^\pi| = |K_B^\pi - \tilde{K}_B^\pi|/\sqrt{B} = o_p(1)$, where the numerator uses $|K - \tilde{K}| = O_p(1)$ in `EtC-CS` and $O_p(B^{1/4})$ in `ACS` by Props. E.12 and E.13.

It remains to characterize L_π. We first show that H has no atoms. Because Σ is nonsingular, G has full support on $\mathbb{R}^2$ and every nonzero linear functional of G is a non-degenerate normal random variable. By construction, H equals one of two such functionals, G_w/τ^2 or $\nabla g' G$, depending on whether the τ^2 clip binds. Hence, for every Lebesgue-null set $\mathcal{n}$,

$$P(H \in \mathcal{n}) \leq P\big(G_w/\tau^2 \in \mathcal{n}\big) + P\big(\nabla g' G \in \mathcal{n}\big) = 0,$$

so H is absolutely continuous. Moreover, H takes both signs with positive probability, since each branch is non-constant on an open set charged by G. In particular, $V_\lambda = \mathrm{Var}(\nabla g' G) > 0$.

The law of L_π then depends only on the λ clip. If $\lambda_0 \in (\lambda_m, \lambda_M)$, the clip's derivative is the identity, so L_π is a nonzero multiple of H and is absolutely continuous. If $\lambda_0 = \lambda_M$, the derivative censors H at zero, so $L_\pi \propto \min\{H, 0\}$. This variable is absolutely continuous on $\mathbb{R} \setminus \{0\}$ and has an atom at zero of mass $P(H \geq 0) \in (0, 1)$. The case $\lambda_0 = \lambda_m$ is symmetric, with $\max\{H, 0\}$ and atom mass $P(H \leq 0) \in (0, 1)$. □

Proof of Theorem 6.9. By Propositions E.12 and E.13 both policies are asymptotically even, non-starving and N-consistent. Let $A_B = \{\Xi_B^\pi \in C\}$ for a Borel $C \subseteq \mathbb{R}$ with $\mathbb{P}(L_\pi \in \partial C) = 0$ and $\mathbb{P}(L_\pi \in C) > 0$. By Lemma E.15 and Remark D.29 the margin condition (10) holds for every C with Lebesgue-null boundary when $\lambda_0 \in (\lambda_m, \lambda_M)$, and for every such C with $0 \notin \partial C$ when $\lambda_0 \in \{\lambda_m, \lambda_M\}$. Theorem E.9 delivers both conditional statements. Intersecting A_B with $\{N_B^\pi = n\}$ is asymptotically innocuous because only $n = N_{\underline{0}}$ carries non-vanishing probability under N-consistency, as discussed extensively in the text. □

Remark E.16. *Theory Informed Remarks on the Simulation Evidence. Three comments. First, whenever $\lambda_0 \in (\lambda_m, \lambda_M)$, the quantile cells of K_B^π used in Section 7 belong to the valid class of asymptotically even events: intervals in Ξ_B^π with Lebesgue-null boundary and limiting probability strictly inside $(0, 1)$. When $\lambda_0 \in \{\lambda_m, \lambda_M\}$, L_π has an atom at zero, so the quantile cells remain in the class only if none of its cut points falls at zero. Second, the variances of the limiting designs are ranked. L_π depends on π only through the scale $1/\sqrt{\nu^\pi}$, hence $L_{\texttt{ACS}} \overset{d}{=} \sqrt{h(\lambda_M)/h(\lambda_0)}\, L_{\texttt{EtC-CS}}$. Since h is strictly decreasing and $\lambda_0 \leq \lambda_M$, $\nu^{\texttt{ACS}} = h(\lambda_0) \geq h(\lambda_M) = \nu^{\texttt{EtC-CS}}$ and $L_{\texttt{ACS}} \leq L_{\texttt{EtC-CS}}$ with equality iff $\lambda_0 = \lambda_M$. By committing around $K^*(\lambda_0)$ rather than at the worst-case pilot K_M, `ACS` estimates the variance ratio on a weakly larger sample and induces a weakly less dispersed design in the limit. Third, Anscombe's theorem enters only for the pilot statistic $\hat{\lambda}_{p_B, N_p}$, where the commitment stage is random. It is silent about the outcome statistic of interest $\hat{\mu}_{K_B^\pi, N_B^\pi}$, whose design-conditional limit is not a random-index problem, what justifies the need for evenness and symmetry.*

F Additional Simulation Evidence

F.1 Performance

Figure 14 traces the median and mean terminal design against the the oracle map (3) evaluated in a grid of variance ratios. Both algorithms track the oracle closely, with a small tendency to oversample depth and undersample width, which can be explained by the right skew of $\hat{\lambda}_{pB}$ documented in Figure

3. Both algorithms display a similar performance with no clear domination. By definition, the static policies induce the same design invariably across λ_0 values.

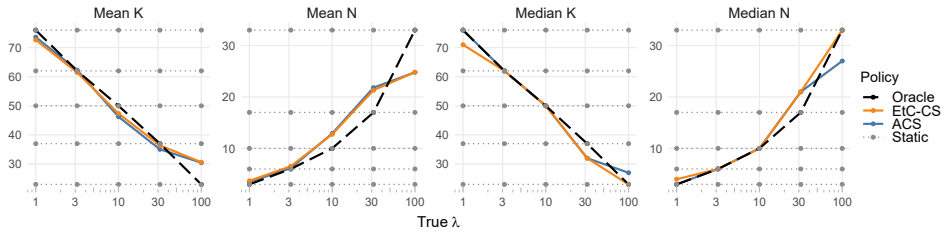


Figure 14: Mean and median terminal design (K_B^π, N_B^π) across λ_0

Notes: Static policies imputed at $\lambda \in \{1, 3.16, 10, 31.6, 100\}$. 10^5 replications. $\mu = 0$, $\tau^2 = 1$, $B = 2{,}000$, $F = 20$, $V = 2$.

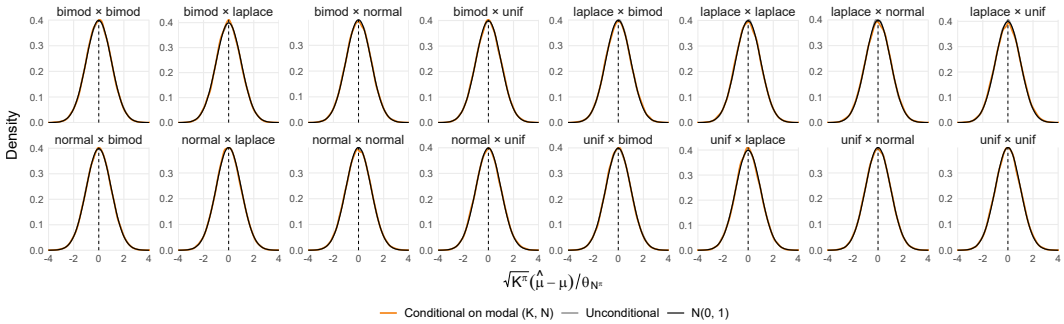


Figure 15: Empirical density of the design-conditional statistic $\sqrt{K_B^\pi}(\hat{\mu}_{K_B^\pi, N_B^\pi} - \mu)/\theta_{N_B^\pi}$ at the modal design across symmetric error distributions

Notes: Distribution of u_k (left) and ε_{ki} (right). 10^5 replications. $\mu = 0$, $\omega = (1, 5)$. $B = 2{,}000$, $F = 20$, $V = 2$.

G Appendix to Empirical Application

G.1 Data and Variable Construction

Sampling frame. All the details below have been obtained from the paper (Fink et al., 2020b), online appendices and pre-register (Fink et al., 2020a), repository data (Fink et al., 2024), and pre-analysis plan (Fink et al., 2014), as well as several communications with the authors and the relevant IPA division in charge of the implementation. FJM began from a random list of 250 villages—five per administrative block, for 50 of the 53 blocks in Chipata District, Eastern Province, Zambia—with populations reported to comprise between 20 and 100 farming households. After inspection and logistical screening, 175 villages entered the experiment. The smallest of these villages contained 22 medium-sized farmers, and so the authors agreed to sample 22 households in all villages. Realized depths, however, ranged from 5 to 22, with an average of 17.9, leading to a baseline sample of 3,139 households, of which 3,028 appear in the year-one survey. Treatment was assigned at the village level after the baseline survey using min-max T-randomization (Bruhn and McKenzie, 2009) resulting in three equally sized arms.

Outcome variables. The experiment's pre-registration lists harvest value as the primary outcome, while the published paper emphasizes reallocation of informal labor (*ganyu*). I therefore target $\log(1+$ total harvest value) and $\log(1 +$ ganyu mandays) as outcome variables. For the second, FJM report the binary variables `did_ganyu` and `plan_ganyu`. Binary outcomes cannot accommodated by the

additive-error model (1), and so I consider a continuous variable alternative. In particular, I use the variable `did_ganyu_mandays_long`, which, despite being available at baseline and in both follow-up rounds, is not reported in the paper. Alternative continuous outcomes like `did_ganyu_hours_yest` and `did_ganyu_hours_short` were not available in the baseline data.

Trimming and residualization. The bottom 0.5% of baseline harvest values (16 observations) is trimmed for implausibly low values. The same absolute threshold (179.5 ZMW) is applied to the year-one data. Both outcomes are then residualized on the covariates entering the harvest value regression in Fink et al. (2020a), Table 4 Panel B, with three exclusions: the outcome's own baseline value, the assignment variable `control_gift`, which is not defined pre-assignment, and—for ganyu mandays only—the mechanically related outcomes `did_ganyu` and `plan_ganyu`. In both settings, harvest and ganyu mandays, I residualize by using the specification of harvest value because the outcome ganyu mandays does not enter any of the reported regressions. All specifications in the authors' analysis share nearly identical controls anyway, so this decision is probably immaterial.

G.2 Homogeneity of the Variance Components

Section 8 identifies the design from baseline data alone. Therefore, the results rest on a variance-homogeneity assumption across survey rounds. I split this assumption in two testable hypotheses. Let y^d_{kit} be the potential outcome d of variable y in period $t \in \{0, 1\}$, then

$$\text{(A1) Time homogeneity} : y^d_{ki1} \overset{d}{=} y^d_{ki0} \text{ for all } d$$

(A1) is only testable on control units, since y^1_{kit} is unobserved at baseline. I run two versions of the test: one using control units in both rounds, which permits a paired bootstrap, i.e. $(\tau^2_{00}, \lambda_{00}) = (\tau^2_{10}, \lambda_{10})$ and one using all baseline units against year-one controls, which gains precision at the cost of pairing, i.e. $(\tau^2_{0,0+1}, \lambda_{0,0+1}) = (\tau^2_{10}, \lambda_{10})$. Secondly,

$$\text{(A2) Arm homogeneity at endline} : y^0_{ki1} \overset{d}{=} y^1_{ki1}$$

(A2) is stronger than required by the framework of Section 3, yet it supplies indirect evidence on the plausibility of (A1) for treated units, which is untestable: if time homogeneity holds for control units and the second moments agree across arms after treatment, then $(\tau^2_{01}, \lambda_{01}) = (\tau^2_{11}, \lambda_{11})$ is likely to hold in the data. Arm homogeneity at baseline holds by construction of the RCT, and so (A1-A2) allow us to drop subscripts entirely from the discussion, simplifying the exposition. Tests are conducted using a cluster bootstrap on log-ratios.

Figure 16 summarizes the results of the tests. Homogeneity can only be rejected marginally in three out of the 20 tests. Two qualifications are worth mentioning. First, $\hat{\tau}^2$ is estimated from roughly 58 villages per arm, so the tests might be low-powered in this dimension. And second, a minority of bootstrap draws produced numerically degenerate cluster variances ($\hat{\tau}^2 \leq 10^{-12}$). These draws (between 0 and 17% of the total) were discarded uniformly across $\hat{\tau}^2$ and $\hat{\lambda}$ estimation.[60]

[60] All estimators $(\hat{\tau}^2, \hat{\lambda})$ use the unclipped counterparts to the estimators in Section 4.1 adapted to unequal village sizes N^k. Replacing the estimators by their standard uniform-depth form, i.e. $N = K^{-1} \sum_k N^k \approx 18$, leaves the input estimates nearly unchanged.

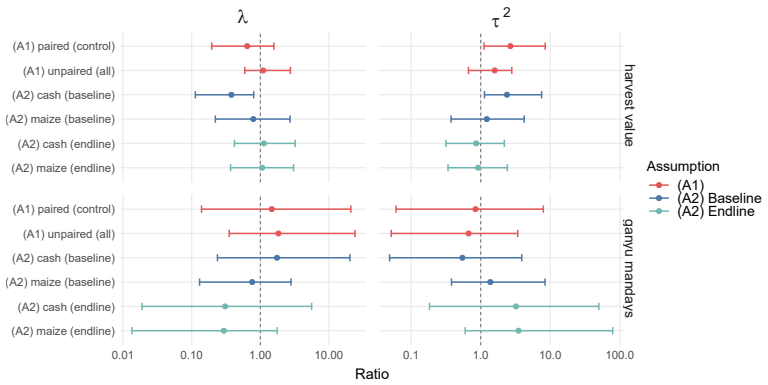


Figure 16: Variance-homogeneity tests

Notes: Cluster bootstrap on log-ratios. Dashed line at 1. 2,000 bootstrap draws.

G.3 Error Symmetry

The inference invariance theory relies on Assumption 6.2 which requires the entire error block $C_k = (u_k, \varepsilon_{k1}, \varepsilon_{k2}, \dots)$ to satisfy sign-flip distributional invariance, i.e. $C_k \overset{d}{=} -C_k$. Unfortunately, when only cluster average residuals $M_{k,N} = u_k + \bar{\varepsilon}_{k,N}$ are available, this assumption is untestable. Testing $M_{k,N} \overset{d}{=} -M_{k,N}$ is necessary but not sufficient for testing $C_k \overset{d}{=} -C_k$. $M_{k,N}$ is an odd linear functional of C_k, so joint symmetry implies $M_{k,N} \overset{d}{=} -M_{k,N}$ at every depth, but the reverse is not true. Testing across a grid of depths is nonetheless informative. For small N, household noise ε_{ik} dominates, while for large N, $M_{k,N} \approx u_k$, and so the cluster-level noise dominates. Therefore, a joint test along the N dimension, i.e. $(M_{k,N})_N = -(M_{k,N})_N$, would examine the two error sources in different proportions. I run such test at $N \in \{4, 8, 12, 16, 20\}$, restricting the sample entering each test to villages with $N^k \geq N$ and randomly drawing N households from each village. Figure 17 plots nearly symmetric distributions for the cluster residuals $\hat{M}_{k,N} = \hat{\mu}^k_N - \hat{\mu}_{K,N}$ at three depths.

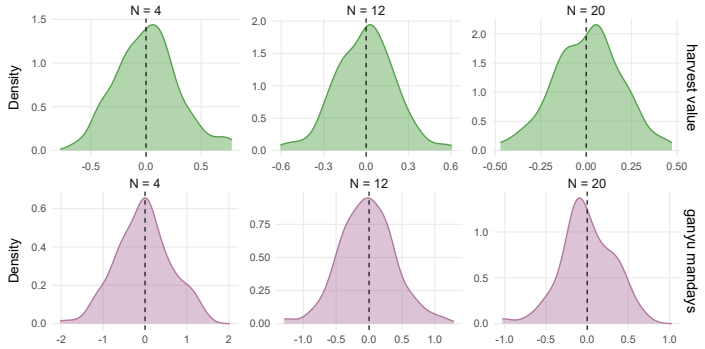


Figure 17: Density of the centered residuals $\hat{M}_{k,N}$

Notes: $N \in \{4, 12, 20\}$ with covariate-residualized data at baseline. Each panel uses a single random draw of N households per village, i.e. unlike Table 7, the results are not pooled across the subsampling draws. Sample includes villages only with $N^k \geq N$. For $N = 20$, only $K = 66$ villages qualify, hence the coarseness of the rightmost panel.

I conduct the symmetry test of Miao et al. (2006), calibrating the null via parametric bootstrap up to an $O(1/K)$ term.[61] The choice of N households per village is not unique, therefore I

[61] The vector of residuals inherits the symmetry properties of the vector of errors, but the per-cluster Rademacher flips used to calibrate the null preserve the joint law only up to an $O(1/K)$ term arising from the common demeaning. At $K = 175$ this adjustment is immaterial.

further bootstrap over sub-sampling draws and report the dependence-robust aggregate $\min\{1, 2 \times \text{median } p\text{-value}\}$ as in Meinshausen et al. (2009). Table 7 reports the results of the test. At standard confidence levels, symmetry cannot be rejected for either outcome at any depth. Although underpowered by construction, the results in this Appendix are conducive to block symmetry and support the implementation of standard inference.

Outcome	N	$k : N^k \geq N$	med(p)	Valid p-value	$Q_{0.10}(p)$	$Q_{0.90}(p)$
log(harvest value + 1)	4	175	0.501	1.000	0.101	0.904
	8	174	0.595	1.000	0.229	0.915
	12	166	0.756	1.000	0.361	0.948
	16	141	0.533	1.000	0.264	0.806
	20	66	0.615	1.000	0.487	0.780
log(ganyu mandays + 1)	4	175	0.461	0.922	0.072	0.864
	8	174	0.457	0.914	0.102	0.911
	12	166	0.496	0.992	0.097	0.894
	16	142	0.589	1.000	0.245	0.923
	20	67	0.500	1.000	0.342	0.854

Table 7: Symmetry test of Miao et al. (2006) on the centered residuals $\hat{M}_{k,N}$

Notes: For each depth, villages with $N^k \geq N$ are subsampled to depth N. The MGG statistic is calibrated by a Rademacher sign-flip null over 2,000 draws. Table reports the median, 10th and 90th percentile for the p-value recomputed over 200 independent sampling draws. Valid p-value $= \min\{1, 2\,\text{med}(p)\}$ is the dependence-robust aggregate of Meinshausen et al. (2009).

G.4 Semi-Synthetic Data Generation

Holding the budget fixed, any counterfactual design will require a population larger than the realized sample, as it will demand either more villages or more households per village than those actually surveyed. Thus, I construct and calibrate a semi-synthetic population for each outcome and each survey round (baseline and year 1).

Starting from the realized data, I residualize the outcome variables using the regression specification. Year-one data follows this procedure exactly. Baseline data omits the outcome's own lag, the assignment variable `control_gift`, and, for ganyu mandays, `did_ganyu` and `plan_ganyu`. I then recover the design input estimates $(\hat{\mu}, \hat{\tau}^2, \hat{\sigma}^2)$ and decompose the residuals into an intercept, a village component and a household component. Residuals are re-scaled and re-centered such that sample moments are matched exactly. Synthetic data are generated under the additive model by sampling u_k and ε_{ki} independently and uniformly from the rescaled empirical residuals. This construction presents two limitations. First, it understates tail mass. Data are constructed by resampling realized residuals and so they cannot extrapolate beyond their support. Second, the model cannot reproduce point masses because it assumes the additive structure in Assumption 3.1, hence the mass at zero in ganyu mandays is smoothed. For instance, the tenth percentile is exactly zero in the real data and -0.26 in the synthetic. Tables 8 and 9 compare the quantile distributions of the synthetic and realized data. Away from the kink at zero, both distributions track closely.

Outcome	Arm	Source	N	Mean	SD	$Q_{0.10}$	$Q_{0.25}$	$Q_{0.50}$	$Q_{0.75}$	$Q_{0.90}$
log(harvest value + 1)	NA	Simulated	36000	0.004	0.515	-0.638	-0.319	0.026	0.338	0.624
		Real	3123	-0.000	0.513	-0.631	-0.315	0.026	0.328	0.611
log(ganyu mandays + 1)	NA	Simulated	36000	0.000	1.193	-1.463	-0.819	-0.064	0.771	1.553
		Real	3136	-0.000	1.190	-1.405	-0.856	-0.080	0.793	1.571

Table 8: Validation of the calibrated DGP against the realized baseline sample

Notes: First and second moments are matched by construction.

Outcome	Arm	Source	N	Mean	SD	$Q_{0.10}$	$Q_{0.25}$	$Q_{0.50}$	$Q_{0.75}$	$Q_{0.90}$
log(harvest value + 1)	cash	Simulated	1044	0.123	0.652	-0.688	-0.259	0.156	0.551	0.913
		Real	1023	0.112	0.645	-0.737	-0.255	0.172	0.517	0.909
	control	Simulated	1044	-0.002	0.647	-0.830	-0.386	0.040	0.427	0.784
		Real	952	-0.000	0.660	-0.860	-0.414	0.044	0.442	0.791
	maize	Simulated	1062	0.023	0.659	-0.817	-0.368	0.062	0.462	0.816
		Real	1016	0.037	0.657	-0.800	-0.338	0.117	0.475	0.812
log(ganyu mandays + 1)	cash	Simulated	1044	-0.175	1.318	-1.877	-1.116	-0.202	0.768	1.559
		Real	600	-0.182	1.293	-1.921	-1.141	-0.214	0.768	1.499
	control	Simulated	1044	-0.004	1.314	-1.690	-0.953	-0.034	0.911	1.742
		Real	575	0.000	1.305	-1.730	-0.979	-0.060	0.990	1.799
	maize	Simulated	1062	-0.223	1.317	-1.910	-1.180	-0.240	0.712	1.518
		Real	636	-0.210	1.347	-1.983	-1.229	-0.285	0.811	1.561

Table 9: Validation of the calibrated DGP against the realized year-one sample.

Notes: First and second moments are matched by construction.

H Appendix to Section 9

Assumption H.1. *Assumptions for the Heterogeneous-Variances Extension. Consider the model with heterogeneous cluster variances in (11).*

(H1) ***Errors and Covariates.*** u_k*:* $\mathbb{E}[u_k \mid X_k] = 0$, $\mathbb{E}[u_k^2 \mid X_k] = \tau_k^2 = \ell(X_k, \gamma)$*;* ε_{ki}*:* $\mathbb{E}[\varepsilon_{ki}] = 0$, $\mathbb{E}[\varepsilon_{ki}^2] = \sigma^2$*, and* $\varepsilon_{ki} \perp\!\!\!\perp X_k$*;* $X \sim F_X \in \mathscr{F}_X$*.*

(H2) ***Compactness.*** $\mathscr{F}_X$ *is the set of distributions in* $\mathcal{X} = [x_m, x_M]$*,* $\gamma \in \Gamma$ *with* $dim(\Gamma) < \infty$*;* $(\mathcal{X}, \Gamma)$ *are both known and compact;* ℓ *continuous and bounded away from zero on* $\mathcal{X} \times \Gamma$*; and* $\sigma^2 \in [\sigma_m^2, \sigma_M^2]$ *with* $\sigma_m^2 > 0$*. Consequently,* $\tau_k^2 \in [\tau_{k,m}^2, \tau_{k,M}^2]$ *with* $\tau_{k,m}^2 = \inf_{\gamma\in\Gamma} \ell(X_k, \gamma)$ *and* $\tau_{k,M}^2 = \sup_{\gamma\in\Gamma} \ell(X_k, \gamma)$*. Let* $\tau_m^2 = \min_k \tau_{k,m}^2$ *and* $\tau_M^2 = \max_k \tau_{k,M}^2$*. The implied* λ_0 *bounds are given by* $\lambda_m = \sigma_m^2/\tau_M^2$ *and* $\lambda_M = \sigma_M^2/\tau_m^2$*. The aggregate pairs* $(\bar{\tau}^2, \bar{\lambda}_0)$ *and* $(\tau_{\mathrm{h}}^2, \lambda_{0\mathrm{h}})$ *lie in the interior of* $[\tau_m^2, \tau_M^2] \times [\lambda_m, \lambda_M]$*.*

(H3) ***Interiority.*** $\dfrac{\tau_m^2}{\tau_M^2}\sqrt{\lambda_m F/V} \geq 2$ *and* $B \geq 2\big(F + V N_{\max}\big)$ *with* $N_{\max} = \dfrac{\tau_M^2}{\tau_m^2}\sqrt{\lambda_M F/V}$*.*

(H4) ***Conditional Subgaussianity.*** $u_k \mid X_k$ *is* κ_u*-subgaussian uniformly in* X_k*,* ε_{ki} *is* κ_ε*-subgaussian.*

The cost structure remains homogeneous across clusters, as in Assumption 3.3, i.e. $KF + V\sum_k^K N^k$.

Proof of Proposition 9.2. In a nutshell, the idea behind the proof is to exploit the convexity of the design variance around the non-adapted minimizer K^* to show that strategic deviations from K^* coming from adaptive cluster selection cannot outweigh the gains coming from selecting the fixed minimizer. Start by fixing $\omega_{\text{het}} \in \Omega_{\text{het}}$ and write $W^* := W^*_{\text{UNW}}$. For some realized design and covariate

sequence, the heterogeneous variance of the unweighted average is given by

$$W_\pi^{\text{het}} = \frac{1}{K^2}\sum_{k=1}^{K}\left(\tau_k^2 + \sigma^2/N^k\right)$$

Consider an oracle policy π which fixes $K = K^*(\bar{\lambda}_0)$, $N^k = N^*(\bar{\lambda}_0)$ for all k before any sampling. The event $\{K \geq k{+}1\}$ is trivially in $\mathcal{I}_k$; depths are deterministic (hence trivially measurable wrt information at the terminal design); and, by definition $K^*(F{+}VN^*) = B$; thus the policy is admissible. Moreover,

$$\mathbb{E}[W_\pi^{\text{het}}] = \frac{\mathbb{E}\left[\sum_{k\leq K^*}\tau_k^2\right]}{K^{*2}} + \frac{\sigma^2}{K^*N^*} = \frac{\bar{\tau}^2}{K^*} + \frac{\sigma^2 V}{B - K^*F} = W_K(K^*) = W^*$$

where the second equality follows from $(\tau_k^2)_k$ being *iid* under (H1)–(H2) and Definition 9.1.(i), and the third equality is used as the definition of the variance as a function of K after solving for an homogeneous depth from the budget constraint. W_K is strictly convex with minimizer K^*.

Step 1: Pathwise lower bound. Define $\xi_k := \tau_k^2 - \bar{\tau}^2$, $T_K := \sum_{k\leq K}\tau_k^2$ and $D_K := T_K - K\bar{\tau}^2 = \sum_{k\leq K}\xi_k$. For any admissible policy with $N^k \geq 2$, induce width K such that $K \leq B/(F{+}2V) < B/F$ a.s., and $\{K \geq k\} \in \mathcal{I}_{k-1}$ for all k. Thus, for any admissible policy K is a bounded $(\mathcal{I}_k)$-stopping time. Next, using Cauchy–Schwarz, $\sum_{k\leq K} 1/N^k \geq K^2/\sum_{k\leq K} N^k \geq K^2V/(B - KF)$. So for any policy π

$$W_\pi^{\text{het}} = \frac{1}{K^2}\sum_{k=1}^{K}\left(\tau_k^2 + \frac{\sigma^2}{N^k}\right) \;\geq\; \frac{T_K}{K^2} + \frac{\sigma^2 V}{B - KF} = W_K(K) + \frac{D_K}{K^2}$$

Step 2: Wald. Re-write $D_K = \sum_{k\leq B/F}\mathbb{1}(K \geq k)\xi_k$. Since $\{K \geq k\} \in \mathcal{I}_{k-1}$ and ξ_k *iid* and $\perp\!\!\!\perp \mathcal{I}_{k-1}$ with $\mathbb{E}[\xi_k] = 0$, then, K is a stopping time wrt the natural filtration generated by $(\xi_k)_k$, and so, by Wald's Identity $\mathbb{E}[D_K] = 0$. Therefore, one may subtract the mean-zero object $\mathbb{E}[D_K]/K^{*2} = 0$ to obtain

$$\mathbb{E}\big[W_\pi^{\text{het}}\big] - W^* \;\geq\; \mathbb{E}[A], \qquad A := W_K(K) - W^* + D_K\Big(\frac{1}{K^2} - \frac{1}{K^{*2}}\Big)$$

This re-write decomposes the gap A into a deterministic optimality gap $W_K(K) - W^*$ and a centered random perturbation, which disappears as $K^2 \to K^{*2}$.

Step 3: Curvature. On $(0, B/F)$, $W_K''(K) \geq 2\bar{\tau}^2/K^3 \geq 2c_1/B^3$ with $c_1 := \bar{\tau}^2F^3$. Since $W_K'(K^*) = 0$, a second-order expansion gives

$$W_K(K) - W^* \;\geq\; \frac{c_1}{B^3}(K - K^*)^2, \qquad K \in (0, B/F)$$

Step 4: Partition. Fix $\varepsilon \in (0, h]$, not depending on B, small enough that $\tau_m^2/\varepsilon > BW^* + \tau_M^2\varepsilon/h^2$, which is possible since $BW^* = (\sqrt{F\bar{\tau}^2} + \sqrt{V\sigma^2})^2$ does not depend on B.

(a) $K < \varepsilon B$. Then $T_K/K^2 \geq \tau_m^2/K > \tau_m^2/(\varepsilon B)$, and, since $D_K \leq T_K \leq \tau_M^2 K$ and $K^* = hB$, $D_K/K^{*2} < \tau_M^2\varepsilon/(h^2B)$. Hence

$$A \;>\; \frac{1}{B}\Big[\frac{\tau_m^2}{\varepsilon} - BW^* - \frac{\tau_M^2\varepsilon}{h^2}\Big] \;>\; 0$$

(b) $K \geq \varepsilon B$. Both K and K^* are at least εB. By the mean value theorem applied to $x \mapsto x^{-2}$,

$|K^{-2} - K^{*-2}| = 2|K - K^*|/\zeta^3 \le 2|K - K^*|/(\varepsilon B)^3$ for some intermediate ζ between K and K^*. Combining with Step 3 and writing $x = |K - K^*|$, $b = |D_K|/\varepsilon^3$,

$$A \;\ge\; \frac{1}{B^3}\big(c_1 x^2 - 2bx\big) \;\ge\; -\frac{b^2}{c_1 B^3} = -\frac{D_K^2}{c_1 \varepsilon^6 B^3},$$

where the last inequality uses $c_1 x^2 - 2bx \ge -b^2/c_1$ for all $x \ge 0$. To bound $\mathbb{E}[D_K^2]$, observer that all cross-terms vanish because for all $j < k$, $\xi_j \mathbb{1}\{K \ge j\}\mathbb{1}\{K \ge k\} \in \mathcal{G}_{k-1}$ and the $(\xi_k)_k$ sequence is conditionally *iid*. Therefore,

$$\mathbb{E}[D_K^2] = \mathrm{Var}(\tau_k^2) \sum_k P(K \ge k) = \mathrm{Var}(\tau_k^2)\, \mathbb{E}[K] \le \mathrm{Var}(\tau_k^2)\, B/F$$

Combining both partition events,

$$\mathbb{E}[A] \;\ge\; \mathbb{E}\big[A\, \mathbb{1}\{K \ge \varepsilon B\}\big] \;\ge\; -\frac{\mathrm{Var}(\tau_k^2)}{c_1 \varepsilon^6 F} \cdot \frac{1}{B^2} =: -\frac{C}{B^2}.$$

with the constant C only depending on $(F, V, \omega_{\mathrm{het}})$ and the domain bounds in (H2), not on B or π. □

Proof of Corollary 9.3. Fix $\omega_{\mathrm{het}} \in \Omega_{\mathrm{het}}$ and recall the notation $(\xi_k, T_K, D_K, W_K, W^*)$ in Proposition 9.2. Let $\hat{\lambda} := \hat{\lambda}_{K_M, N_m}$ be the `EtC-CS` pilot estimate, $\mathcal{P}$ the σ-algebra generated by the pilot data, and $(K, N) := (K^*(\hat{\lambda}), N^*(\hat{\lambda}))$ the terminal `EtC-CS` design.

Step 0: Verifying the Conditions in Lemma C.5. By (H1), $(X_k, u_k, (\varepsilon_{ki})_i)$ is *iid* across k. Hence the pairs $(S_{k,N}^2, M_{k,N})$ are *iid* across clusters, with $\mathbb{E}[S_{k,N}^2] = \sigma^2$, $\mathbb{E}[M_{k,N}] = 0$ and $\mathbb{E}[M_{k,N}^2] = \bar{\tau}^2 + \sigma^2/N$. By (H4) and the tower property, $\mathbb{E}[e^{su_k}] = \mathbb{E}\big[\mathbb{E}[e^{su_k} \mid X_k]\big] \le e^{s^2\kappa_u^2/2}$, so u_k is κ_u-subgaussian and Lemma C.5 therefore applies verbatim with (τ^2, λ_0) replaced by $(\bar{\tau}^2, \bar{\lambda}_0)$.

Step 1: Decomposition. `EtC-CS` is trivially admissible, hence the decomposition in Proposition 9.2 applies, i.e.

$$W^{\mathrm{het}}_{\texttt{EtC-CS}} = W_K(K) + \frac{D_K}{K^2}, \qquad \mathbb{E}\left[W^{\mathrm{het}}_{\texttt{EtC-CS}}\right] - W^* = \mathbb{E}\left[W_K(K) - W^*\right] + \mathbb{E}\left[\frac{D_K}{K^2}\right]$$

The first term is bounded as in Theorem 5.4, i.e.

$$\mathbb{E}\big[W_K(K) - W^*\big] \le \frac{\tau_M^2 \sqrt{FV}}{4\lambda_m^{3/2}} \cdot \frac{\mathbb{E}[(\hat{\lambda} - \bar{\lambda}_0)^2]}{B} \le \frac{C}{B^2}$$

Step 2: Composition term. Write $D_K = D_{K_M} + \sum_{K_M < k \le K} \xi_k$. Since K is $\mathcal{P}$-measurable and $(\xi_k)_{k > K_M}$ is independent of $\mathcal{P}$ with mean zero, $\mathbb{E}\big[K^{-2} \sum_{K_M < k \le K} \xi_k \,\big|\, \mathcal{P}\big] = 0$. For the pilot term, K_M is deterministic, so $\mathbb{E}[D_{K_M}] = 0$ and $\mathbb{E}[D_{K_M}^2] = K_M \mathrm{Var}(\tau_k^2)$. Mimicking the previous proof device: since $K, K_0 \ge K_M$, the mean value theorem and the Lipschitz property of h give $|K^{-2} - K_0^{-2}| \le 2|K - K_0|/K_M^3 \le 2BL_h|\hat{\lambda} - \bar{\lambda}_0|/K_M^3$. Cauchy–Schwarz concludes via,

$$\left|\mathbb{E}\left[\frac{D_K}{K^2}\right]\right| = \left|\mathbb{E}\left[D_{K_M}\left(\frac{1}{K^2} - \frac{1}{K_0^2}\right)\right]\right| \le \frac{2BL_h}{K_M^3} \sqrt{K_M \mathrm{Var}(\tau_k^2)} \sqrt{\frac{C_\lambda}{K_M}} \le \frac{C}{B^2}$$

where I used that $K_M \ge Bh(\lambda_M)/2$ and $\mathrm{Var}(\tau_k^2) \le (\tau_M^2 - \tau_m^2)^2/4$.

Bringing both terms together delivers the first statement. The second statement follows directly from Proposition 9.2. □

Proof of Proposition 9.4. Fix $\omega_{\text{het}} \in \Omega_{\text{het}}$ and some admissible policy π (with $K \geq 1$, $N^k \geq 2$, and weights $(\vartheta_k)_k$ satisfying $\sum_k \vartheta_k \neq 0$). By the same argument as in Proposition 9.2, $K < B/F$ is a bounded $(\mathcal{I}_k)$-stopping time. Write $W^* := W^*_{\text{GLS}}$, and let $\rho_k := 1/\tau_k^2 \in [1/\tau_M^2, 1/\tau_m^2]$, so that $\mathbb{E}[\rho_k] = 1/\tau_{\text{h}}^2$.

Step 1: Pathwise bound. By Cauchy–Schwarz, $\left(\sum_k \vartheta_k\right)^2 = \left(\sum_k (\vartheta_k \theta_k)\, \theta_k^{-1}\right)^2 \leq \sum_k \vartheta_k^2 \theta_k^2 \cdot \sum_k \theta_k^{-2}$, so $W_\pi^{\text{het}} \geq 1/\sum_{k\leq K} \theta_k^{-2}$ for any weights. Next, I use a parallel sum argument. For $a, b \geq 0$ with $a + b > 0$, the parallel sum satisfies $a : b := ab/(a+b) = \min_{x\in[0,1]}\{ax^2 + b(1-x)^2\}$. Hence, evaluating every summand at the minimizer x of the aggregate problem with nonnegative $(a_k, b_k)_k$ gives

$$\sum_k a_k : b_k \;\leq\; \Big(\sum_k a_k\Big) : \Big(\sum_k b_k\Big)$$

Replace $a_k = \rho_k$ and $b_k = N^k/\sigma^2$, such that $\theta_k^{-2} = (\tau_k^2 + \sigma^2/N^k)^{-1} = a_k : b_k$. Define $S_K := \sum_{k\leq K} \rho_k = K/\tau_{\text{h},K}^2$. From the budget constraint $\sum_k N^k \leq (B - KF)/V$, thus

$$W_\pi^{\text{het}} \;\geq\; \left(S_K : \tfrac{1}{\sigma^2} \sum_k N^k\right)^{-1} = \frac{1}{S_K} + \frac{\sigma^2}{\sum_k N^k} \;\geq\; \frac{1}{S_K} + \frac{\sigma^2 V}{B - KF}$$

Step 2: Wald. Following Proposition 9.2, write $S_K = \sum_{k\leq B/F} \mathbb{1}\{K \geq k\}\rho_k$. By Definition 9.1.(i), $\{K \geq k\} \in \mathcal{I}_{k-1}$ and $\rho_k \perp\!\!\!\perp \mathcal{I}_{k-1}$, so $\mathbb{E}[S_K] = \sum_k \mathbb{E}[\rho_k] P(K \geq k) = \mathbb{E}[K]/\tau_{\text{h}}^2$.

Step 3: Jensen. Define $W_K^{\text{h}}(K) := \frac{\tau_{\text{h}}^2}{K} + \frac{\sigma^2 V}{B - KF}$ for $K \in (0, B/F)$. By Cauchy–Schwarz, $W_K^{\text{h}}(K) \geq \left(\sqrt{F\tau_{\text{h}}^2} + \sqrt{V\sigma^2}\right)^2/B$ for all K, with equality only if $K = K^*_{\text{GLS}}$. $s \mapsto 1/s$ is convex on $(0, \infty)$, and $K \mapsto \sigma^2 V/(B - KF)$ is convex on $(0, B/F) \supset [K_M, K_m]$. Then,

$$\mathbb{E}\left[W_\pi^{\text{het}}\right] \;\geq\; \frac{1}{\mathbb{E}[S_K]} + \frac{\sigma^2 V}{B - F\,\mathbb{E}[K]} = W_K^{\text{h}}\left(\mathbb{E}[K]\right) \;\geq\; W^*$$

where the first inequality follows from Jensen and Step 2, and the second inequality from Step 1. The first statement follows. For the second statement, I now show that a policy π which fixes width $K := K^*_{\text{GLS}}$, selects X^K-measurable depth $N^k := (\tau_{\text{h},K}^2/\tau_k^2)\,\bar{N}$, where $\bar{N} := \sqrt{\lambda_{0\text{h}} F/V}$, and uses GLS weights attains the lower bound W^* up to an $O(B^{-2})$ term. First, such policy is trivially admissible: K is deterministic, so $\{K \geq k+1\} \in \mathcal{I}_k$ trivially. The depths are functions of X^K, which Definition 9.1.(ii) allows through free revisits. Importantly, the policy is budget feasible a.s.,

$$\sum_{k\leq K} N^k = \bar{N}\,\tau_{\text{h},K}^2 S_K = K\bar{N} \implies KF + V\sum_k N^k = K\big(F + \sqrt{\lambda_{0\text{h}} FV}\big) = B, \quad \text{a.s.}$$

For some realized X^K, note that $\tau_k^2 N^k = \tau_{\text{h},K}^2 \bar{N}$ for all k, hence $\theta_k^2 = \tau_k^2\big(1 + \sigma^2/(\tau_{\text{h},K}^2 \bar{N})\big)$, and therefore

$$W_\pi^{\text{het}} = \frac{1}{\sum_k \theta_k^{-2}} = \frac{1}{S_K} + \frac{\sigma^2}{\tau_{\text{h},K}^2 S_K \bar{N}} = \frac{1}{S_K} + \frac{\sigma^2}{K\bar{N}} = \frac{1}{S_K} + \frac{\sigma^2 V}{B - KF}$$

i.e. Step 1 holds with equality (conditional on the realized X^K). I finally bound the expectation of the random deviations. For any fixed K, let $Z_K := S_K - K/\tau_{\text{h}}^2$, so that $\mathbb{E}[Z_K] = 0$ and $\mathbb{E}[Z_K^2] = K\,\text{Var}(\rho_k)$.

The identity

$$\frac{1}{S_K} = \frac{\tau_{\mathrm{h}}^2}{K} - \frac{\tau_{\mathrm{h}}^4 Z_K}{K^2} + \frac{\tau_{\mathrm{h}}^4 Z_K^2}{K^2 S_K}$$

and $S_K \geq K/\tau_M^2$ give $\mathbb{E}[1/S_K] \leq \tau_{\mathrm{h}}^2/K + \tau_{\mathrm{h}}^4 \tau_M^2 \mathrm{Var}(\rho_k)/K^2$. Hence

$$\mathbb{E}\big[W_\pi^{\mathrm{het}}\big] \;\leq\; W_K^{\mathrm{h}}(K_{\mathrm{GLS}}^*) + \frac{\tau_{\mathrm{h}}^4 \tau_M^2 \mathrm{Var}(\rho_k)}{K_{\mathrm{GLS}}^{*2}} = W^* + \frac{C}{B^2}$$

with $C := \tau_{\mathrm{h}}^4 \tau_M^2 \mathrm{Var}(\rho_k)\big(F + \sqrt{\lambda_{0\mathrm{h}} F V}\big)^2$ not depending on B. □

Proof of Proposition 9.5. Expanding $W_{\mathrm{UNW}}^* = \bar{\tau}^2(\sqrt{V\bar{\lambda}_0} + \sqrt{F})^2/B$ using $\bar{\tau}^2\bar{\lambda}_0 = \sigma^2$ and $\bar{\tau}^2\sqrt{V\bar{\lambda}_0} = \bar{\tau}\sqrt{V\sigma^2}$,

$$B \cdot W_{\mathrm{UNW}}^* = V\sigma^2 + 2\sqrt{VF\sigma^2}\,\bar{\tau} + F\bar{\tau}^2, \qquad B \cdot W_{\mathrm{GLS}}^* = V\sigma^2 + 2\sqrt{VF\sigma^2}\,\tau_{\mathrm{h}} + F\tau_{\mathrm{h}}^2$$

and differencing gives (14). Nonnegativity of both terms follows from the arithmetic–harmonic mean inequality, $\bar{\tau}^2 \geq \tau_{\mathrm{h}}^2$, with equality if and only if τ_k^2 is degenerate. Both gaps are increasing in the dispersion of τ_k^2 in the usual convex-order sense. □

Estimation of the Variance Link. The unit-level variance is estimated as in the main text after accommodating for heterogeneous depths, i.e.

$$\hat{w}_K^2 = \frac{1}{K}\sum_k \frac{1}{N^k - 1} \sum_{i \leq N^k} \big(y_{ki} - \hat{\mu}_{N^k}^k\big)^2, \qquad \hat{w}_K^2 \overset{\text{a.s.}}{\underset{K}{\to}} \sigma^2$$

Each cluster contributes a single draw of u_k, so τ_k^2 is identified only by pooling through the link's parametric reduction. The estimation strategy resembles that of non-linear least squares with pseudo-outcomes $(q_{k,K})_k$ in place of u_k. Define

$$q_{k,K} = \frac{K(\hat{\mu}_k - \hat{\mu})^2 - \hat{b}_K^2}{K-2} - \frac{\hat{w}_K^2}{N^k} = \frac{K(M_k - \bar{M})^2 - \hat{b}_K^2}{K-2} - \frac{\hat{w}_K^2}{N^k} \tag{17}$$

where $\hat{\mu} = K^{-1}\sum_k \hat{\mu}_{N^k}^k$ is the unweighted grand-mean, $\hat{\mu}_k = \hat{\mu}_{N^k}^k$, $M_k = u_k + \bar{\varepsilon}_{k,N^k}$ and $\hat{b}_K^2 = (K-1)^{-1}\sum_k(\hat{\mu}_k - \hat{\mu})^2$. The pseudo-outcomes are conditionally unbiased for $K \geq 3$,

$$\begin{aligned}
\mathbb{E}[q_{k,K} \mid X^K] &= \frac{1}{K-2}\left(K\mathbb{E}[(M_k - \bar{M})^2 \mid X^K] - \bar{\theta}^2\right) - \frac{\sigma^2}{N^k} \\
&= \frac{1}{K-2}\left(K\left(1 - \frac{1}{K}\right)^2 \theta_k^2 + \frac{1}{K}\sum_{j \neq k} \theta_j^2 - \bar{\theta}^2\right) - \frac{\sigma^2}{N^k} \\
&= \theta_k^2 - \frac{\sigma^2}{N^k} = \tau_k^2
\end{aligned}$$

where the first equality uses $\mathbb{E}[\hat{b}_K^2 \mid X^K] = \bar{\theta}^2$ and $\mathbb{E}[\hat{w}_K^2 \mid X^K] = \sigma^2$, similarly to the results in Appendix B; and the second equality uses cross-cluster independence. Next, consider the nonlinear least squares estimator of γ,

$$\hat{\gamma} = \arg\min_{\gamma \in \Gamma} \; \frac{1}{K}\sum_k \big(\ell(X_k, \gamma) - q_{k,K}\big)^2$$

$\hat{\gamma}$ is $\sqrt{K}$-consistent by standard M-estimation arguments under the usual regularity conditions: (i) identification, the map $\gamma \mapsto \mathbb{E}[(\ell(X,\gamma) - \ell(X,\gamma_0))^2]$ is uniquely minimized at γ_0 with $\mathbb{E}[\nabla_\gamma \ell \nabla_\gamma \ell']$ nonsingular; (ii) compactness as in (H2); and (iii) finite conditional fourth moments of the outcomes. There is one non-standard argument in this process: the pseudo-outcomes are not independent across k, as they share the terms $\hat{\mu}$, $\hat{w}^2_K$ and $\hat{b}^2_K$. However, the induced perturbation of the NLS score should be $O_p(1/K)$, which does not affect rates.

Algorithm 3. Finally, I introduce the algorithm and discuss its properties alongside Conjecture 9.6. Define the clipped plug-ins as expected,

$$\hat{\tau}^2_k = \ell(X_k, \hat{\gamma}), \qquad \hat{\tau}^2_{\mathrm{h},K} = \frac{K}{\sum_{k \le K} 1/\hat{\tau}^2_k}, \qquad \hat{\lambda}_{\mathrm{h}} = \Pi_{[\lambda_m,\, \lambda_M]} \left(\frac{\hat{w}^2_{K_M}}{\hat{\tau}^2_{\mathrm{h},K_M}} \right)$$

Since $\hat{\gamma} \in \Gamma$, $\hat{\tau}^2_k \in [\tau^2_{k,m}, \tau^2_{k,M}]$ and $\hat{\tau}^2_{\mathrm{h},K} \in [\tau^2_m, \tau^2_M]$, so no further clipping is required.

Algorithm 3 `wEtC-CS`

Input $B, F, V, \Gamma, [x_m, x_M], \ell(\cdot,\cdot), [\sigma^2_m, \sigma^2_M]$
compute $\tau^2_m, \tau^2_M, \lambda_m, \lambda_M, \quad K_M = \lfloor K^*(\lambda_M) \rfloor$
Stage 1 (pilot)
for $1 \le k \le K_M$: **recruit** cluster k, **observe** X_k, **sample** $N^k_p = \left\lfloor \frac{\tau^2_m}{\tau^2_{k,M}} \sqrt{\lambda_m F/V} \right\rfloor$ units
estimate $\hat{w}^2_{K_M}$, $\hat{\gamma}$, $\hat{\tau}^2_{\mathrm{h},K_M}$, $\hat{\lambda}_{\mathrm{h}}$ from the pilot data
commit to the width $K_B^{\texttt{wEtC-CS}} = K^*(\hat{\lambda}_{\mathrm{h}})$
Stage 2 (exploration)
for $K_M < k \le K_B^{\texttt{wEtC-CS}}$: **recruit** cluster k, **observe** X_k, **sample** N^k_p units
Stage 3 (allocation)
re-compute $\hat{\tau}^2_k$ for all $k \le K_B^{\texttt{wEtC-CS}}$ and $\hat{\tau}^2_{\mathrm{h},K_B^{\texttt{wEtC-CS}}}$ ($\hat{\gamma}$ is not re-computed)
top up every cluster $k \le K_B^{\texttt{wEtC-CS}}$ to $N_B^{k,\texttt{wEtC-CS}} = \frac{\hat{\tau}^2_{\mathrm{h},K_B^{\texttt{wEtC-CS}}}}{\hat{\tau}^2_k} \sqrt{\hat{\lambda}_{\mathrm{h}} F/V}$ units
return $\hat{\mu}_{\hat{\vartheta}}$ with weights $\hat{\vartheta}_k = \left(\hat{\tau}^2_k + \hat{w}^2_{K_M}/N^k\right)^{-1}$

I further note that the algorithm satisfies the budget constraint a.s.. By definition of the realized harmonic mean at width $K_B^{\texttt{wEtC-CS}} = K$, $\sum_{k \le K} N^k = \hat{\tau}^2_{\mathrm{h},K} \sqrt{\hat{\lambda}_{\mathrm{h}} F/V} \sum_{k \le K} 1/\hat{\tau}^2_k = K\sqrt{\hat{\lambda}_{\mathrm{h}} F/V}$, hence

$$KF + V \sum_{k \le K} N^k = K^*(\hat{\lambda}_{\mathrm{h}}) \left(F + \sqrt{\hat{\lambda}_{\mathrm{h}} FV} \right) = B \qquad \text{a.s.}$$

Because the design is nested throughout, interim cost never exceeds the terminal cost, and the budget constraint holds along the entire sampling path. The realized normalization is feasible because depths are set only after all covariates have been observed, which is granted under free revisit. In particular, compared to `EtC-CS`, all clusters—and not just pilot clusters—are visited twice, once at recruitment and once at the top-up.